

Mental Healthcare Facilities and Mortality: Evidence from Local Access and Insurance Expansion

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Abstract

Do mental health treatment facilities save lives? Using county-level variation in facility openings from County Business Patterns data matched to the universe of U.S. mortality records (1999–2016), I estimate heterogeneity-robust event studies of a county’s first facility: all-cause deaths fall by roughly 3 percent, about 1 percent of deaths in population-weighted per-person terms, with no differential pre-trends. In the full county panel, each additional facility prevents 1.56 deaths per 100,000 residents annually. A cause-of-death decomposition attributes the effect to mental health related deaths and to chronic-disease deaths, with placebo causes unchanged, direct evidence that treatment spills over onto physical health. Medicaid expansion amplifies facility effects, indicating that insurance coverage and treatment infrastructure are complements: facilities in high-uninsurance expansion counties prevent 1.80 all-cause and 0.16 MBD deaths per 100,000. Facilities operate partly by connecting patients to disability benefits and prescription treatment. Conservative estimates imply \$11.7 million in annual net benefits per facility.

JEL Classifications: H51, I10, I12, I13, I18, J24, R12;

Keywords: Mental Health, Healthcare Access, Treatment Facilities, Mortality, Medicaid Expansion, Insurance Coverage, Healthcare Infrastructure, Asymmetric Effects.

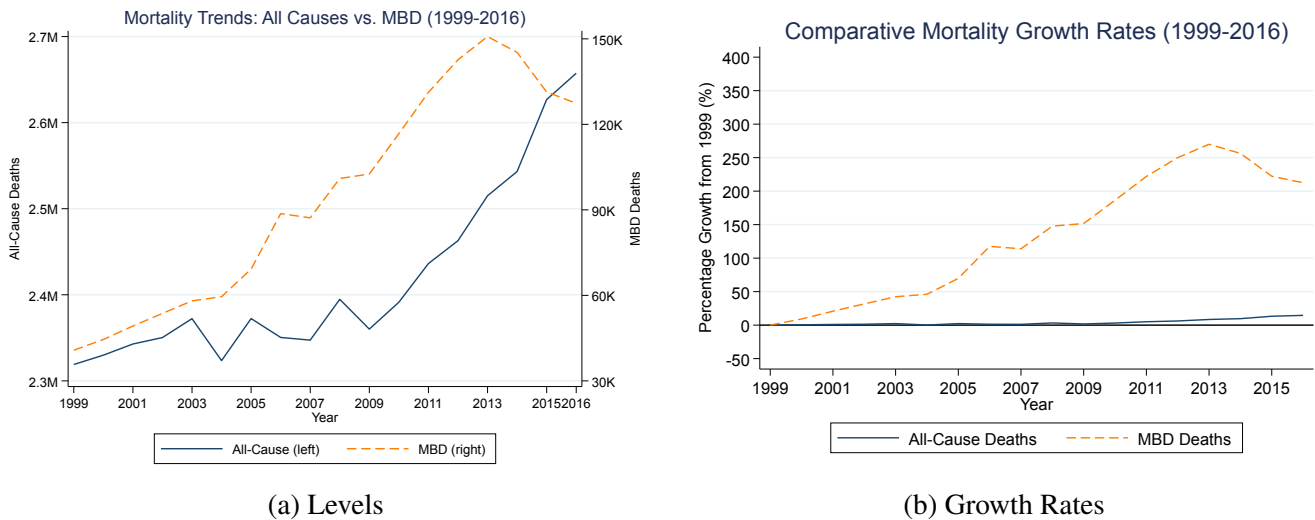
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1 Introduction

Does local access to mental health treatment save lives? Deaths from mental and behavioral disorders (MBD) in the United States increased nearly fivefold between 1999 and 2016, from roughly 30,000 to more than 120,000 annually, making them one of the fastest growing mortality categories (Figure 1). This surge occurred even as the number of mental health treatment facilities grew by 50 percent, from about 12,000 to more than 18,000. The coexistence of expanding treatment capacity and deteriorating outcomes raises a first order policy question: does mental health infrastructure actually reduce mortality, and under what conditions? This paper provides causal evidence that it does. In event studies of counties gaining their first mental health treatment facility, all-cause deaths fall by roughly 3 percent, about 1 percent of deaths in population-weighted per-person terms, and in the full county panel each additional facility prevents 1.56 deaths per 100,000 residents annually. A cause-of-death decomposition shows that the effect extends well beyond deaths directly attributed to mental illness, but the benefits depend critically on complementary insurance coverage.

Figure 1: Divergent Mortality Trends: The Mental Health Crisis



Notes: Panel (a) shows total deaths from all causes (left axis) and mental and behavioral disorder (MBD) deaths (right axis) from 1999-2016. Panel (b) shows the percentage growth from the 1999 baseline. While all-cause mortality increased modestly by 15 percent over this period, MBD deaths surged nearly 300 percent from 40,000 to over 120,000 annually. Data from restricted-access Multiple Cause of Death files. MBD deaths are defined as ICD-10 codes F00-F79.

The stakes are large. Mental illness affects roughly one in five American adults, and mental health disorders cost the global economy approximately 4 percent of GDP annually, with projections suggesting these costs will double by 2030 (Bloom et al., 2012; Chisholm et al., 2016). In the United States, productivity losses alone exceed \$193 billion annually through reduced labor productivity, wage penalties, longer unemployment spells, and the underutilization of human capital (WHO, 2021; Ettner et al., 1997; Currie et al., 2024). Yet despite these stakes, causal evidence on whether treatment infrastructure reduces mortality has been scarce. Recent work establishes that access to mental health care matters for specific

populations and outcomes: [Costantini \(2025\)](#) shows that delays in follow up care after mental health emergencies raise two year mortality among veterans, and [Shakya and RoyChoudhury \(2026\)](#) finds that improved proximity to mental health providers reduces county suicide rates. What remains unknown is whether the physical infrastructure of mental health treatment affects population mortality broadly, across all causes of death, and how this effect interacts with insurance coverage.

I answer these questions using establishment level data from County Business Patterns matched with the universe of U.S. mortality records over 1999 to 2016. The empirical strategy exploits within county variation in the number of mental health treatment facilities, in a difference-in-differences framework with county and state-by-year fixed effects. To address concerns about heterogeneous treatment effects in staggered settings ([Goodman-Bacon, 2021](#); [de Chaisemartin and D’Haultfoeuille, 2020](#); [Callaway and Sant’Anna, 2021](#)), I implement the [Sun and Abraham \(2021\)](#) interaction weighted estimator, the first application of this method to mental health infrastructure. Because the siting of facilities is plausibly endogenous to local health conditions, I lead with the event-study evidence rather than the dose-response panel estimates, and I report joint Wald tests of parallel pre-trends alongside every design; the tests never reject in the headline specifications. Effects are reported in two sets of units throughout: death counts, which are estimated most precisely, and population-weighted deaths per 100,000, which give the per-person magnitude. I validate that facility counts capture real treatment capacity: each facility raises county mental health employment by 3 percent, or 8 to 10 workers ([Eckert et al., 2020](#)). Placebo tests on causes of death that mental health treatment should not plausibly affect, such as complications of medical and surgical care, show null effects.

The results establish four findings. First, facilities reduce mortality broadly, and the event studies show it most cleanly: a county’s first facility reduces all-cause deaths by 3.0 percent ($p < 0.01$) in the years after opening, about 1 percent of deaths in population-weighted per-person terms, with joint tests never rejecting parallel pre-trends. In the full panel, each additional facility lowers all-cause mortality by 1.56 deaths per 100,000 residents. I reconcile these magnitudes explicitly: the panel estimate is an intensive-margin dose response (one more facility in a county that on average already has five), while the event-study estimate is the extensive margin of a first facility in an underserved county, and the two align once units, weighting, and samples are held fixed. Second, the mortality reduction is not confined to deaths coded to mental illness. A cause-of-death decomposition on the event sample attributes 60 percent of the all-cause effect to mental health related causes and 40 percent to chronic and other internal causes, with placebo causes flat, direct evidence that mental health treatment spills over onto physical health. Third, insurance coverage and treatment infrastructure are complements. Using Medicaid expansion under the ACA as quasi-experimental variation, facilities in expansion states prevent 1.53 all-cause and 0.11 MBD deaths per 100,000 after expansion, rising to 1.80 and 0.16 in counties with high baseline uninsurance, where expansion binds most. Fourth, facilities operate through mechanisms beyond direct treatment: they serve as gateways to disability programs, raising SSI participation by 1.4 percent, and, combined with insur-

ance, they enable pharmaceutical treatment, generating up to 280 additional antidepressant prescriptions per 100,000 residents. Two further patterns discipline the interpretation. In the dose-response panel, the association between facility churn and MBD mortality is carried by closings rather than openings, with the formal test rejecting symmetry in the richest specification; and county populations decline by about 1.4 percent after a first facility opens, which makes the per-person estimates conservative and motivates conducting the cause decomposition on death counts.

This paper makes four contributions to the health economics literature. First, I provide the first causal evidence that mental health infrastructure reduces overall mortality, extending the healthcare access literature beyond physical health facilities. Prior work on facilities and mortality focused on general healthcare (Buchmueller et al., 2006; Gujral and Basu, 2019) or substance abuse treatment (Swensen, 2015; Bondurant et al., 2018): Bailey and Goodman-Bacon (2015) show that Community Health Centers reduced age adjusted mortality by 2 percent over ten years, and Buchmueller et al. (2006) find that hospital closures increased deaths among infants and the elderly. The closest recent evidence on mental health care examines single outcomes or specific populations, suicide in Shakya and RoyChoudhury (2026) and veteran mortality after treatment delays in Costantini (2025). My finding that mental health facilities reduce all cause county mortality, with effects far exceeding mental health specific deaths, places mental health infrastructure alongside hospitals and primary care as essential health infrastructure (Currie et al., 2023; Alexander and Richards, 2023; Miller et al., 2021).

Second, I provide the first evidence on how mental healthcare infrastructure churn relates to mortality, and I report it with the caution the data demand. In joint estimation, facility closings are associated with higher MBD mortality (0.139 deaths per 100,000, $p < 0.01$) while openings show no dose-response effect; the test of asymmetry rejects in the most demanding specification ($p = 0.021$), though not in sparser ones, and dedicated closings event studies are underpowered. This pattern is consistent with the irreversible disruption of patient and provider relationships emphasized by Corredor-Waldron and Currie (2022) for hospital closures and Frank (2006) for mental healthcare, and it cautions that dose-response estimates of churn confound siting decisions, which is precisely why the paper leads with clean first-opening event studies.

Third, I establish that insurance and infrastructure are complements, not substitutes, in reducing mortality. I implement the first interaction analysis of how insurance modifies mental health treatment facility effectiveness. The complementarity advances the insurance literature (Finkelstein, 2007; Miller, 2012; Sommers et al., 2013) by showing that coverage without capacity yields limited benefits, and extends infrastructure research (Swensen, 2015; Bondurant et al., 2018) by demonstrating that facilities without coverage cannot maximize health gains. Recent work has increasingly recognized interactions between coverage and provision (Finkelstein and Hendren, 2020; Bradford and Maclean, 2023; Ortega, 2023); this paper is the first to quantify their mortality implications for mental health treatment facilities.

Fourth, I identify mechanisms that explain why the mortality effects extend beyond mental health specific causes. Facilities connect vulnerable populations to disability programs and the income support and healthcare access those programs provide (Messel et al., 2023; Deshpande, 2016). With insurance, facilities enable pharmaceutical treatment. These pathways align with evidence that mental health treatment improves the management of chronic physical conditions (Walker et al., 2015; Druss et al., 2011), and with recent evidence that mental healthcare utilization responds to the surrounding institutional environment (Salcedo et al., 2026).

The heterogeneity of the effects reveals important disparities. Elderly populations experience the largest benefits, consistent with higher baseline risk and mobility constraints (Currie et al., 2024). Less educated individuals see twice the mortality reduction of the college educated, suggesting that geographic proximity matters most for resource constrained populations (Salcedo et al., 2026; Deza et al., 2022a). These patterns extend evidence on healthcare proximity gradients (Currie et al., 2023; Fadlon et al., 2025) to mental health, which exhibits even stronger distance decay than physical health services.

The welfare implications are substantial. Each facility generates \$15.1 million in annual mortality benefits against \$3.4 million in costs, yielding net benefits of \$11.7 million and a benefit-cost ratio above 4 to 1. These returns justify public investment, particularly given market failures in mental healthcare provision, including externalities, information asymmetries, and insurance imperfections (Frank and McGuire, 2000; Bloom et al., 2012).

Beyond health economics, the findings speak to several strands of the broader literature. They extend research on geography and healthcare utilization (Wennberg and Gittelsohn, 1973; Baicker and Chandra, 2006; Chandra and Staiger, 2007; Skinner, 2011; Finkelstein et al., 2016; Molitor, 2018) by showing that mental health exhibits stronger proximity gradients than physical healthcare, and they inform work on insurance and provider interactions (Clemens and Gottlieb, 2014; Cabral et al., 2021; Gottlieb et al., 2018) by quantifying complementarities between coverage and capacity. By revealing spillover effects on overall mortality, the paper connects mental health treatment to the broader determinants of population health (Chetty et al., 2016; Dwyer-Lindgren et al., 2017; Case and Deaton, 2015a,b), showing that mental healthcare infrastructure deserves consideration alongside traditional medical facilities in explaining geographic mortality variation.

The remainder of the paper proceeds as follows. Section 2 describes the institutional and policy background. Section 3 presents the conceptual framework. Sections 4 and 5 describe the data and empirical strategy. Section 6 presents the main results, Section 7 the insurance interaction analysis, Section 8 the mechanisms, and Section 9 the welfare analysis. Section 10 concludes.

2 Background

2.1 Mental Health Crisis and Mental Behavioral Disorder Mortality

The United States faces an unprecedented crisis in mental health outcomes, characterized by a striking empirical puzzle: mortality from mental and behavioral disorders (MBD) has risen sharply even as treatment infrastructure has expanded (Online Appendix Figures A1 and A2).¹

While all-cause mortality increased modestly by 15 percent from 1999 to 2016, MBD deaths surged 300 percent (Figure 1). This divergence suggests that standard healthcare improvements have failed to address mental health.

This apparent disconnect between expanding treatment capacity and deteriorating health outcomes suggests potential mismatches in both the geographic distribution of mental health services and insurance coverage for accessing them. While national trends show infrastructure growth, significant local variation exists in facility access. Figure 2a reveals substantial heterogeneity across counties, with some areas experiencing facility closures despite the overall upward trend. This pattern aligns with Buchmueller et al. (2006)'s finding that local access to healthcare facilities critically influences both utilization and outcomes, particularly when combined with adequate insurance coverage.

2.2 Geographic Access, Insurance Coverage, and Treatment Utilization

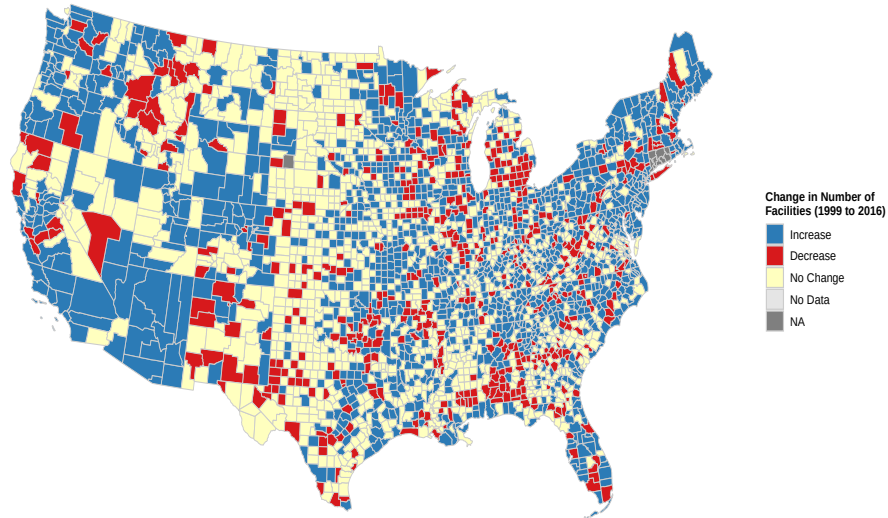
The distribution of mental health facilities shows substantial geographic and temporal variation that interacts critically with insurance coverage patterns. The average U.S. county has 5.3 facilities per 100,000 residents (Table 1), but this masks considerable heterogeneity: rural counties average 2.1 facilities per 100,000 while urban counties average 7.8. This variation creates a unique setting for studying how treatment availability affects health outcomes, particularly when combined with insurance expansions that remove financial barriers to care.

The demographic pattern of MBD deaths indicates potential mismatches between service availability, insurance coverage, and population needs. While individuals aged 65 and older experience the highest mortality rates (26.51 per 100,000), working-age adults face significant barriers to both treatment access and insurance coverage. Prior to the ACA, approximately 20 percent of adults with mental illness were uninsured, with rates exceeding 30 percent in some states (Grooms and Ortega, 2019). These disparities

¹These statistics are derived from International Classification of Diseases (ICD-10) codes specifically referencing mental and behavioral disorder deaths. According to the National Institute of Mental Health, mental illness affects approximately one in five U.S. adults, with mortality consequences extending beyond direct mental health causes to include excess deaths from suicide, substance abuse, and comorbid physical conditions (Walker et al., 2015; Druss et al., 2011).

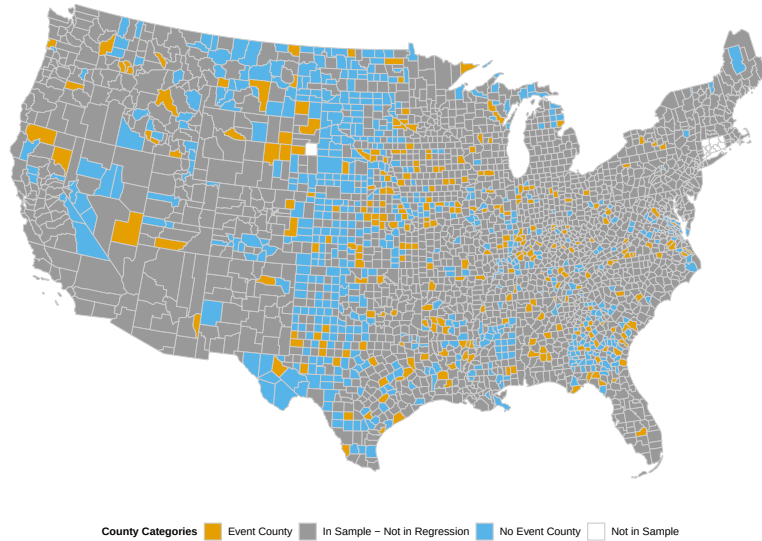
Figure 2: Spatial Variation in Access to Mental Health Treatment Facilities

Changes in Mental Health Treatment Facilities by County
Based on Number of Establishments (1999–2016) – Continental US



(a) Spatial Variation in Mental Health Treatment Facilities

Event Study Sample
Counties by Facility Change Categories (Continental US)



(b) Spatial Variation in Mental Health Treatment Facilities: Event Study Sample

Note: Data on mental health treatment facilities comes from the County Business Patterns. The top panel (a) displays the spatial distribution of changes in mental health treatment facilities across U.S. counties between 1999 and 2016. Section 4.1 presents more information on these data. Counties are categorized by whether they experienced an increase (blue), decrease (red), or no change (yellow) in the number of facilities over this period. The bottom panel (b) shows the event study sample classification, where “Event Counties” (orange) represent those that experienced exactly one change in facilities during the study period, “No Event Counties” (light blue) had no changes, and “In Sample - Not in Regression” (gray) experienced multiple changes. The sample consists of 275 event counties, 653 no-event counties, and 2,202 counties with multiple changes, while 92 counties were not in the sample. Alaska and Hawaii are included in the analysis but shown as insets due to their geographic separation from the continental United States.

Table 1: Summary Statistics

	(Mean)	(Std. dev.)
<i>Mental health treatment facilities</i>		
Total facilities	53.32	95.10
Net openings	0.35	0.48
Net closings	0.24	0.43
Outpatient facilities	28.09	50.85
Inpatient facilities	24.66	44.24
Facilities (per 100,000)	5.16	3.73
<i>MBD Deaths (per 100,000)</i>		
All	29.42	21.40
Age under 19	0.03	0.17
Ages 20–34	0.27	0.58
Ages 35–49	0.99	1.26
Ages 50–64	1.63	1.72
Ages 65+	26.51	20.60
White	26.49	20.35
Black	2.50	3.99
Other	0.42	1.25
Counties	3,199	
County-year observations	63,650	

Notes: All statistics are weighted by county population. The sample includes all U.S. counties with available data from 1999–2016. MBD refers to Mental and Behavioral Disorder deaths.

reflect differences in both physical access to facilities and financial ability to utilize available services.

2.3 Treatment Delivery and Insurance Market Interactions

Mental health treatment in the United States operates through a complex network of outpatient and residential facilities whose effectiveness depends critically on patients' ability to pay for services. Treatment episodes typically include diagnostic assessment, medication management, psychotherapy, and support services. Most care (90 percent) occurs in outpatient settings, where the availability of insurance coverage determines treatment intensity and duration.²

The complementarity between physical infrastructure and insurance coverage creates important policy implications. [Sommers et al. \(2016\)](#) find that Medicaid expansion increased mental health treatment utilization by 25 percent in areas with adequate facility capacity but had minimal effects in underserved areas. This suggests that neither insurance expansion nor facility development alone maximizes health benefits; coordinated policies addressing both dimensions of access are necessary for improving population mental health outcomes.

2.4 Mental Health Treatment Infrastructure and Employment

The mental health treatment sector has evolved significantly since the era of deinstitutionalization, shifting toward community-based care delivery ([Frank, 2006](#)). This transition produced substantial variation in facility locations over time, with distinct waves of openings and closures across counties. Importantly, these facility changes represent real expansions in treatment capacity: each additional facility increases mental health employment by approximately 3 percent, translating to 8 to 10 additional mental health workers including psychiatrists, psychologists, social workers, and support staff ([Eckert et al., 2020](#)).³

The industry's structure exhibits important features that influence service provision. Treatment facilities are predominantly privately owned (87 percent), split between non-profit (60 percent) and for-profit (27 percent) organizations, with public facilities comprising the remaining 13 percent. The Affordable Care Act's Medicaid expansion fundamentally altered this financing landscape by providing comprehensive mental health coverage to millions of previously uninsured Americans ([Glier and Frank, 2016](#)), though capacity constraints remain a significant barrier to care delivery even with expanded insurance coverage.

²According to SAMHSA (2020), insured patients receive 2.5 times more outpatient mental health visits than uninsured patients, even when facilities are geographically accessible. The ACA's essential health benefits provisions specifically require coverage for mental health and substance abuse services at parity with medical care.

³Following [Swensen \(2015\)](#) and [Bondurant et al. \(2018\)](#), this variation provides identification for estimating causal effects while accounting for underlying mental health trends. The employment response validates that facility counts capture meaningful changes in treatment capacity rather than administrative reorganizations.

2.5 Policy Environment: From Parity Laws to the Affordable Care Act

The policy landscape for mental health has undergone a dramatic transformation, shifting from state-level parity laws to comprehensive federal reform under the Affordable Care Act. While earlier state parity legislation studied by [Popovici et al. \(2017\)](#); [Lang \(2013\)](#) attempted to equalize coverage between mental and physical health services, implementation varied widely, and many individuals remained uninsured. The ACA fundamentally restructured mental health access through two mechanisms: Medicaid expansion, which extended coverage to adults below 138 percent of the federal poverty level, and essential health benefits requirements mandating mental health coverage in all insurance plans.

The staggered implementation of Medicaid expansion across states, following the Supreme Court's 2012 ruling making expansion optional, creates quasi-experimental variation for identifying policy effects. By 2016, 31 states had expanded Medicaid, with expansion particularly benefiting individuals with mental illness who previously faced both geographic and financial barriers to treatment ([Maclean and Saloner, 2019](#)). Evidence from early expansion states suggests that insurance coverage amplifies the mortality-reducing effects of mental health facilities, with the largest benefits in counties with high pre-expansion uninsurance rates.

3 Conceptual Framework

This section develops a framework for understanding how mental health facilities affect mortality through multiple pathways and why facility closures may have larger effects than openings in Online Appendix Figure [A3](#).

3.1 Primary Pathway: Local Access to Treatment

Mental health facilities affect mortality primarily through providing local access to treatment. Geographic proximity reduces travel costs, time burdens, and logistical barriers that prevent individuals from receiving care. This local access effect operates independently of insurance status; even uninsured individuals benefit from nearby facilities through sliding scale fees, charity care, and crisis intervention services.

However, the effectiveness of local access varies with insurance coverage. Insurance amplifies the baseline access effect by removing financial barriers and enabling more intensive treatment. While facilities provide benefits to all residents through improved local access, those with insurance can utilize services more frequently and consistently. This suggests two distinct but related effects: a direct effect of geographic access and an additional complementary effect when access combines with insurance coverage.

3.2 Asymmetric Adjustment Costs

The framework incorporates asymmetric adjustment costs between facility entry and exit. When facilities open, individuals gain an additional treatment option but may continue with existing care arrangements or self-management strategies. The marginal benefit depends on the quality differential between the new facility and current alternatives.

When facilities close, however, all current patients face immediate disruption. Finding new providers involves search costs, waiting periods, and rebuilding therapeutic relationships. These adjustment costs are particularly high for individuals experiencing acute symptoms who may struggle with the cognitive and emotional demands of navigating care transitions. These frictions suggest closures should generate larger mortality effects than openings, though the magnitude remains an empirical question.

3.3 Multiple Mechanisms

Beyond direct treatment provision, facilities may affect mortality through several indirect channels. First, mental health providers serve as gateways to social support programs, supplying documentation necessary for disability applications and connecting patients to wraparound services. Second, facilities enable prescription access even for those with limited ability to pay, through pharmaceutical assistance programs and sample medications. Third, if facilities represent real treatment capacity rather than administrative entities, we should observe corresponding changes in local mental health employment. These mechanisms suggest facilities affect mortality through both direct treatment and broader community health infrastructure.

4 Data

4.1 County Business Patterns

I measure access to mental health treatment using establishment counts from the U.S. Census Bureau's County Business Patterns (CBP). The CBP data are derived from annual tax returns submitted to the Internal Revenue Service (IRS) and capture nearly all establishments operating during the week of March 12th each year. These data have been extensively used in recent economic research to study the effects of healthcare access (Deza et al., 2022a,b; Swensen, 2015; Bondurant et al., 2018).

Following the literature on treatment facilities (Swensen, 2015; Bondurant et al., 2018; Bradford and Maclean, 2023), I identify mental health treatment facilities using North American Industry Classifica-

tion System (NAICS) codes. Specifically, I include establishments classified under NAICS codes 621420 (outpatient mental health and substance abuse centers) and 623220 (residential mental health and substance abuse facilities). Each establishment represents a single physical location where mental healthcare services are provided, and establishments can only be assigned one NAICS code.

While these establishments primarily focus on mental healthcare, they may also provide substance use disorder (SUD) treatment services. This comprehensive classification allows me to capture the full spectrum of mental health treatment facilities available in each county. Consistent with previous studies (Swensen, 2015; Bondurant et al., 2018), I lag facility counts by one year in my regression analyses to account for the lag between IRS reporting and CBP publication, the time required for facility establishment, patient access, and potential improvements in mental health outcomes.

My analytical sample includes 53,498 county-year observations spanning 1999 to 2016. This extensive coverage enables me to examine how changes in local access to mental health treatment facilities affect mortality outcomes while controlling for time-invariant county characteristics and common temporal shocks that might influence both facility location decisions and mortality patterns.

4.2 Mental Health Treatment Facilities Employment Data

To establish the relationship between facility openings and actual healthcare provision, I utilize employment data from the County Business Patterns following the imputation methodology developed by Eckert et al. (2020). The CBP employment data present unique challenges as the majority of county-industry cells have suppressed values to protect establishment confidentiality. Eckert et al. (2020) address this limitation through a linear programming method that exploits the hierarchical adding-up constraints implicit in the data structure to impute missing employment values.⁴

For my analysis, I extract employment data for mental health establishments using NAICS codes 621420 and 623220, consistent with my facility count measures. These employment figures capture the lower bound of full-time and part-time employees working in mental health treatment facilities during the payroll period that includes March 12th of each year. The employment data serve as my first-stage outcome, allowing me to verify that facility openings translate into increased mental healthcare capacity as measured by sectoral employment. This validation is crucial for establishing that changes in facility counts represent meaningful expansions in treatment availability rather than merely administrative changes or establishment reorganizations.

⁴Eckert et al. (2020) develop a comprehensive imputation procedure that leverages the nested structure of industry classifications and geographic aggregations in the CBP data. Their method ensures that imputed values satisfy all disclosed totals and subtotals while minimizing deviations from observed patterns.

4.3 Vital Statistics Mortality Data

My primary outcomes are mortality rates constructed from the Multiple Cause of Death data files from the National Center for Health Statistics (NCHS) for the period 1999 to 2016. These restricted access data capture all recorded deaths in the United States and provide detailed geographic identifiers at the county level, allowing me to track mortality patterns across space and time.

I examine three categories of mortality outcomes. First, I measure *overall mortality* from all causes of death to capture the comprehensive health effects of mental health facility access. This broad measure includes both direct effects on mental health related mortality and indirect effects through improved management of comorbid conditions, reduced substance abuse, and other pathways (Walker et al., 2015; Druss et al., 2011).

Second, I focus on *mental and behavioral disorder (MBD) mortality*, defined from ICD-10 Chapter V (Mental and Behavioral Disorders, codes F00-F99).⁵ I examine deaths attributed to eleven distinct diagnostic categories: organic mental disorders (F00-F09), substance use disorders (F10-F19), schizophrenia and psychotic disorders (F20-F29), mood disorders (F30-F39), anxiety disorders (F40-F48), behavioral syndromes (F50-F59), personality disorders (F60-F69), intellectual disabilities (F70-F79), developmental disorders (F80-F89), childhood onset disorders (F90-F98), and unspecified mental disorders (F99). I construct the aggregate measure of total MBD mortality by summing deaths with an underlying cause in codes F00 through F79; the remaining Chapter V blocks (developmental disorders, childhood onset disorders, and unspecified mental disorders, F80-F99) are excluded because they are rarely recorded as the underlying cause of death. This aggregate provides the overall effect on mental health related mortality, while the diagnostic subcategories allow examination of heterogeneous effects across condition types.⁶

Third, I examine *component causes of mortality* to understand mechanisms and rule out alternative explanations. These include mental health related causes (intentional self harm via ICD-10 codes X60 to X84, and substance use disorders), external causes potentially affected by mental health (assault, transport accidents, and other accidental injuries), and placebo outcomes not expected to respond to mental health treatment (deaths from medical and surgical care complications, and other natural causes). This categorization allows me to test whether effects are specific to mental health related causes or reflect broader changes in mortality reporting or healthcare quality.

To construct my dependent variables, I aggregate death counts at the county year level for each cause category and normalize them by county population (per 100,000 residents), using annual population estimates from the U.S. Census Bureau. This normalization ensures that my mortality rates are comparable

⁵The ICD-10 classification system is documented at <https://icd.who.int/browse10/2019/en?blockCode=V>. Classification includes dementia in Alzheimer disease (F00) and dementia in other diseases classified elsewhere (F02), based on ICD-10 Chapter V: Mental and Behavioral Disorders (World Health Organization, 2019).

⁶Online Appendix Table A25 provides the complete classification structure.

across counties of different sizes and over time.⁷

4.4 Medicare Part D Prescription Data

To examine mechanisms through which access to mental health facilities affects outcomes, I incorporate Medicare Part D prescription drug data from 2013 to 2016. These administrative data, provided by the Centers for Medicare and Medicaid Services (CMS), contain comprehensive information on prescription drug claims for Medicare beneficiaries, covering approximately 70 percent of all Medicare enrollees during this period (Centers for Medicare & Medicaid Services, 2023).

I focus on prescriptions for nine major antidepressant medications that account for the vast majority of depression treatment in the United States: sertraline, escitalopram, trazodone, bupropion, fluoxetine, citalopram, duloxetine, venlafaxine, and amitriptyline. These medications span multiple pharmacological classes, including selective serotonin reuptake inhibitors (SSRIs), serotonin-norepinephrine reuptake inhibitors (SNRIs), and tricyclic antidepressants, representing both first-line and adjunctive treatments for depression and anxiety disorders (Cipriani et al., 2018).

The Medicare Part D data are aggregated to the state-year level to ensure sufficient statistical power while maintaining geographic variation. I construct my prescription access measure as the total number of antidepressant claims per state-year, which provides a direct measure of realized access to pharmacological mental health treatment. This outcome complements my mortality analysis by capturing an intermediate mechanism through which facility expansion may improve mental health outcomes. The prescription data are particularly valuable as they reflect actual treatment utilization rather than merely potential access, allowing me to trace the causal chain from facility availability through treatment uptake to ultimate health outcomes.

4.5 Social Security Disability Data

To examine whether mental health facilities affect disability program participation, I obtain county-level data on Supplemental Security Income (SSI) recipients from the Social Security Administration for 2009–2016.⁸ The data provide counts of individuals receiving SSI disability payments, which require

⁷For county year cells with zero deaths in a particular category, I assign a mortality rate of zero. The NCHS suppresses county level data when death counts are small (typically fewer than 10 deaths) to protect privacy; I treat suppressed values as missing. Data suppression is minimal for overall mortality and aggregate MBD mortality due to aggregation across multiple cause categories, but more common for specific diagnostic subcategories and rare causes of death, particularly in small rural counties.

⁸SSA publishes annual county-level counts of SSI recipients by age group. I use recipients aged 18–64, the working-age population eligible for disability benefits based on mental health conditions. SSA county-level data are available at https://www.ssa.gov/policy/docs/statcomps/ssi_sc/

documented medical conditions that prevent substantial gainful activity.⁹

I merge the SSA disability data with my main county-year panel by FIPS code, restricting the analysis to 2009–2016 to match the SSA data availability.¹⁰ I normalize disability recipients by county population to create a rate per 100,000 residents. This shorter panel (2009–2016) yields approximately 25,000 county-year observations for the mechanism analysis, compared to the full mortality panel (1999–2016) with 56,000 observations.

4.6 Additional Data Sources

I use data from several sources to account for differences between counties that experience changes in mental health treatment facility access and those that do not. I gather time-varying county-level information from the National Institute for Health Surveillance (SEER) ([Surveillance, Epidemiology, and End Results](#), SEER) and the Regional Economic Information System (REIS) ([Bureau of Economic Analysis, BEA](#)). I obtain data on personal income from REIS, including per capita net earnings, welfare receipts, and employment to population ratios, to capture each county’s economic conditions that might influence both mental health outcomes and facility locations. SEER provides population counts by age group, which is crucial for controlling for demographic factors that could affect mental health treatment needs and mortality patterns. I also use the 1993 rural-urban continuum codes from the U.S. Department of Agriculture to identify rural and urban counties, as access to mental health facilities may vary systematically with urbanization levels.

To examine heterogeneous treatment effects by insurance coverage, I incorporate county-level health insurance data from the Small Area Health Insurance Estimates (SAHIE) program administered by the U.S. Census Bureau ([U.S. Census Bureau, 2021](#)). The SAHIE provides model-based estimates of health insurance coverage for all counties, utilizing data from the American Community Survey, administrative records, and demographic information to produce reliable estimates even for areas with small populations. I use the 2013 baseline uninsurance rates to capture pre-Affordable Care Act variation in insurance coverage across counties. This baseline measure is particularly valuable as it reflects longstanding differences in healthcare access that predate both Medicaid expansion and the individual mandate, allowing me to identify counties where expanded mental health treatment facilities might have the greatest impact due to previously limited access to care.

To examine heterogeneous treatment effects and policy interactions, I incorporate information on state

⁹SSI provides cash assistance to disabled individuals with limited income and resources. Unlike SSDI, SSI does not require prior work history, making it particularly relevant for individuals with severe mental illness who may have limited labor force attachment.

¹⁰The SSA data end in 2016. I process the raw Excel files to extract county-level recipient counts, handling suppressed values (coded as missing when counts are below disclosure thresholds) by setting them to zero. See the data appendix for details on the cleaning procedure.

Medicaid expansion decisions under the Affordable Care Act. Following Carey et al. (2020), I classify states based on their expansion status and timing, creating indicators for states that expanded Medicaid between 2014 and 2016. This policy variation allows me to explore whether the effects of mental health facility access differ based on the insurance coverage environment, as Medicaid expansion substantially increased coverage for mental health and substance abuse treatment services (Maclean and Saloner, 2019). The combination of baseline uninsurance rates and Medicaid expansion timing creates rich variation that helps identify how insurance access modulates the relationship between treatment facility availability and health outcomes.

4.7 Analytical Sample Construction

My analysis employs two complementary samples to balance statistical power with clean identification. The full TWFE sample maximizes observations for detecting average treatment effects, while the event study sample provides cleaner identification of dynamic treatment patterns.

Full TWFE Sample. The baseline analysis includes all U.S. counties (excluding Alaska and Hawaii) with consistent identifiers from 1999 to 2016. I measure treatment using counts of mental health facilities identified through NAICS codes 621420 (outpatient mental health and substance abuse centers) and 623220 (residential mental health and substance abuse facilities). After excluding counties that never report facilities during the sample period, the full sample contains 53,498 county-year observations, and 53,480 with the full control set, across 2,972 counties.

Event Study Sample. Following Messel et al. (2023), the event study analysis restricts attention to counties with clear treatment timing: treatment counties experience exactly one facility increase, with no facility changes before the event year, and control counties experience no facility increases over the sample period. Counties with multiple facility increases are excluded because they lack clean treatment timing. Each event-study exhibit reports its own estimation sample: the population-weighted rate specifications use 26,395 county-year observations, and the count-based cause-of-death specifications use 23,742 across 1,319 counties, the difference reflecting suppression of cause-specific death counts in small counties.¹¹

Figure 2 illustrates the spatial distribution of both samples. Panel (a) shows the full TWFE sample, with counties colored by whether they experienced increases (blue), decreases (red), or no change (yellow) in facilities between 1999 and 2016. Panel (b) displays the event study sample classification, distinguishing event counties (orange) from no event counties (light blue) and excluded multiple change counties (gray). The geographic dispersion of treated and control counties across all regions helps alleviate concerns about

¹¹Online Appendix Table A13 presents detailed summary statistics for the event study sample. Treatment and control counties are well balanced in terms of demographic characteristics and baseline mortality rates. However, treatment counties are somewhat larger and more urban, consistent with facilities locating in areas with sufficient demand to support operations.

regional confounding.

For heterogeneity analyses examining insurance interactions, I classify counties by 2013 baseline uninsurance rates from SAHIE, using within-state medians to define high and low coverage groups. This yields balanced subsamples for examining how insurance environments moderate facility effects. Mechanism analyses using SSI disability data (2009 to 2016) and Medicare Part D prescriptions (2013 to 2016) employ corresponding restricted panels matched to data availability.

The dual sample approach allows me to present robust average effects from the full sample while providing credible dynamic evidence from the restricted event study sample. Small differences between TWFE and event study estimates reflect compositional differences, with event study counties representing more stable rural areas without existing mental health infrastructure.

5 Methods

5.1 Empirical Strategy

I exploit within-county variation in mental health treatment facility presence to identify their effect on mortality outcomes. This approach builds on established research examining the health effects of treatment access (Swensen, 2015; Bondurant et al., 2018).¹² The empirical strategy leverages temporal within-county variation from facility openings and closings while accounting for time-invariant county characteristics and aggregate shocks.

5.2 Baseline Specification

My baseline specification estimates the relationship between mental health treatment facilities and MBD mortality using a two-way fixed effects (TWFE) model:

$$\text{MBD Mortality}_{ct} = \alpha_c + \eta_t + \beta \text{MHT Facilities}_{c,t-1} + \varepsilon_{ct} \quad (1)$$

where $\text{MBD Mortality}_{ct}$ is the mortality rate from mental and behavioral disorders per 100,000 residents in county c in year t . The main explanatory variable, $\text{MHT Facilities}_{c,t-1}$, measures the number of mental health treatment facilities in county c in year $t - 1$; the one-year lag allows for delayed mortality responses

¹²This identification strategy has been effectively employed in related healthcare contexts. Swensen (2015) uses it to study substance abuse treatment facilities, while Bondurant et al. (2018) examines access to drug treatment.

to changes in treatment access.¹³ The model includes county fixed effects (α_c) to absorb time-invariant county characteristics that might correlate with both facility locations and mortality outcomes, such as persistent differences in healthcare and year-fixed effects (η_t) to control for common temporal shocks. Standard errors are clustered at the county level to account for serial correlation in outcomes within counties over time.

5.3 Extended Specification with County-Level Controls

To account for potential confounding from state-specific policy changes, local economic conditions and time-varying county characteristics, I extend the baseline model:

$$\text{MBD Mortality}_{ct} = \alpha_c + \gamma_{st} + \beta \text{MHT Facilities}_{c,t-1} + \mathbf{X}'_{ct} \boldsymbol{\theta} + \varepsilon_{ct} \quad (2)$$

State-by-year fixed effects (γ_{st}) absorb time-varying state-level factors like Medicaid expansion or mental health parity laws that could affect both treatment access and mortality outcomes. The vector $\mathbf{X}_{ct}$ includes county-year varying controls for demographic composition and economic conditions that might independently influence mental health outcomes.¹⁴

5.4 First-Stage Validation: Facilities and Healthcare Employment

A fundamental concern in interpreting facility counts as meaningful changes in healthcare access is whether these establishments represent actual increases in mental healthcare resources rather than administrative reorganizations or empty storefronts. To validate that facility openings translate into expanded treatment capacity, I estimate the relationship between mental health facilities and sectoral employment:

$$\ln(\text{MH Employment per 100k})_{ct} = \alpha_c + \gamma_{st} + \delta \text{Mental Health Facilities}_{ct} + \mathbf{X}'_{ct} \boldsymbol{\phi} + \nu_{ct} \quad (3)$$

where $\ln(\text{MH Employment per 100k})_{ct}$ represents the natural log of mental health employment per 100,000 residents in county c and year t , constructed using the imputed County Business Patterns data from [Eckert et al. \(2020\)](#).¹⁵ This specification parallels the mortality analysis but uses contemporaneous

¹³The one-year lag structure follows [Swensen \(2015\)](#) and helps mitigate potential simultaneity bias between facility locations and current-period health outcomes.

¹⁴Demographic controls from SEER include population shares by age groups, race/ethnicity, and gender. Economic indicators from BLS and CBP capture local business conditions that could affect both healthcare access and mental health.

¹⁵The log transformation addresses the skewed distribution of employment across counties while preserving zeros through the $\ln(x+1)$ transformation. The coefficient δ can be interpreted as the percentage change in employment from one additional

facility counts since employment adjustments should occur immediately upon facility opening, unlike health outcomes which require time to manifest.¹⁶

The first-stage analysis serves three critical functions in the research design. First, it validates that the facility measure captures real changes in healthcare provision capacity rather than merely counting administrative entities. Second, it provides a quantitative benchmark for the magnitude of treatment expansion represented by each additional facility. Third, it strengthens the causal interpretation by demonstrating that the mechanism operates through increased healthcare resources, consistent with standard models of healthcare production (Grossman, 1972; Acemoglu and Finkelstein, 2008; Finkelstein, 2007).

For the event study framework, I extend the first-stage analysis to examine dynamic employment responses:

$$\ln(\text{MH Employment per 100k})_{ct} = \alpha_c + \eta_t + \sum_{k=-5, k \neq -1}^2 \delta_k \mathbf{1}[\text{Years Since Opening} = k]_{ct} + \mathbf{X}'_{ct} \phi + v_{ct} \quad (4)$$

This specification traces employment changes around facility openings, where $k = 0$ represents the opening year and $k = -1$ serves as the omitted reference category. The pre-treatment coefficients ($k < 0$) test for anticipatory hiring or differential trends, while post-treatment coefficients ($k \geq 0$) reveal the persistence of employment effects.¹⁷

5.5 Event Study Analysis

To strengthen the causal interpretation of my results, I complement the TWFE analysis with an event study framework that examines the dynamic effects of changes in mental health treatment access. The event study specification is:

facility, scaled by 100 for log models.

¹⁶Mental health employment in these facilities encompasses a diverse workforce essential for comprehensive treatment delivery. This includes psychiatrists who provide medical management and psychopharmacology (Olfson et al., 2014; Glied and Frank, 2016), clinical psychologists offering psychotherapy and psychological assessments (Norcross and Karpiak, 2017). These psychiatric nurse practitioners increasingly serve as primary prescribers particularly in underserved areas (Phoenix, 2019). These licensed clinical social workers provide the majority of mental health services in the United States (Heisler et al., 2022), mental health counselors specializing in various therapeutic modalities (Hill, 2007), psychiatric nurses managing inpatient and intensive outpatient care (Ameel et al., 2019), peer support specialists who improve treatment engagement and recovery outcomes (Zuccarini and Stiller, 2024), and administrative staff coordinating care and insurance authorization (Cutler et al., 2010). The employment measure thus captures the full complement of personnel required for mental health facilities to deliver effective treatment, from direct clinical care to essential support services.

¹⁷Following Jacobson et al. (1993) and Autor et al. (2007), examining employment dynamics provides insights into both the timing of resource expansions and potential adjustment frictions in healthcare labor markets.

$$\text{MBD Mortality}_{ct} = \alpha_c + \eta_t + \gamma_{st} + \sum_{k=-5, k \neq -1}^5 \beta_k \text{MHT Facilities}_{c,t-k} + \mathbf{X}'_{ct} \boldsymbol{\theta} + \varepsilon_{ct} \quad (5)$$

where the coefficients β_k trace out the temporal path of mortality responses to changes in mental health treatment access relative to the year before a facility opens ($k = -1$), which serves as the baseline period. This specification serves two key purposes. First, it allows me to assess whether mortality trends differ between counties that experience changes in facility access and those that do not prior to facility openings, providing a test of the parallel trends assumption underlying my main specification. Second, it reveals the dynamic pattern of treatment effects, showing how mortality responses evolve in the years following changes in facility access.

The event study approach advances the baseline TWFE model by providing a richer understanding of treatment effect dynamics. By examining effects up to five years before and after changes in facility access, I can evaluate both the immediate impact of mental health treatment facilities and any adaptation in mortality responses over time. Importantly, this specification also helps address potential concerns about anticipatory effects or delayed responses that might not be captured in the baseline model.

The event study sample is restricted to two specific county groups. Following [Messel et al. \(2023\)](#) treatment counties experience exactly one increase in mental health establishments, with no decreases over the study period. Control counties maintain the same number of mental health establishments throughout, experiencing no changes. This clear distinction between treatment and control counties helps establish a clean comparison for analyzing the causal effects of changes in mental health facility availability.¹⁸

5.6 Units, Weights, and Inference

Three estimation choices govern how the event-study results are reported, and I make them explicit because they matter for magnitudes. First, units: I report effects both on death *counts* and on deaths per 100,000 residents. Count regressions are estimated most precisely in the event sample, which consists disproportionately of small counties gaining their first facility, where a one-death change moves the per-

¹⁸I also estimate a contemporaneous event study specification:

$$\text{MBD Mortality}_{ct} = \alpha_c + \eta_t + \gamma_{st} + \sum_{k=-5, k \neq -1}^5 \beta_k \text{MHT Facilities}_{c,t+k} + \mathbf{X}'_{ct} \boldsymbol{\theta} + \varepsilon_{ct} \quad (6)$$

where the key difference from the main specification is that facilities are now measured at time $t + k$ rather than $t - k$, allowing me to examine both anticipatory effects ($k < 0$) and contemporaneous impacts ($k \geq 0$). This complements the main lagged specification by testing for potential anticipatory responses to facility changes. Following [Malani and Reif \(2015\)](#) and [Angrist and Pischke \(2009\)](#), examining both contemporaneous and lagged effects provides a more complete picture of the treatment dynamics.

capita rate by a large and noisy amount. Second, weighting: per-100,000 regressions are weighted by county population, so the estimates describe the effect on the average *person* rather than the average *county*, the standard choice in the county mortality literature. Third, inference on pre-trends: rather than relying on visual inspection, I report a joint Wald test of the four pre-period coefficients for every event study, and I treat a rejection as disqualifying for causal interpretation. The main event-study specifications use the parsimonious control set of the dose-response tables (share under 19 and the employment-population ratio); the full control set and the no-controls specification are reported as robustness, because several of the additional controls, such as earnings, transfers, and hospital establishments, are plausibly affected by facility openings themselves, and conditioning on them degrades pre-period balance. Finally, because facility openings may affect county population, I estimate the population response directly and account for it when interpreting per-capita results (Online Appendix Table A7).

5.7 Identification and Threats to Validity

Following Deza et al. (2022a), the identifying assumption of Equation 2 can be expressed as:

$$Cov(\text{MHT Facilities}_{c,t-1}, \varepsilon_{ct} \mid \mathbf{X}_{ct}, \alpha_c, \gamma_{st}) = 0 \quad (7)$$

The number of mental health facilities per county is uncorrelated with the error term after conditioning on county-level characteristics, county fixed-effects, and state-by-year fixed-effects. While this specification addresses several potential confounders, I discuss key threats to identification and empirical strategies to address them.

A primary threat is reverse causality: changes in mental health and behavioral disorders could drive changes in facility openings through demand-side effects. Healthcare providers might strategically locate in areas experiencing increasing MBD prevalence.¹⁹ I address this by lagging the facility measure by one year, which helps mitigate simultaneous feedback effects. The first-stage employment results provide additional evidence against pure demand-driven explanations, as immediate employment responses suggest supply-side expansions rather than gradual demand accumulation.

A second threat to validity is that changes in both mental health facilities and MBD rates could be responding to unobserved time-varying county-level factors. For instance, counties experiencing economic growth might simultaneously see increases in healthcare infrastructure and changes in mental health outcomes through various channels (e.g., stress levels, substance use, or access to insurance). The rich set of time-varying controls $\mathbf{X}_{ct}$ partially addresses this concern by accounting for observable economic and

¹⁹Similar concerns have been raised in studies of healthcare access. See Shannon et al. (1986) and Bailey and Goodman-Bacon (2015).

demographic shifts.²⁰

The inclusion of state-by-year fixed effects γ_{st} addresses concerns about state-level policy changes that might affect both facility operations and mental health outcomes. These fixed effects flexibly absorb any state-specific shocks or policy changes that could confound my relationship of interest at the county level.

Finally, I note that the county-level analysis captures the relevant geographic market for mental healthcare services, as evidence suggests that most individuals seek mental health treatment within their county of residence (Schmitt et al., 2003; Lindrooth et al., 2006). This helps ensure that the facility measure appropriately captures meaningful variation in access to mental healthcare.

5.8 Summary Statistics

Table 1 presents population-weighted summary statistics for mental health treatment facilities and mental health-related mortality across U.S. counties from 1999-2016, describing availability and mortality in the average resident's county. The average resident lives in a county with 53.8 facilities, reflecting that most of the population is concentrated in large counties with many establishments; the average county itself operates five facilities. Per capita, counties average 5.19 facilities per 100,000 residents, with a net opening rate of 0.37 and a net closing rate of 0.25.

Mental and behavioral disorder (MBD) deaths per 100,000 population exhibit stark demographic patterns. The elderly population (ages 65+) experiences substantially higher mortality rates at 29.17, while younger age groups show markedly lower rates: 0.03 for those under 19, 0.29 for ages 20-34, 1.08 for ages 35-49, and 1.79 for ages 50-64. The overall MBD death rate averages 32.38 across all groups.

Racial differences are notable in the data, with White populations showing an average MBD death rate of 29.16, compared to 2.75 for Black populations and 0.47 for other racial groups. The summary statistics sample encompasses 3,199 counties and 57,440 county-year observations. Regression samples reported in subsequent tables are slightly smaller because the one-year lag of facilities and the requirement of non-missing outcome and control data restrict the estimation sample; Online Appendix Table A1 tracks the estimation samples used across all exhibits.

²⁰Following Dave et al. (2021), I include comprehensive economic indicators to capture local market conditions that might influence both healthcare provision and mental health outcomes.

6 Results

6.1 First-Stage Evidence: Facilities and Healthcare Employment

Before examining mortality outcomes, I first validate that mental health facility counts represent meaningful expansions in treatment capacity rather than administrative reorganizations. Table 2 presents estimates examining the relationship between facility counts and mental health employment using the specification described in Section 5.4.

The results provide strong evidence that facilities translate into real healthcare resources. Column 1 shows that each additional mental health facility increases sectoral employment by 3.35 percent in the baseline specification with county and year fixed effects. This effect remains stable across increasingly demanding specifications. Adding state-by-year fixed effects (Column 2) leaves the estimate unchanged, while incorporating demographic controls (Column 3) yields a coefficient of 2.94 percent. The full specification with economic controls (Column 4) produces an estimate of 2.90 percent, statistically significant at the 1 percent level across all specifications.

Figure 3 presents the dynamic employment response around facility openings. The event study reveals no pre-treatment trends in employment, supporting the exogeneity of facility timing. Importantly for interpreting the mortality dynamics below, employment begins rising in the year of the opening itself (event year 0) and persists through the post-treatment period. The immediate employment response validates that facility openings represent actual expansions in treatment capacity, with the average facility increasing mental health employment by approximately 3 percent, translating to roughly 8-10 additional mental health workers per facility based on average establishment sizes in the sector.²¹

6.2 Overall Mortality Effects

Mental health facilities generate substantial reductions in overall mortality. Because facility siting is plausibly endogenous to local health conditions, I lead with the event-study evidence from counties gaining their first facility, then present the dose-response panel estimates, and then reconcile the two magnitudes explicitly.

Event-study evidence. Figure 4 presents Sun and Abraham (2021) interaction-weighted estimates of the effect of a county's first facility opening on all-cause mortality. Panel (a) reports effects on death counts:

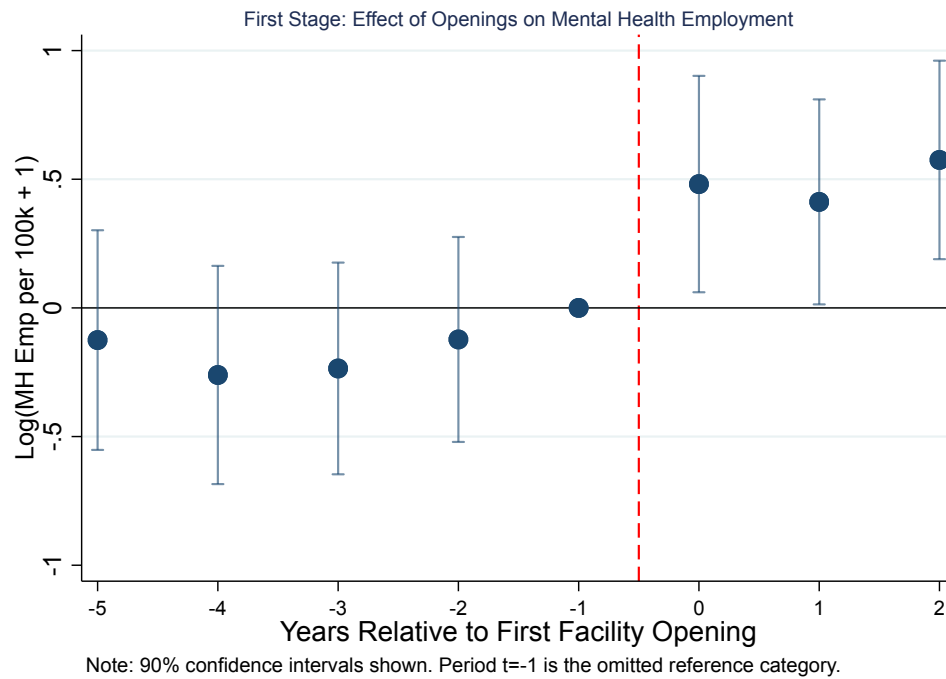
²¹The employment elasticity of approximately 0.03 aligns with industry reports suggesting that outpatient mental health facilities employ 8-15 workers on average, while residential facilities employ 20-30 workers (U.S. Bureau of Labor Statistics, 2022). The persistence of employment effects indicates these are permanent expansions rather than temporary staffing adjustments.

Table 2: First Stage: Mental Health Facilities and Sectoral Employment

	(1) Basic FE	(2) State×Year FE	(3) Add Demographics	(4) Full Controls
Number of MH Facilities	0.03351*** (0.00778) [3.35%]	0.03351*** (0.00778) [3.35%]	0.02941*** (0.00829) [2.94%]	0.02903*** (0.00825) [2.90%]
<i>Fixed Effects:</i>				
County	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
State × Year	No	Yes	Yes	Yes
<i>Controls:</i>				
Demographics	No	No	Yes	Yes
Economic	No	No	No	Yes
Observations	43,904	43,904	43,904	43,886
Counties	2,440	2,440	2,440	2,439
Adjusted R ²	0.642	0.642	0.642	0.642
Mean of Dep. Var.			5.307	

Notes: This table presents first-stage estimates validating that mental health (MH) treatment facilities represent real increases in mental healthcare resources. The dependent variable is the natural log of mental health employment per 100,000 county residents, constructed from County Business Patterns data for NAICS codes 621420 (outpatient mental health and substance abuse centers) and 623220 (residential mental health and substance abuse facilities). All specifications include county and year fixed effects. Column (1) presents the baseline specification. Column (2) adds state-by-year fixed effects to account for state-specific trends and policies. Column (3) adds demographic controls (share of population under 19). Column (4) adds economic controls (employment-to-population ratio). Standard errors clustered at the county level are in parentheses. Numbers in brackets show the percentage increase in employment from one additional facility (coefficient × 100 for log models). Sample period is 1999-2016. Counties with zero employment are included with $\ln(0+1)$ transformation. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

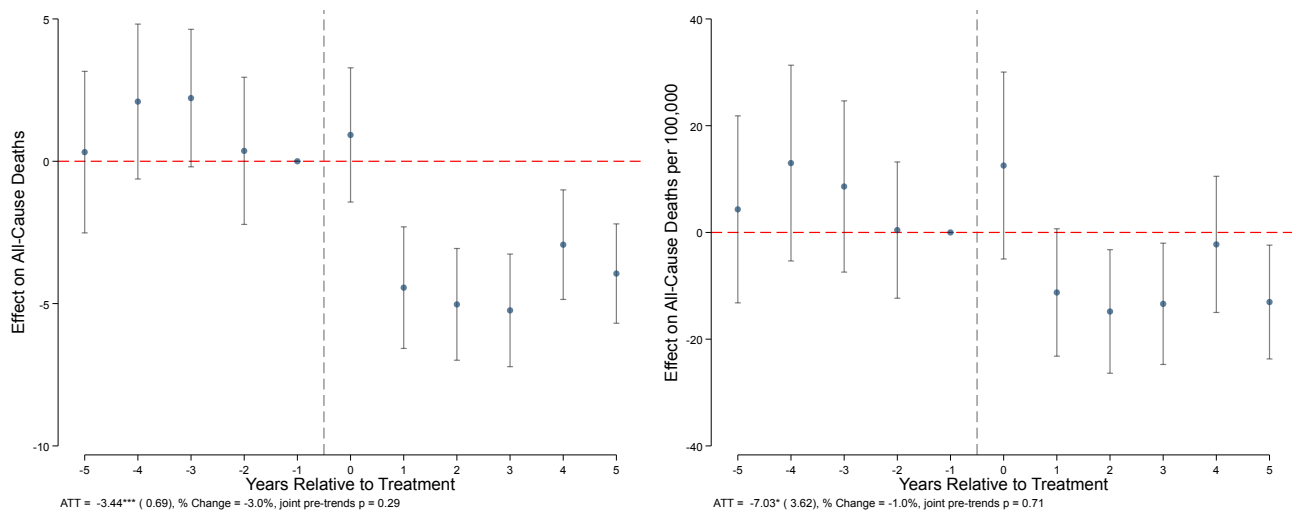
Figure 3: First Stage: Effect of Facility Openings on Mental Health Employment



Notes: This figure plots event study estimates of mental health facility openings on sectoral employment. The dependent variable is the natural log of mental health employees per 100,000 residents plus one, constructed from County Business Patterns employment as imputed by [Eckert et al. \(2020\)](#). The event is the first opening of a mental health facility in a county, with $t = -1$ the omitted reference period; 90 percent confidence intervals shown. The specification includes county and year fixed effects and demographic and economic controls. Employment rises beginning in event year 0, consistent with the CBP March 12 reference date: a facility first observed in year t was already operating by March of that year.

mortality falls by an average of 3.44 deaths per event county per year over the five post-opening years ($p < 0.01$), a 3.0 percent reduction relative to the pre-opening mean, and the joint test of the four pre-period coefficients does not reject parallel trends ($p = 0.29$). Panel (b) reports the population-weighted per-capita analogue: deaths per 100,000 residents fall by 7.03 (SE 3.62, $p = 0.052$), about 1.0 percent, with a joint pre-trends p-value of 0.71. The count estimates are more precise because the event sample consists disproportionately of small counties, where rare-event noise inflates per-capita variance; the two panels tell the same substantive story in different units. Estimates without controls are similar (Online Appendix Figure A21), and a full specification grid across units, weights, and fixed effects shows the all-cause effect is negative and significant in every cell (Online Appendix Table A6).

Figure 4: Dynamic Effects of a First Mental Health Treatment Facility on Overall Mortality: Sun-Abraham Estimates



Notes: Sun and Abraham (2021) interaction-weighted event studies of a county's first mental health treatment facility opening. The sample includes counties experiencing a single facility increase with no prior changes plus never-changing control counties; $t = -1$ is the omitted reference period and standard errors are clustered by county, with 95% confidence intervals shown. Panel (a): the outcome is the annual count of all-cause deaths; the specification includes county, year, and state fixed effects with the full control set. Panel (b): the outcome is all-cause deaths per 100,000 residents, weighted by county population; the specification includes county and state-by-year fixed effects with demographic (share under 19) and economic (employment-population ratio) controls. Each panel's note reports the post-period average treatment effect, the percent change relative to the pre-period mean, and the p-value from a joint Wald test of the four pre-period coefficients. Data: County Business Patterns and Vital Statistics, 1999–2016.

Timing of the response. Effects emerge in the year after opening rather than the opening year itself. Three institutional facts explain this pattern rather than a flaw in the design. First, CBP records establishments as of the week of March 12, so a facility first observed in year t began operating at some point before March of year t and provides at most a partial year of treatment exposure during t . Second, the first stage shows that hiring begins in event year 0 (Figure 3): capacity is demonstrably being built during the opening year even though mortality has not yet responded. Third, treatment for the conditions driving the mortality response requires time to work: medication titration, therapy engagement, and the

stabilization of chronic conditions all generate clinically expected lags between treatment initiation and averted deaths.

Dose-response estimates. Table 3 presents the complementary panel estimates of facility effects on all-cause mortality across specifications. The baseline specification (Column 1) shows that an additional mental health facility reduces overall mortality by 1.73 deaths per 100,000 population. This effect remains stable and precisely estimated as controls are added: 1.56 deaths per 100,000 with state-by-year fixed effects (Column 2), maintaining similar magnitude with demographic controls (Column 3), and 1.56 deaths per 100,000 in the full specification (Column 4). At the sample mean of 927 deaths per 100,000, the preferred estimate represents a 0.17 percent reduction; for the average county with 97,000 residents, each facility saves approximately 1.5 lives annually. This effect size is comparable to other major healthcare interventions: [Miller et al. \(2021\)](#) finds that Medicaid expansion reduces mortality by 0.13 percent annually, while [Bailey and Goodman-Bacon \(2015\)](#) documents a 2 percent reduction from community health centers.

Table 3: Effect of Mental Health Treatment Facilities on Overall Mortality

	(1)	(2)	(3)	(4)
Number of Facilities	-1.73977*** (0.22019) [-0.19]	-1.37340*** (0.22417) [-0.15]	-1.55872*** (0.25482) [-0.17]	-1.55855*** (0.25512) [-0.17]
Fixed Effects				
County	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
State-by-Year	No	Yes	Yes	Yes
Controls				
Demographic	No	No	Yes	Yes
Economic	No	No	No	Yes
Adj. R ²	0.874	0.881	0.882	0.882
N	53,498	53,498	53,498	53,480

Notes: This table presents estimates of the effect of mental health treatment facilities on overall mortality rates. The dependent variable is total deaths per 100,000 county residents from all causes. Standard errors clustered at the county level are reported in parentheses (* p<0.10, ** p<0.05, *** p<0.01). Numbers in square brackets show the percentage change relative to the pre-treatment mean of 927.41 deaths per 100,000. Column (1) includes only county and year fixed effects. Column (2) adds state-by-year fixed effects to account for differential state-level trends in mortality and mental health policy. Column (3) incorporates demographic controls including age distribution. Column (4) adds economic controls including the employment-to-population ratio. The identification strategy exploits within-county variation in facility openings and closings over time. The sample covers 1999-2016 and includes all U.S. counties with non-missing data. These results provide evidence that increased access to mental health treatment facilities is associated with reductions in overall mortality, suggesting important spillover effects beyond mental health-specific outcomes.

Reconciling the magnitudes. The event-study and dose-response estimates measure different objects, and Table 4 shows that the gap between them is fully accounted for by three transparent differences: units and weighting, sample, and margin. Column (1) reproduces the preferred dose-response estimate (−1.56 per 100,000, unweighted); Column (2) shows the population-weighted analogue (−0.87); Column (3) restricts the weighted dose response to the event sample (−3.09, imprecise); and Column (4) reports the event-study post-period ATT (−7.03). The dose response is an intensive-margin object, the effect of one more facility in a county that on average already has five, while the event study captures the extensive margin of a first facility arriving in a previously unserved county, where the marginal value of access is naturally largest. Both are policy-relevant: the intensive margin prices marginal capacity in served

markets, and the extensive margin prices entry into underserved ones.

Table 4: Reconciling Dose-Response and Event-Study Estimates of the Effect of Facilities on All-Cause Mortality

	(1) Dose response full sample	(2) Dose response full sample	(3) Dose response event sample	(4) Event study event sample
Number of Facilities _{<i>t</i>-1}	-1.559*** (0.255)	-0.872*** (0.161)	-3.085 (1.891)	
First facility (SA post-period ATT)				-7.032* (3.621)
Percent of pre-treatment mean	-0.17%	-0.09%	-0.45%	-1.02%
Joint pre-trends test (<i>p</i>)	—	—	—	0.71
Population weighted	No	Yes	Yes	Yes
Sample	Full	Full	Event	Event
Estimator	TWFE	TWFE	TWFE	Sun-Abraham
Observations	53,480	53,480	23,742	26,395

Notes: The dependent variable is all-cause deaths per 100,000 county residents. Columns (1) through (3) report the coefficient on the lagged count of mental health treatment facilities, an intensive-margin dose response (the average sample county holds five facilities); Column (1) reproduces the preferred specification of Table 3. Column (4) reports the [Sun and Abraham \(2021\)](#) post-period average treatment effect of a county's *first* facility opening, an extensive-margin effect, from the event-study design of Figure 4, Panel (b). All specifications include county fixed effects; Columns (1) and (2) include state-by-year fixed effects and Columns (3) and (4) include state-by-year fixed effects within the event sample. Demographic (share under 19) and economic (employment-population ratio) controls included throughout. The event sample restricts to counties with a single facility increase and no prior changes, plus never-changing controls. Standard errors clustered by county in parentheses (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). The comparison shows why dose-response and event-study magnitudes differ: sample (full versus event counties), weighting (per county versus per person), and margin (marginal facility versus first facility). The count-based analogue of Column (4) is -3.44 deaths per event county (-3.0 percent, $p < 0.01$; Figure 4, Panel (a)).

Population response. Facility openings also affect the denominator of per-capita rates: county population declines by about 1.4 percent after a first opening, with no differential pre-trends (Online Appendix Table A7). Because death counts decline proportionally more (about 3 percent), per-capita mortality falls on net, so the population response cannot explain the mortality results; but it implies that count-based estimates are the cleaner basis for decomposing effects across causes of death, and that the per-person estimates in Panel (b) are, if anything, conservative. I return to the population response as a finding of independent interest in Section 10.

6.3 From Overall Mortality to Mental and Behavioral Disorder (MBD) Mortality

While the overall mortality reduction demonstrates the comprehensive health value of mental health facilities, a detailed examination of mental and behavioral disorder (MBD) mortality is essential for understanding treatment effectiveness and designing targeted policy responses.

Figure 1 reveals why MBD mortality demands specific attention. Panel (a) shows that while all-cause deaths increased modestly from 2.3 to 2.65 million (15 percent growth), MBD deaths exploded from 40,000 to over 120,000. Panel (b) makes this divergence explicit: MBD mortality grew nearly 300 percent over 1999-2016, representing one of the fastest-growing mortality categories in the United States. This dramatic growth occurred despite substantial expansion in treatment infrastructure, suggesting a crisis that standard healthcare improvements have failed to address.

I focus detailed analysis on MBD mortality for four reasons. First, the 300 percent growth makes it an urgent policy priority requiring targeted intervention. Second, MBD provides the cleanest identification of facility effects, as these deaths result from conditions explicitly treated in mental health settings. Third, examining MBD allows precise estimation of treatment mechanisms and heterogeneity where the causal chain from treatment to outcome is most direct. Fourth, no prior economic study has examined MBD as a primary outcome despite its growing importance for population health.

The remainder of my analysis, therefore, examines MBD mortality in detail, exploring heterogeneous effects, asymmetric responses, and mechanisms, while using the overall mortality effects for welfare calculations. This approach provides both the broad health impacts needed for cost-benefit analysis and the specific mental health insights essential for policy design.

6.4 Mental and Behavioral Disorder Mortality: Detailed Analysis

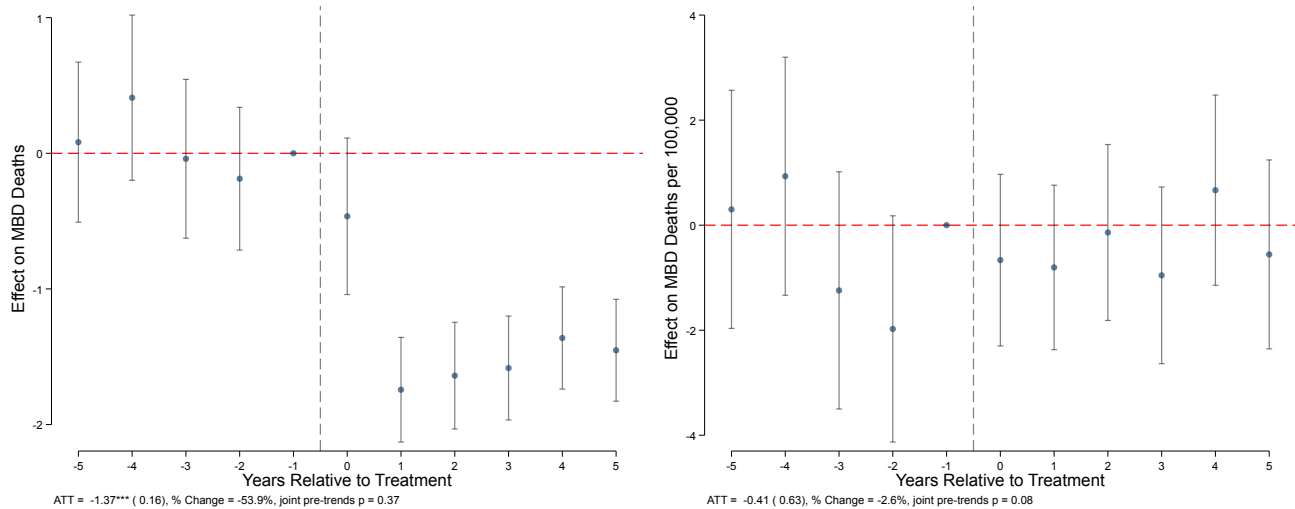
Online Appendix Table A2 examines effects on MBD mortality across diagnostic categories. Column 1 shows that an additional facility reduces overall MBD mortality by 0.066 percent, with effects concentrated in organic mental disorders (F00-F09) and schizophrenia/psychotic disorders (F20-F29). These conditions require consistent medication management and therapeutic support, explaining their responsiveness to improved facility access.

Online Appendix Table A3 demonstrates robustness across specifications. The baseline TWFE model yields a 0.066 percent reduction, increasing slightly to 0.079 percent in the full specification with all controls. At the sample mean of 30.9 MBD deaths per 100,000, this implies each facility prevents 0.024 deaths per 100,000 residents annually.

Figure 5 presents the dynamic evidence for MBD mortality in the same two-panel format as the overall mortality results. Panel (a) shows effects on MBD death counts: a first facility reduces MBD deaths by an average of 1.37 per event county per year over the post-opening period ($p < 0.01$), with clean pre-trends (joint $p = 0.38$). Because event counties are small, averaging roughly 2.5 MBD deaths per year before opening, this is a large proportional decline, consistent with a first facility transforming access to treatment in previously unserved communities. Panel (b) shows the population-weighted per-capita

analogue, which is negative but imprecise (-0.41 deaths per 100,000, SE 0.63): MBD deaths are rare events, and per-capita rates in small counties are too noisy to resolve the effect in per-person units. I therefore treat the count-based estimates as primary for cause-specific outcomes and report per-capita results for transparency, a choice Section 5.6 motivates and the population response in Online Appendix Table A7 reinforces.

Figure 5: Dynamic Effects of a First Mental Health Treatment Facility on MBD Mortality: Sun-Abraham Estimates



Notes: Sun and Abraham (2021) interaction-weighted event studies of a county's first mental health treatment facility opening on mental and behavioral disorder (MBD, ICD-10 F codes) mortality. Sample, reference period, clustering, and panel-specific specifications are identical to Figure 4: Panel (a) uses MBD death counts with county, year, and state fixed effects and full controls; Panel (b) uses population-weighted MBD deaths per 100,000 with county and state-by-year fixed effects and parsimonious controls. Each panel's note reports the post-period average treatment effect, the percent change relative to the pre-period mean, and the joint pre-trends p-value. Data: County Business Patterns and Vital Statistics, 1999–2016.

The contrast between the dose-response estimates and the event-study magnitudes for MBD mortality follows the same logic as Table 4 for overall mortality: the panel coefficient prices the marginal facility in served counties, while the event study prices the first facility in unserved ones, in different units and samples. Rather than treating the difference as a puzzle, the paper reports both margins and the mapping between them.

6.5 Robustness and Validation

I conduct extensive robustness checks to validate the causal interpretation of my findings. The main results prove remarkably stable across alternative specifications, sample restrictions, and estimation methods.

Progressive Specifications. Tables 3 and A3 demonstrate that effects remain stable as controls are added progressively. For overall mortality, the coefficient ranges from -1.74 to -1.56 deaths per 100,000 across specifications, with the full model including state-by-year fixed effects and all controls yielding -1.56 (SE = 0.26). The stability across specifications with increasingly rich controls suggests that selection on unobservables is unlikely to drive the results.

Alternative Event Study Specifications. Online Appendix Figure A16 addresses potential concerns about dynamic selection and pre-existing trends. Panel (a) includes lagged mortality rates as controls, while Panel (b) controls for lagged facility counts. Both specifications show no pre-trends and similar post-treatment dynamics to the main results, confirming that facilities are not endogenously locating in counties with improving or deteriorating mortality trends.

Cause-of-Death Decomposition. Where does the all-cause mortality reduction come from? Table 5 partitions all deaths into three exhaustive categories and runs each through the identical event-study design on death counts: mental health related causes (MBD, intentional self-harm, assault, and other external injuries), placebo causes (transport accidents, complications of medical and surgical care, and other external causes), and the chronic and other internal cause residual. Because the categories partition all deaths, the component effects sum exactly to the all-cause effect, providing the explicit accounting behind the paper's spillover claim: of the 2.98 deaths averted per event county per year, 1.78 (60 percent) come from mental health related causes and 1.22 (40 percent, $p < 0.05$) from chronic and other internal causes such as cardiovascular and metabolic disease, while placebo causes are flat (0.01, SE 0.07). The significant chronic-cause component is direct evidence that mental health treatment improves physical health, consistent with clinical evidence on the management of comorbid conditions (Walker et al., 2015; Druss et al., 2011), and the placebo null simultaneously addresses the concern that facility openings proxy for correlated expansions of physical healthcare capacity: if they did, placebo causes should fall too.

Figure 6 shows the dynamics behind the first and third rows: mental health related deaths decline sharply beginning the year after opening, while placebo-cause deaths never move.

One classification choice deserves comment. Transport accidents are included in the placebo set for continuity with the prior literature, but they are an imperfect placebo: impaired driving links transport fatalities to substance use disorders, so mental health treatment could plausibly reduce them. Empirically the choice is innocuous for the decomposition, because the count-based transport effect is small and insignificant (0.040, SE 0.063). In population-weighted per-capita terms, transport deaths do decline after openings (Online Appendix Table A6), a pattern consistent with an SUD treatment channel; I therefore do not lean on transport accidents as evidence of design validity, and the placebo conclusion rests on medical and surgical complications and other external causes.

Online Appendix Table A19 reports the corresponding dose-response estimates for individual placebo

Table 5: Decomposition of the Mortality Effect by Cause of Death

Cause of death	ATT (deaths)	SE	Pre-period mean (deaths)	Percent change
All causes	−2.979***	(0.712)	113.5	−2.6%
Mental health related	−1.777***	(0.213)	7.4	−24.0%
Chronic and other internal causes	−1.215**	(0.615)	102.8	−1.2%
Placebo causes	0.013	(0.065)	3.4	0.4%
Sum of components	−2.979			

Notes: Each row reports the [Sun and Abraham \(2021\)](#) post-period average treatment effect of a county's first mental health treatment facility on annual death counts by cause, estimated on the event sample (single-increase counties plus never-changing controls) with county and state-by-year fixed effects, demographic and economic controls, and standard errors clustered by county (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). Mental health related deaths combine mental and behavioral disorder (MBD) mortality, intentional self-harm, assault, and other external injuries. Placebo causes combine transport accidents, complications of medical and surgical care, and other external causes; the transport component alone is 0.040 (SE 0.063). Chronic and other internal causes are the residual: all-cause deaths minus the two categories above. Joint pre-trend tests do not reject parallel trends for any row (p-values: all causes 0.30, mental health related 0.29, chronic residual 0.52, placebo 0.96). Because the categories partition all deaths and the specification is linear, the component effects sum exactly to the all-cause effect, attributing 60 percent of the total to mental health related causes and 40 percent to chronic and other internal causes, with placebo causes flat.

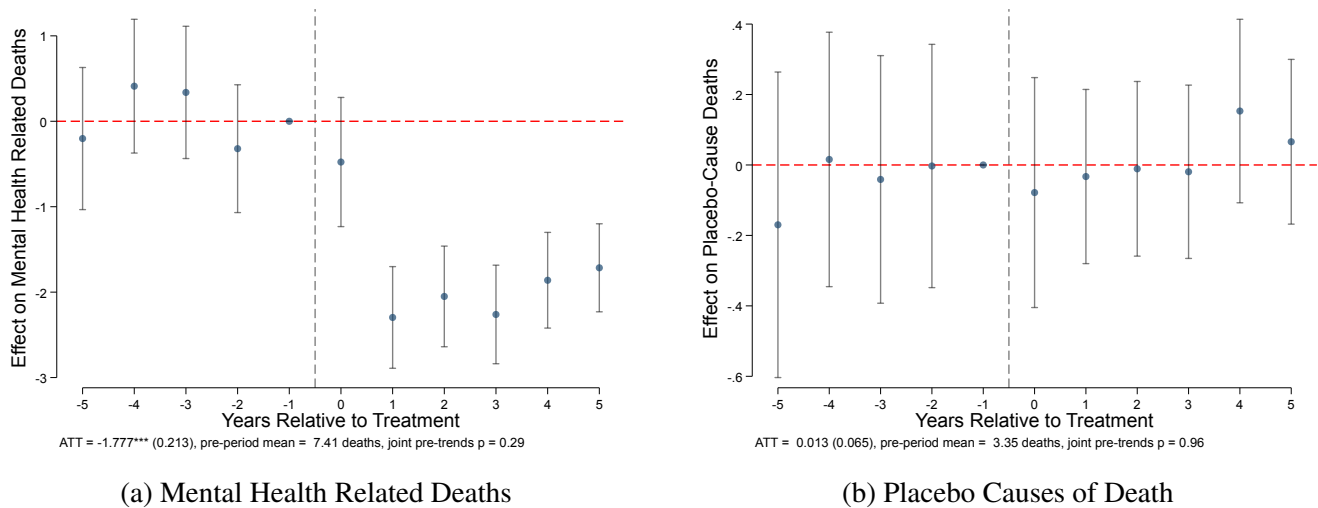
outcomes, which are likewise null: deaths from medical and surgical complications show no response to facility counts (−0.001, SE 0.001). The specificity of effects to mental health related and chronic causes, with placebo causes flat in both the event-study and dose-response designs, rules out spurious correlation with general mortality trends or healthcare quality improvements.

Healthcare Infrastructure Controls. A critical concern is whether mental health facilities proxy for broader healthcare expansion. Online Appendix Table [A18](#) addresses this by controlling for hospital establishments. The MBD mortality effect remains virtually unchanged at -0.079 percent (SE = 0.030), confirming that mental health facilities have distinct impacts beyond general healthcare access.

Additional Robustness Checks. Appendix Section [C.5](#) reports extensive additional analyses. Inverse probability weighting following [Fischer et al. \(2024\)](#) yields similar estimates after balancing observable characteristics between treated and control counties. The [Sun and Abraham \(2021\)](#) estimator, which addresses potential bias from staggered treatment timing, produces comparable results to the main TWFE specifications. Sample restrictions excluding small rural counties or large urban areas do not meaningfully alter conclusions.

The consistency of results across specifications, the absence of pre-trends, the specificity to mental health-related causes, and the robustness to alternative estimators provide strong evidence for a causal interpretation. Mental health facilities reduce mortality through targeted treatment of mental health conditions rather than through general healthcare improvements or spurious correlation with local trends.

Figure 6: Effect of a First Facility on Mental Health Related versus Placebo Causes of Death



Notes: Sun and Abraham (2021) event study estimates of a county's first facility opening on annual death counts by cause category, corresponding to the first and third component rows of Table 5. Panel 6a: mental health related deaths, combining mental and behavioral disorder (MBD) mortality, intentional self-harm, assault, and other external injuries. Panel 6b: placebo causes, combining transport accidents, complications of medical and surgical care, and other external causes; these are deaths that mental health treatment should not plausibly reduce through correlated physical-healthcare expansion. The vertical dashed line separates the pre- and post-opening periods, with $t = -1$ the omitted reference year. Specifications include county and state-by-year fixed effects with the full control set; the event sample and clustering follow Figure 4, and each panel's note reports the post-period ATT and joint pre-trends p-value. Data: County Business Patterns and Vital Statistics, 1999–2016.

6.6 Heterogeneous Effects

The average treatment effects mask important heterogeneity across population subgroups and facility types. Understanding this variation helps identify vulnerable populations and inform targeted policy interventions.

Facility Type and Treatment Setting. The two NAICS industries underlying the facility measure differ in the setting and intensity of care, and the distinction matters for interpreting the treatment. Outpatient centers (NAICS 621420) deliver ongoing community-based care, psychotherapy and medication management on a weekly or biweekly cadence, and serve the large majority of patients in treatment; residential facilities (NAICS 623220) provide intensive, structured stays, typically 30 to 90 days, for patients with more severe conditions or following a crisis. Because CBP assigns each establishment exactly one NAICS code based on its primary activity, the facility counts capture standalone mental health treatment establishments rather than mental health services embedded in physical-health clinics, which addresses the concern that facility openings proxy for co-located physical healthcare expansion; the placebo-cause results in Table 5 corroborate this. Online Appendix Table A4 shows both facility types reduce MBD mortality, with larger effects for outpatient centers (0.128 versus 0.094 deaths per 100,000 per facility), consistent with outpatient care carrying the sustained treatment relationships, continuing therapy and

refilled prescriptions, through which facility access plausibly reduces mortality, and Online Appendix Figure A5 confirms the patterns dynamically, with outpatient facilities showing immediate and persistent effects. The presence of effects in both settings suggests that the mortality response operates through both continuing community-based treatment and intensive stabilization rather than either alone.

Age Heterogeneity. Effects vary substantially across the age distribution (Online Appendix Table A5). The largest impacts appear among elderly populations, with facilities reducing MBD mortality by 0.27 percent for those over 65. Online Appendix Figure A6 reveals interesting dynamics: while elderly populations show immediate mortality reductions, working age adults (35 to 64) experience gradually increasing benefits over time. Youth under 19 show minimal effects, possibly reflecting lower baseline mortality or different treatment needs. The pronounced elderly response aligns with greater vulnerability to untreated mental health conditions and higher barriers to accessing distant care.

Education. Online Appendix Table A8 documents stark differences by educational attainment. Facilities reduce MBD mortality by 0.43 percent among those with a high school education or less, compared to insignificant effects for those with some college. The event studies in Online Appendix Figure A7 show persistent benefits for less educated populations emerging immediately after facility entry, while college-educated populations show no clear pattern. This gradient suggests geographic access particularly constrains less educated individuals who may face transportation barriers or limited awareness of alternative treatment options.

Race and Gender. Racial heterogeneity analysis (Online Appendix Table A9) shows significant effects for White populations (0.35 percent reduction) with similar magnitude but imprecise estimates for Black populations. Online Appendix Figure A8 reveals comparable dynamic patterns across racial groups, with the precision differences reflecting sample composition. White individuals comprise 90 percent of observations in the data (Table 1). The imprecision for minority populations reflects limited statistical power rather than differential treatment effectiveness. Gender effects (Online Appendix Table A10), and Online Appendix Figure A9 are remarkably similar, with facilities reducing male and female MBD mortality by 0.26 and 0.25 percent, respectively, suggesting geographic access equally constrains both genders despite documented differences in help-seeking behavior.

Openings versus Closings. A policy-relevant question is whether losing a facility harms more than gaining one helps. Table 6 estimates the effects of facility openings and closings on MBD mortality *jointly*, entering year-over-year increases and decreases in the same regression in every column so that the coefficients and the test of asymmetry share one specification, with the test reported per column. In the most demanding specification with state-by-year fixed effects, closings are associated with 0.139 additional MBD deaths per 100,000 ($p < 0.01$) while openings show no effect (0.024, insignificant); the formal test rejects symmetry in this specification ($p = 0.021$) but not in less saturated ones, where the closing coefficients are smaller and only marginally significant. Two further pieces of evidence counsel

caution. First, the same joint regression for overall mortality is uninformative: both coefficients are small relative to their standard errors, so the churn design lacks the power to detect effects on the broad outcome, and dose-response estimates of churn in any case confound siting decisions with treatment effects, precisely the concern that motivates leading with clean first-opening event studies. Second, a dedicated closings event study, constructed as the mirror of the openings design using counties with a single facility decrease and no other changes, is too underpowered to be informative (Online Appendix Figure A22): clean single-closure counties are rare, and the confidence intervals admit both large harms and zero. The honest summary is that the dose-response association between churn and MBD mortality is carried by closings rather than openings, consistent with adjustment frictions when established care relationships are severed (Corredor-Waldron and Currie, 2022; Frank, 2006), but the data cannot yet deliver a sharp causal estimate of closure harm, which I leave as a priority for future work with richer closure data.

Table 6: Facility Openings and Closings and MBD Mortality: Joint Estimation

	(1)	(2)	(3)	(4)
Facilities Opened	−0.0545 (0.0353)	−0.0346 (0.0350)	−0.0337 (0.0350)	0.0241 (0.0359)
Facilities Closed	0.0916* (0.0501)	0.0847* (0.0495)	0.0861* (0.0496)	0.1385*** (0.0499)
Asymmetry test p -value	0.592	0.473	0.453	0.021
County and Year FE	Yes	Yes	Yes	Yes
Demographic Controls	No	Yes	Yes	Yes
Economic Controls	No	No	Yes	Yes
State-by-Year FE	No	No	No	Yes
Observations	53,517	53,517	53,499	53,499
Counties	2,974	2,974	2,973	2,973

Notes: The dependent variable is MBD deaths per 100,000 county residents (sample mean 35.06). Openings (year-over-year facility increases) and closings (decreases, in absolute value) enter the *same* regression in every column, so the coefficients and the asymmetry test share one specification. The asymmetry test is H_0 : opening effect = −closing effect, reported per column. Positive closing coefficients indicate that mortality rises when facilities close. Standard errors clustered by county in parentheses (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). Sample: U.S. counties, 1999–2016.

These heterogeneous effects highlight that mental health facility benefits concentrate among vulnerable populations: the elderly, the less educated, and residents of previously unserved counties. The patterns suggest targeting expansion toward areas with older, less educated populations, and they flag closures in underserved communities as a first-order concern for future research and monitoring.

7 Insurance Expansion, Mental Health Treatment Access, and Mortality

While my county-level analysis establishes that mental health treatment facilities reduce mortality from mental and behavioral disorders, an important question remains: how do insurance coverage expansions affect this mortality-reducing impact of treatment infrastructure? This section examines whether Medicaid expansion under the Affordable Care Act amplifies the life-saving effects of mental health facilities, providing evidence on critical complementarities between insurance access and treatment availability in preventing deaths from mental and behavioral disorders.

7.1 Medicaid Expansion Background

The Affordable Care Act's Medicaid expansion, implemented beginning in January 2014, represents one of the largest insurance coverage expansions in U.S. history. The expansion extended eligibility to all adults with incomes below 138 percent of the federal poverty level, regardless of family status or disability status (Sommers et al., 2013; Wherry and Miller, 2016). Crucially for mental health mortality, the expansion included comprehensive coverage for mental health and substance abuse treatment as essential health benefits, with strong parity provisions ensuring coverage comparable to physical health services (Glied and Frank, 2016; Beronio et al., 2014). This comprehensive mental health coverage is particularly important given that untreated mental illness is a leading cause of preventable death through suicide, substance abuse complications, and chronic disease interactions (Walker et al., 2015; Colton and Rodin, 2009).

However, the Supreme Court's 2012 decision in *National Federation of Independent Business v. Sebelius* made expansion optional for states, creating substantial variation in implementation timing. As illustrated in Online Appendix Figure A10, by 2016, 31 states and the District of Columbia had expanded Medicaid, while 19 states had not.^{22,23}

This staggered implementation creates plausibly exogenous variation in insurance coverage that I exploit to examine how expanded insurance access affects the mortality-reducing impact of mental health treatment infrastructure. The expansion is particularly relevant for preventing mental health-related deaths given the high rates of uninsurance among individuals with mental illness prior to the ACA and evidence

²²Online Appendix Table A26 details the expansion timing, with most expansion states implementing in January 2014, though several states expanded later, including Pennsylvania (January 2015), Indiana (February 2015), Alaska (September 2015), Montana (January 2016), and Louisiana (July 2016)

²³Early expansion states that expanded under Section 1115 waivers before 2014 are coded as expansion states throughout the sample period, following Wherry and Miller (2016). These include California, Connecticut, Delaware, Massachusetts, Minnesota, New Jersey, New York, Vermont, and Washington, plus the District of Columbia.

that insurance coverage improves mental health treatment utilization and reduces mortality (Guth et al., 2020; Ortega, 2023; Sommers et al., 2013).

7.2 Facility–Expansion Interaction Specification

To identify how insurance expansion affects the relationship between mental health facilities and mortality, I implement an interaction specification that exploits three sources of variation: (1) within-county changes in mental health facilities over time, (2) state-level variation in Medicaid expansion decisions, and (3) cross-sectional variation in baseline uninsurance rates that determines treatment intensity. This approach allows me to test whether insurance expansion enhances the mortality-reducing effects of mental health facilities by removing financial barriers to accessing available treatment.

The baseline interaction specification, whose key parameter is a triple interaction, takes the form:

$$\begin{aligned} \text{MBD Mortality}_{ct} = & \beta_0(\text{MH Facilities}_{c,t-1} \times \text{Post-Expansion}_{st}) \\ & + \beta_1(\text{MH Facilities}_{c,t-1} \times \text{High Uninsured}_c) \\ & + \beta_2(\text{MH Facilities}_{c,t-1} \times \text{Post-Expansion}_{st} \times \text{High Uninsured}_c) \\ & + \mathbf{X}'_{ct} \boldsymbol{\theta} + \alpha_c + \gamma_{st} + \varepsilon_{ct} \end{aligned} \quad (8)$$

Where $\text{MBD Mortality}_{ct}$ represents mental and behavioral disorder deaths per 100,000 residents, $\text{MH Facilities}_{c,t-1}$ is the lagged count of mental health treatment facilities, $\text{Post-Expansion}_{st}$ indicates post-2014 periods in states that expanded Medicaid, and High Uninsured_c indicates counties with above-median baseline uninsurance rates within their state.²⁴

β_0 captures the effect of mental health facilities on mortality in expansion states after Medicaid implementation, averaging across all counties regardless of baseline uninsurance. This coefficient identifies the overall impact of facilities when insurance coverage has been expanded. β_1 identifies heterogeneity in the facility-mortality relationship by baseline uninsurance rates, pooling across expansion and non-expansion states. This tests whether facilities are more effective in areas with historically limited insurance access.

β_2 is the coefficient of primary interest, capturing the additional mortality reduction from mental health facilities in high-uninsurance counties within expansion states after Medicaid implementation. A negative coefficient indicates that the life-saving effects of mental health facilities are amplified when combined

²⁴Using within-state variation in uninsurance rates ensures comparisons between counties facing similar state policies and healthcare markets. The median split provides a transparent, nonparametric approach that avoids functional form assumptions about the relationship between baseline uninsurance and treatment effects (Finkelstein, 2007; Miller, 2012).

with expanded insurance coverage in areas with the greatest coverage gaps. This complementarity could arise through multiple channels: insurance enables individuals to afford treatment at existing facilities, reduces delays in seeking care that could prove fatal, and facilitates ongoing treatment adherence crucial for managing chronic mental health conditions (Baicker et al., 2014; Wherry et al., 2018).

The specification includes county fixed effects (α_c) and state-by-year fixed effects (γ_{st}), ensuring identification comes from within-county variation in facilities interacted with policy-induced insurance changes. The vector $\mathbf{X}_{ct}$ includes time-varying demographic and economic controls. Standard errors are clustered at the state level to account for serial correlation and the state-level nature of the expansion decision (Bertrand et al., 2004; Cameron et al., 2008).

7.3 Event Study Specification

To examine the dynamic evolution of the complementarity between insurance and mental health facilities, and to test for differential pre-trends, I estimate a complementary event study design that interacts mental health facility counts with Medicaid expansion status across years relative to expansion:

$$\begin{aligned} \text{MBD Mortality}_{ct} = & \sum_{k=-5, k \neq -1}^2 \delta_k (\text{MH Facilities}_{c,t-1} \times \text{Expansion}_s \times \mathbf{1}[\text{Year} = 2014 + k]_t) \\ & + \mathbf{X}'_{ct} \boldsymbol{\theta} + \alpha_c + \gamma_{st} + \varepsilon_{ct} \end{aligned} \quad (9)$$

where $\mathbf{1}[\text{Year} = 2014 + k]_t$ indicates year t is k years from 2014, with $k = -1$ (year 2013) as the omitted reference category. The coefficients δ_k trace out how the differential effect of facilities in expansion versus non-expansion states evolves around Medicaid implementation.²⁵

The pre-treatment coefficients ($k < 0$) provide a test of the identifying assumption that the mortality effects of facilities would have evolved similarly in expansion and non-expansion states absent Medicaid expansion. Statistically insignificant pre-treatment coefficients support this parallel trends assumption. The post-treatment coefficients ($k \geq 0$) reveal whether the complementarity between insurance and facilities in preventing deaths emerges immediately or strengthens over time as newly insured individuals establish care relationships and receive consistent treatment for previously untreated conditions.

²⁵This specification provides a more conservative complement to the interaction results in Equation 8. While the interaction specifications in Tables 7 and 8 isolate additional facility effects in expansion states and high-uninsurance expansion counties (where the complementarity should be largest), the event study estimates the average differential effect across all expansion counties, providing a transparent dynamic test of the parallel trends assumption. Standard errors in the dynamic specification are clustered at the state level, reflecting the state-level treatment. The event study approach follows Goodman-Bacon (2021) and Sun and Abraham (2021) in addressing concerns about heterogeneous treatment effects and negative weighting in two-way fixed effects models with staggered treatment timing.

7.4 Identifying Assumptions and Robustness

The identification strategy relies on three key assumptions. First, in the absence of Medicaid expansion, the relationship between mental health facilities and mortality would have evolved similarly in expansion and non-expansion states. The event study’s flat pre-treatment coefficients support this parallel trends assumption, indicating no differential mortality trends prior to expansion.

Second, baseline uninsurance rates must be uncorrelated with other factors that differentially affect the facility-mortality relationship over time. This assumption is plausible given that 2013 uninsurance rates reflect longstanding differences in state policies, labor markets, and demographics that predate the ACA (Courtemanche et al., 2017). Moreover, using within-state variation in uninsurance rates helps control for state-level confounders that might affect mortality patterns.

7.5 Results: Insurance Expansion Amplifies Facility Effects

Tables 7 and 8 present the interaction estimates for all-cause and MBD mortality, measured in deaths per 100,000 residents and weighted by population. Responding to the natural question of what facilities do *outside* expansion environments, every column retains the baseline facility effect, so the estimates separate the facility effect in non-expansion settings from the additional effect once coverage expands.

Three results stand out. First, insurance coverage substantially amplifies facility effectiveness: combining the baseline and interaction terms, the total effect of a facility in expansion states after expansion is -1.53 all-cause deaths per 100,000 (SE 0.24) and -0.108 MBD deaths per 100,000 (SE 0.029), both significant at the 1 percent level. Second, the amplification is concentrated precisely where expansion bound: in counties with above-median baseline uninsurance within expansion states, the post-expansion facility effect rises to -1.80 all-cause (SE 0.36) and -0.164 MBD deaths per 100,000 (SE 0.039), roughly 18 and 50 percent larger than the average expansion-state effects. Relative to pre-expansion means, these are reductions of 0.24 percent of all-cause and 0.59 percent of MBD mortality per facility. Third, the ordering of effects across environments, largest where uninsurance was highest, matches the complementarity mechanism: coverage converts local treatment capacity into realized care by removing the financial barrier at exactly the margin where it previously bound.²⁶

The dynamic version of this analysis is reported in Online Appendix Figures A23 and A24, and both deserve a transparent accounting. The first reproduces the dynamic exhibit of earlier drafts of this paper

²⁶Restricting the estimation window to 2011–2016, so that the pre-expansion years exclude the Great Recession recovery and the earliest ACA provisions, leaves the complementarity estimates intact: the all-cause facility $\times$ post-expansion interaction is -0.176 (SE 0.030) and the MBD triple interaction is -0.029 (SE 0.012), both statistically significant. The baseline facility coefficients, identified from only three pre-expansion years of within-county variation in the short window, become imprecise, so the total-effect linear combinations are not directly comparable to the full-sample estimates.

Table 7: Medicaid Expansion, Mental Health Treatment Facilities, and All-Cause Mortality

	(1) Medicaid Expansion	(2) Uninsurance Heterogeneity	(3) Triple Interaction
Number of Facilities _{t-1}	- 1.215*** (0.273)	- 1.099*** (0.218)	- 0.940*** (0.219)
× Post-Expansion	- 0.311*** (0.080)		- 0.191** (0.092)
× High Uninsurance		- 0.478 (0.462)	- 0.468 (0.455)
× Post-Expansion × High Uninsurance			- 0.198 (0.145)
<i>Total facility effect (linear combinations):</i>			
In expansion states, post-expansion	- 1.527*** (0.244) [-0.20%]		
In high-uninsurance expansion counties, post			- 1.797*** (0.360) [-0.24%]
County FE	Yes	Yes	Yes
State-by-Year FE	Yes	Yes	Yes
Demographic and Economic Controls	Yes	Yes	Yes
Population Weighted	Yes	Yes	Yes
Pre-expansion mean	764.65	764.65	764.65
Observations	43,886	43,886	43,886
Counties	2,439	2,439	2,439

Notes: The dependent variable is all-cause deaths per 100,000 county residents, and regressions are weighted by county population. Every column retains the baseline facility effect so that the estimates separate the effect of facilities outside expansion environments from the additional effect once coverage expands. Post-Expansion equals one for expansion states after their expansion date; High Uninsurance indicates counties above their state's median 2013 uninsurance rate (SAHIE); the sample excludes early expansion states. The total-effect rows report linear combinations of the coefficients above: Column (1), baseline plus the expansion interaction; Column (3), the sum of all four terms. Numbers in square brackets are percentage changes relative to the population-weighted pre-expansion (through 2013) mean. Standard errors clustered by county in parentheses (* p<0.10, ** p<0.05, *** p<0.01). Sample period 1999–2016.

Table 8: Medicaid Expansion, Mental Health Treatment Facilities, and MBD Mortality

	(1) Medicaid Expansion	(2) Uninsurance Heterogeneity	(3) Triple Interaction
Number of Facilities _{t-1}	- 0.0797** (0.0326)	- 0.0342 (0.0328)	- 0.0419 (0.0352)
× Post-Expansion	- 0.0280** (0.0113)		0.0174 (0.0151)
× High Uninsurance		- 0.1008* (0.0590)	- 0.0707 (0.0616)
× Post-Expansion × High Uninsurance			- 0.0693*** (0.0227)
<i>Total facility effect (linear combinations):</i>			
In expansion states, post-expansion	- 0.1077*** (0.0287) [-0.38%]		
In high-uninsurance expansion counties, post			- 0.1644*** (0.0391) [-0.59%]
County FE	Yes	Yes	Yes
State-by-Year FE	Yes	Yes	Yes
Demographic and Economic Controls	Yes	Yes	Yes
Population Weighted	Yes	Yes	Yes
Pre-expansion mean	28.09	28.09	28.09
Observations	43,886	43,886	43,886
Counties	2,439	2,439	2,439

Notes: The dependent variable is mental and behavioral disorder (MBD) deaths per 100,000 county residents, and regressions are weighted by county population. Every column retains the baseline facility effect so that the estimates separate the effect of facilities outside expansion environments from the additional effect once coverage expands. Post-Expansion equals one for expansion states after their expansion date; High Uninsurance indicates counties above their state's median 2013 uninsurance rate (SAHIE); the sample excludes early expansion states. The total-effect rows report linear combinations of the coefficients above: Column (1), baseline plus the expansion interaction; Column (3), the sum of all four terms. Numbers in square brackets are percentage changes relative to the population-weighted pre-expansion (through 2013) mean. Standard errors clustered by county in parentheses (* p<0.10, ** p<0.05, *** p<0.01). Sample period 1999–2016.

exactly: facility counts interacted with event-time dummies defined relative to each state’s own expansion date, over 2009–2015, with the baseline facility effect included, death counts as outcomes, and county-clustered standard errors. In that design, facilities in expansion states generate additional post-expansion mortality reductions averaging 0.32 all-cause and 0.08 MBD deaths per facility over the expansion year and the year after, as reported in the figure. The second form re-estimates the differential in population-weighted deaths per 100,000 with state-clustered standard errors over 2009–2016: there the all-cause post-expansion differential is negative and significant (-0.093 , SE 0.030) but the joint test rejects parallel pre-trends for this outcome ($p < 0.01$), driven by secular drift in the facility-mortality gradient, while for MBD mortality the differential pre-trends are marginal ($p = 0.07$) and the post-expansion differential is negative but insignificant. Neither dynamic variant is strong enough to carry the section on its own, so I treat both as suggestive corroboration and rest the complementarity conclusion on the levels specifications above, whose identifying variation is absorbed neither by state-by-year fixed effects nor by the gradient’s secular drift. The immediate post-expansion response in the levels estimates is consistent with pent-up demand: individuals who could not afford treatment at existing facilities gain access when financial barriers fall, and the convergence of the average expansion-state effects and the high-uninsurance effects strengthens the conclusion that insurance coverage and treatment infrastructure are policy complements rather than substitutes.

8 Mechanisms

The large mortality reductions documented above, particularly the spillover effects beyond direct mental health causes, raise questions about underlying pathways. This section examines two mechanisms through which facilities affect mortality: connecting individuals to social safety nets and enabling pharmaceutical treatment when combined with insurance coverage.

8.1 Gateway to Disability Benefits

Mental health facilities may reduce mortality by serving as gateways to disability programs that provide both income support and healthcare coverage. For individuals with severe mental illness, SSI/SSDI benefits can prevent homelessness and ensure treatment continuity, both critical for survival (Shinn et al., 2007; Silverman et al., 2017).

I estimate the effect of facilities on disability claims using SSA county-level data on SSI recipients aged 18–64 for 2009–2016:

$$\text{Disability}_{ct} = \sum_{k=-5}^1 \beta_k \mathbf{1}[\text{EventTime}_{ct} = k] + \alpha_c + \gamma_t + \delta_{st} + X'_{ct} \theta + \varepsilon_{ct} \quad (10)$$

where Disability_{ct} is SSI recipients per 100,000 population in county c and year t , EventTime_{ct} measures years relative to the first facility opening, α_c are county fixed effects, γ_t are year fixed effects, δ_{st} are state-by-year fixed effects, and X_{ct} includes demographic and economic controls. Standard errors are clustered at the county level.

Online Appendix Figure A11 presents event study estimates of facility effects on SSI disability claims. Panel (a) shows that facility openings increase the number of disability recipients by 8 to 9 per 100,000 population, with effects emerging immediately and persisting through the post-treatment period. The log specification in Panel (b) confirms a 1.4 percent increase that stabilizes after facility entry. Online Appendix Table A11 provides corresponding regression estimates across specifications.

This increase in disability claims represents improved access to existing benefits rather than deteriorating mental health. Mental health providers supply the diagnostic documentation and longitudinal treatment records essential for successful disability applications (Messel et al., 2023). Without provider documentation, individuals with severe mental illness often cannot navigate the complex application process despite qualifying conditions. The immediate response suggests pent-up demand, as individuals with existing conditions gain access to benefits they previously could not obtain.

The mortality implications are substantial. SSI recipients gain Medicaid coverage in most states, ensuring continued access to mental health treatment and medications. The income support (averaging \$600 monthly) provides housing stability, reducing exposure to violence, substance abuse, and untreated medical conditions associated with homelessness. This gateway function may explain part of the spillover mortality effects documented earlier.

8.2 Prescription Access and Insurance Complementarity

A second mechanism operates through prescription drug access, particularly when facilities combine with insurance coverage. Antidepressant utilization is a widely used marker of mental healthcare engagement, and recent evidence shows that it responds to changes in the surrounding institutional environment (Salcedo et al., 2026).

I examine whether mental health facilities combined with Medicaid expansion increase antidepressant utilization, using Medicare Part D prescriber data aggregated to the state level for 2013–2016. The interaction specification compares facility effects in Medicaid expansion versus non-expansion states:

$$\text{Prescriptions}_{st} = \sum_{k=-1}^2 \beta_k^{DDD} (\text{Facilities}_{st} \times \text{Expansion}_s \times \mathbf{1}[\text{EventTime}_{st} = k]) + \alpha_s + \gamma_t + Z'_{st} \theta + \varepsilon_{st} \quad (11)$$

where $\text{Prescriptions}_{st}$ is antidepressant claims per 100,000 population in state s and year t , Facilities_{st} is the count of mental health facilities, Expansion_s indicates Medicaid expansion status, EventTime_{st} measures years relative to 2014 when expansion began, α_s are state fixed effects, γ_t are year fixed effects, and Z_{st} includes controls. Standard errors are clustered at the state level.

Online Appendix Figure A12 presents interaction event studies examining how Medicaid expansion affects the relationship between facilities and antidepressant prescriptions.

Panel (a) reveals striking complementarity: facilities in expansion states generate 200 to 280 additional antidepressant claims per 100,000 population post-expansion, compared to minimal effects in non-expansion states. The effect emerges immediately in 2014 and grows over time, consistent with gradual Medicaid enrollment. Panel (b) confirms this pattern in logs, showing a 3 to 5 percent increase in prescriptions. Online Appendix Table A12 reports the triple interaction coefficient of 217.9 ($p < 0.01$), indicating that insurance amplifies facility effects on pharmaceutical treatment by over 200 percent.

This complementarity shows why insurance expansion magnifies mortality reductions. Facilities can prescribe medications, but uninsured patients often cannot afford them. Antidepressants cost \$50 to \$200 monthly without insurance, which is prohibitive for many individuals with mental illness who have limited employment. Medicaid coverage reduces copayments to \$1 to \$3, transforming prescription access.

The mortality consequences are direct: antidepressant treatment reduces suicide risk by 20 to 30 percent (Craig Nelson, 2018) and improves management of comorbid conditions through better sleep, reduced inflammation, and enhanced treatment adherence. The immediate prescription response following Medicaid expansion, combined with the gradual mortality improvements documented in Section 7, traces a clear causal pathway from insurance to prescriptions to reduced mortality.

8.3 Implications for Understanding Treatment Effects

These mechanisms reveal that mental health facilities operate as multidimensional interventions rather than simple treatment providers. They connect vulnerable populations to social insurance programs, enable pharmaceutical treatment when financial barriers fall, and create cascading benefits through improved economic and health stability.

The mechanisms also explain the heterogeneous effects documented earlier. Elderly populations show larger mortality reductions partly because they qualify for Medicare Part D, ensuring prescription access regardless of Medicaid expansion. Less educated populations benefit more because they face higher barriers to navigating disability applications without provider assistance. The larger closing than opening associations in the panel are consistent with the disruption of established provider relationships crucial for maintaining disability benefits and prescription regimens.

Most importantly, these mechanisms validate the complementarity between supply-side and demand-side interventions. Facilities without insurance enable some benefits through disability connections, while insurance without facilities provides coverage but limited access. The combination of facilities with insurance generates the full mortality reductions documented in this paper, underscoring that comprehensive mental health policy requires coordinated investments in both infrastructure and coverage.

9 Welfare Analysis

The mortality reductions documented in this paper have substantial welfare implications. In this section, I quantify the net social benefits of mental health treatment facilities by comparing the value of lives saved against facility operating and capital costs. I employ standard welfare analysis methods used in health economics (Cutler and Richardson, 1998; Murphy and Topel, 2006), valuing mortality reductions using the Value of Statistical Life (VSL) approach commonly applied by regulatory agencies and in the academic literature (Viscusi and Aldy, 2003; Viscusi, 2018; Finkelstein and Hendren, 2020).

9.1 Framework

The welfare gain per facility can be expressed as:

$$W = \theta \times \text{VSL} - C \quad (12)$$

where θ represents lives saved per facility per year, VSL denotes the value of statistical life, and C captures total annual costs (operating plus annualized capital expenditures). From the mortality estimates in Table 3, Column 4, each additional mental health facility reduces deaths by 1.559 per 100,000 residents (SE = 0.255). I deliberately anchor the welfare analysis to this dose-response estimate rather than the larger event-study effects: it is the intensive-margin effect of a marginal facility in the full sample, the most conservative and most policy-portable of the estimates in Table 4, so the net benefits below should be read as a lower bound on the returns to a first facility in an underserved county. Given the average county population of 97,051 in my sample, this translates to 1.51 lives saved per facility annually.

I value mortality reductions using a baseline VSL of \$10 million, consistent with the EPA’s 2014 estimate (Agency, 2010) and widely used in benefit-cost analyses (Ashenfelter, 2006; Kniesner et al., 2012). For robustness, I also consider a range from \$7 million to \$13 million. Annual facility costs include operating expenses of \$3 million per year and annualized capital costs. Drawing on data from the Montana Legislative HJR 16 Study (2014),²⁷ I use construction costs of \$4.5 to \$7.6 million for a 16-bed facility, annualized over a 20-year facility lifetime at a 3 percent discount rate (Viscusi, 2006). This yields total annual costs of \$3.30 to \$3.51 million per facility.

9.2 Main Results

Table 9 presents the cost-benefit analysis. Under baseline assumptions (VSL = \$10 million, midpoint costs), each mental health facility generates annual benefits of \$15.1 million from mortality reductions alone. Against total annual costs of \$3.4 million, this implies net benefits of \$11.7 million per facility per year, yielding a benefit-cost ratio of 4.44. Online Appendix Figure A13 illustrates these benefit-cost ratios across all nine combinations of VSL and cost assumptions, with all ratios substantially exceeding the break-even threshold of 1.0. The lower bound estimate, using the 95 percent confidence interval for mortality effects and low facility costs, still delivers substantial net benefits of \$11.8 million annually (benefit-cost ratio of 4.58). Even the upper bound scenario, combining the confidence interval upper limit with high costs, generates net benefits of \$11.6 million (benefit-cost ratio of 4.31).

These estimates are conservative for several reasons. First, I measure only mortality benefits, excluding substantial morbidity reductions and quality-of-life improvements from mental health treatment (Chisholm et al., 2016; Bloom et al., 2012). Second, the analysis focuses on overall mortality without separately valuing the particularly high toll of suicide and overdose deaths, which disproportionately affect younger individuals with longer remaining life expectancies (Zeckhauser and Shepard, 1980; Aldy and Viscusi, 2008). Third, I do not account for reduced healthcare utilization, criminal justice costs, or productivity gains associated with improved mental health (Kessler, 2011). Fourth, the mortality effects likely understate the true impact, as my difference-in-differences design captures only local treatment effects and may miss broader spillovers to neighboring areas or untreated populations (Goodman-Bacon, 2021).

²⁷Montana Legislative Children, Families, Health, and Human Services Interim Committee, “HJR 16 Study: State-Operated Institutions Building and Operating a 16-Bed Inpatient Facility” (May 2014). Available at: <https://archive.legmt.gov/content/Committees/Interim/2013-2014/Children-Family/Committee-Topics/HJR16/hjr16-building-operating-16-bed-facilities-may2014.pdf>.

Table 9: Welfare Analysis: Cost-Benefit of Mental Health Facilities

	Scenario		
	Baseline	Lower Bound	Upper Bound
<i>Panel A: Lives Saved and Benefits</i>			
Lives saved per facility per year	1.51	2.00	1.03
Annual benefit (VSL = \$10M)	\$ 15125951	\$ 19978864	\$ 10273038
<i>Panel B: Costs</i>			
Annual operating cost	\$3,000,000	\$3,000,000	\$3,000,000
Annualized capital cost	\$ 406,655	\$ 302,471	\$ 510,839
Total annual cost	\$3,406,655	\$3,302,471	\$3,510,839
<i>Panel C: Net Benefits</i>			
Net annual benefit	\$ 11719296	\$ 11823480	\$ 11615111
Benefit-cost ratio	4.44	4.58	4.31

Notes: This table presents the welfare analysis of mental health treatment facilities based on the mortality effects estimated in Table 3, Column 4. The baseline scenario uses a Value of Statistical Life (VSL) of 10 million dollars (EPA 2014 estimate) and midpoint facility costs. The lower bound uses the 95% confidence interval lower bound for mortality effects and low facility costs; the upper bound uses the 95% CI upper bound and high facility costs. Lives saved per facility per year is calculated as absolute value of coefficient times (county population / 100,000), where coefficient = -1.559 (SE = 0.255) from Table 3, Column 4, and mean county population is 97,051. Annual benefits equal lives saved times VSL. Operating costs (3 million dollars per year) and capital costs (4.5M to 7.6M one-time) are from the Montana Legislative HJR 16 Study (May 2014). Capital costs are annualized over 20 years at a 3% discount rate. Net annual benefit equals annual benefit minus total annual cost. Benefit-cost ratio is annual benefit divided by total annual cost. All monetary values are in 2014 dollars.

9.3 Sensitivity Analysis

Online Appendix Table A23 reports net benefits under nine combinations of VSL and cost assumptions, with all scenarios yielding positive net benefits ranging from \$7.1 million to \$16.4 million annually per facility. Online Appendix Figure A18 visualizes this pattern, showing that results are more responsive to VSL assumptions than to cost variation, consistent with findings in the broader VSL literature (Robinson and Hammitt, 2019). Moving from the conservative \$7 million VSL to the upper bound \$13 million VSL more than doubles net benefits, while cost uncertainty produces relatively modest variation.

As an alternative approach, I calculate benefits using Value of Statistical Life-Year (VSLY) estimates, which may better capture the age distribution of prevented deaths (Aldy and Viscusi, 2008; Murphy and Topel, 2006). Assuming 25 years of remaining life expectancy at the average age of mental health-related mortality and applying VSLY estimates ranging from \$100,000 to \$400,000 per life-year,²⁸ I find present values of \$2.6 to \$10.5 million per facility (discounted at 3 percent annually). The midpoint VSLY estimate of \$250,000 yields net benefits of \$3.2 million per facility per year, reinforcing the finding of positive welfare effects even under more conservative valuation approaches.

9.4 Aggregate National Effects

Scaling these facility-level estimates to the national level implies substantial aggregate welfare gains. The sample contains 268,699 facility-year observations over the eighteen-year panel, corresponding to an average stock of approximately 14,900 facilities operating in a given year across the sample counties. Online Appendix Table A24 applies the per-facility estimates to this annual stock. At baseline parameters, the facilities operating in a given year collectively save approximately 22,500 lives annually, generating total benefits of roughly \$225 billion against total costs of \$51 billion. Net annual benefits amount to approximately \$175 billion, equivalent to roughly \$550 per capita for the US population. As a check on magnitudes, 22,500 averted deaths represent just under one percent of annual deaths in the sample, in line with the reduced-form estimates aggregated over the observed facility stock. These aggregate calculations provide a lower bound for national welfare effects, as my sample does not include all US counties and facilities.²⁹ Moreover, the estimates exclude the substantial benefits from reduced morbidity, improved quality of life, and other positive externalities of mental health treatment (Cuijpers et al., 2013).

²⁸These VSLY values correspond to VSL estimates of \$2-10 million divided by remaining life expectancy, consistent with the approach in Murphy and Topel (2006) and Aldy and Viscusi (2008).

²⁹The sample covers counties with available data from both County Business Patterns and Vital Statistics for the period 1999-2016. Counties with missing or suppressed data are excluded from the analysis.

9.5 Interpretation

The welfare analysis delivers a clear message: mental health treatment facilities generate large positive net social benefits through mortality reductions alone. With benefit-cost ratios consistently exceeding 4 to 1 across all specifications, each dollar invested in mental health facilities yields more than four dollars in mortality-related benefits. These findings inform ongoing policy debates about mental health infrastructure investment (Frank, 2006) and suggest that expanding access to mental health treatment may be welfare-enhancing even before accounting for the full range of benefits beyond mortality reduction.

The results also speak to the underinvestment in mental health services documented in prior research (Frank and McGuire, 2000). If markets functioned efficiently, we would expect investment to continue until marginal benefits equal marginal costs. The substantial net benefits I document suggest we remain far from this margin, consistent with significant market failures in mental health care provision, including externalities, information asymmetries, and insurance market imperfections (Arrow, 1978; Frank, 2006). Policies that increase access to mental health treatment facilities appear to address these market failures while generating significant social returns.

10 Discussion

This study provides the first causal evidence that mental health treatment facilities reduce mortality broadly. A county's first facility reduces all-cause deaths by 3.0 percent, about 1 percent of deaths in per-person terms, and each additional facility in the full panel prevents 1.56 deaths per 100,000 residents annually. The effects extend well beyond deaths coded to mental illness, showing that mental healthcare improves overall population health.

Three findings stand out. First, the spillover effects are large and now directly documented: the cause-of-death decomposition attributes 60 percent of the all-cause effect to mental health related causes and a significant 40 percent to chronic and other internal causes, with placebo causes flat. This places mental health facilities alongside hospitals as essential healthcare infrastructure. While previous work showed that geographic access affects utilization (Buchmueller et al., 2006; Currie and Stabile, 2006), and that treatment reduces suicide (Currie, 2025; Ortega, 2023), no prior study demonstrated that mental health facilities reduce overall mortality, nor traced the reduction into physical-health causes of death.

Second, the association between facility churn and MBD mortality in the panel is carried by closings rather than openings: in joint estimation, closings predict 0.139 additional MBD deaths per 100,000 while openings show no dose-response effect; the asymmetry test rejects at the 5 percent level in the most demanding specification, though dedicated closings event studies remain underpowered. Read alongside

Corredor-Waldron and Currie (2022) on hospital closures, this pattern suggests that severed patient-provider relationships are costly to rebuild (Frank, 2006), and it makes the mortality consequences of mental health facility closures a first-order open question that the present design flags but cannot yet settle.

Third, insurance and infrastructure work together. Facilities in Medicaid expansion states prevent 1.53 all-cause and 0.11 MBD deaths per 100,000 after expansion, rising to 1.80 and 0.16 in high-uninsurance counties where expansion bound most. This complementarity advances work by Sommers et al. (2016) and Maclean and Saloner (2019), showing that coverage without facilities or facilities without coverage both fall short. States that expanded Medicaid should prioritize building facilities, while non-expansion states may see limited gains from new facilities without addressing coverage gaps. A fourth pattern deserves note as a finding in its own right: county populations decline by roughly 1.4 percent after a first facility opens, with deaths falling proportionally more, a demographic response to local healthcare infrastructure that merits dedicated study.

The welfare benefits are substantial. Each facility generates \$15.1 million annually in mortality benefits against \$3.4 million in costs, yielding net benefits of \$11.7 million per year, a benefit-cost ratio exceeding 4 to 1. Beyond mortality, facilities increase mental health employment by 3 percent, creating 8-10 jobs, including psychiatrists and support staff (Eckert et al., 2020). These employment effects validate that new facilities represent real treatment capacity, not just reorganization.

The mechanisms extend beyond direct treatment. Facilities connect people to disability benefits, increasing SSI participation by 1.4 percent (Messel et al., 2023). With insurance, facilities enable medication access, generating 280 additional antidepressant prescriptions per 100,000 residents. These pathways explain why elderly and less-educated populations benefit most; they face the highest barriers to navigating healthcare alone, or driving to far facilities when they are not available locally (Salcedo et al., 2026).

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Online Appendix

Mental Healthcare Facilities and Mortality: Evidence from Local Access and Insurance Expansion

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Supplementary Appendix

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A. Mental Health Treatment Facilities: Context and Institutional Details

B. Data

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A Mental Health Treatment Facilities: Context and Institutional Details

This section provides context on mental health treatment facilities in the United States, documenting trends in facility availability during the sample period and describing the services these facilities provide. Understanding the institutional details helps interpret the estimated treatment effects and welfare calculations.

A.1 Trends in Facility Availability (1999–2016)

Figure A2 displays the aggregate trend in mental health treatment facilities from 1999 to 2016. The total number of facilities increased substantially over this period, rising from approximately 10,500 facilities in 1999 to nearly 18,000 by 2016. This represents a 71 percent increase in aggregate facility availability, corresponding to an average annual growth rate of approximately 3.3 percent.

This aggregate expansion masks important heterogeneity in where facility growth occurred. The increase in facilities was concentrated primarily in counties that already had existing mental health infrastructure. Counties without any baseline mental health facilities were significantly less likely to gain access during the sample period. This pattern is consistent with market forces driving facility location decisions: providers locate where demand is established, reimbursement rates are favorable, and complementary healthcare infrastructure exists (Frank and McGuire, 2000).

The concentration of facility growth in areas with existing infrastructure has important implications for access disparities. Rural counties, which are less likely to have baseline mental health services, experienced slower growth in facility availability compared to urban and suburban counties. This geographic pattern contributes to persistent disparities in mental healthcare access documented in prior research (Cummings et al., 2017; Andrilla et al., 2018). My analysis exploits within county variation in facility openings and closings over time, treating the timing of these changes as plausibly exogenous conditional on county fixed effects, time fixed effects, and observable county characteristics.

The variation I exploit comes from discrete changes in facility counts: new facilities opening, existing facilities closing, or facilities changing their service offerings. These changes reflect a combination of market forces (demand growth, competition), policy changes (state licensing requirements, Medicaid reimbursement rates), and economic conditions (local labor markets, real estate costs). The event study analysis in Section 5.5 shows no evidence of differential pre trends between counties that experience facility changes and those that do not, supporting the parallel trends assumption underlying the difference in differences design.

A.2 Treatment Services Provided by Mental Health Facilities

Mental health treatment facilities in my sample include both outpatient and residential settings, identified using NAICS codes 621420 (outpatient mental health and substance abuse centers) and 623220 (residential mental health and substance abuse facilities). Understanding the services these facilities provide helps interpret the estimated mortality effects and clarifies the mechanisms through which facility access affects health outcomes.

Outpatient Services. The majority of facilities in my sample provide outpatient mental health services. Outpatient facilities offer several core treatment modalities:

- *Psychotherapy:* Individual, group, and family therapy addressing emotional and behavioral symptoms. Common therapeutic approaches include cognitive behavioral therapy (CBT), dialectical behavior therapy (DBT), and trauma informed therapies (Hofmann et al., 2012; Linehan, 1993).
- *Medication management:* Psychiatrists, nurse practitioners, and physician assistants prescribe and monitor psychotropic medications, including antidepressants, mood stabilizers, and antipsychotic medications. Medication management typically involves regular follow up appointments to assess treatment response and adjust dosing (Glier and Frank, 2016).
- *Crisis intervention:* Many outpatient facilities provide crisis services for individuals experiencing acute psychiatric symptoms, including suicidal ideation, severe anxiety, or psychotic episodes. Crisis intervention aims to stabilize patients and connect them with appropriate ongoing care (Guo et al., 2001).
- *Case management:* Coordination of care across providers and connection to community resources, including housing assistance, employment services, and social support programs (Ziguras and Stuart, 2000).

Table A4, Panel A shows that outpatient facilities have significant effects on mental health related mortality, with each additional outpatient facility reducing deaths by 0.126 per 100,000 residents (approximately 0.4 percent relative to baseline).

Residential Services. A smaller subset of facilities provide residential or inpatient mental health treatment. Residential facilities offer more intensive, structured care for individuals who require 24 hour supervision and support. Services at residential facilities include:

- *Structured daily programming:* Scheduled therapeutic activities, group sessions, and skills building exercises throughout the day.

- *Medication management*: Daily medication administration and close monitoring by clinical staff, particularly important for medication titration and managing side effects.
- *Skills development*: Training in coping strategies, stress management, social skills, and activities of daily living. These programs help individuals develop tools for managing symptoms after discharge (Drake et al., 2001).
- *Stabilization and safety*: Residential settings provide a safe, supervised environment for individuals at high risk of harm to themselves or others. This is particularly important for suicide prevention and crisis stabilization (Zalsman et al., 2016).

Table A4, Panel B shows that residential facilities also reduce mental health mortality, with effects of similar magnitude to outpatient facilities (coefficient of 0.090 per 100,000, approximately 0.3 percent relative to baseline).

Treatment Pathways and Mortality Reduction. The services provided by these facilities can reduce mortality through several pathways. First, effective treatment of depression, anxiety, bipolar disorder, and other mental health conditions directly reduces suicide risk (Mann et al., 2005). Second, mental health treatment may reduce mortality from substance use disorders through integrated treatment approaches and referrals to specialized substance abuse services (Drake et al., 2001). Third, improved mental health can lead to better self care and treatment adherence for co occurring physical health conditions, reducing mortality from cardiovascular disease and other chronic conditions (Kupfer et al., 2012). Fourth, case management and social support services connect individuals with housing, employment, and community resources that address social determinants of health (Marmot, 2005).

The welfare calculations in Section 9 value only the mortality reductions documented in this paper. These estimates exclude substantial additional benefits from improved quality of life, reduced morbidity, decreased healthcare utilization, and positive spillovers to family members and communities. The mortality effects I estimate therefore represent a lower bound on the total welfare gains from improved mental health facility access.

A.3 Facility Types and Treatment Intensity

Mental health treatment facilities vary substantially in treatment intensity, staffing levels, and patient populations served. This heterogeneity has important implications for interpreting the estimated effects.

Outpatient Facilities (NAICS 621420) typically serve patients who can safely live in the community while receiving treatment. Patients may attend therapy sessions weekly or biweekly, with additional appointments for medication management as needed. Treatment episodes can range from a few months

for acute conditions to years for chronic mental health conditions requiring ongoing management. Outpatient facilities are the primary source of mental healthcare for most individuals, serving patients with depression, anxiety disorders, bipolar disorder, and less severe manifestations of schizophrenia and other psychotic disorders.

Residential Facilities (NAICS 623220) serve patients requiring more intensive, structured care. Length of stay typically ranges from 30 to 90 days, though some programs extend longer for chronic conditions. Residential facilities serve individuals with more severe symptoms, those who have not responded to outpatient treatment, or those requiring intensive stabilization following a crisis. After discharge, patients typically transition to outpatient care for ongoing management.

The similar magnitudes of estimated effects for outpatient and residential facilities (Table A4) suggest that both types of care contribute to mortality reduction. This is consistent with a treatment model where different facility types serve complementary roles: residential facilities provide intensive stabilization and initiate treatment, while outpatient facilities provide ongoing maintenance care and prevent relapse.

Geographic Distribution and Access Barriers. Access to mental health facilities varies substantially across geographic areas. Urban counties tend to have multiple facilities offering diverse treatment options, while rural counties often have limited or no local mental health services. Travel distance to the nearest facility represents a significant access barrier, particularly for individuals without reliable transportation (Cummings et al., 2017). The within county variation I exploit in the empirical analysis captures changes in local access, holding constant county specific factors that affect baseline mental health needs and healthcare infrastructure. The estimated effects therefore represent the impact of improving local access for individuals who previously faced travel barriers or had no proximate source of care.

B Data

This section provides detailed documentation of the data sources, variable construction, and sample selection criteria used in the analysis. All data processing follows standard practices in the literature and ensures consistency with previous studies on healthcare access and mortality outcomes.

B.1 County Business Patterns

I measure access to mental health treatment using establishment counts from the U.S. Census Bureau's County Business Patterns (CBP). The CBP data are derived from annual tax returns submitted to the Internal Revenue Service (IRS) and capture nearly all establishments operating during the week of March 12th each year.² These data have been extensively used in recent economic research to study the effects of healthcare access (Deza et al., 2022a,b; Swensen, 2015; Bondurant et al., 2018).

Facility Identification. Following the literature, I identify mental health treatment facilities using North American Industry Classification System (NAICS) codes. Specifically, I include establishments classified under:

- NAICS 621420: Outpatient mental health and substance abuse centers
- NAICS 623220: Residential mental health and substance abuse facilities

Each establishment represents a single physical location where mental healthcare services are provided. Establishments can only be assigned one NAICS code, ensuring no double-counting of facilities. While these establishments primarily focus on mental healthcare, they may also provide substance use disorder (SUD) treatment services, allowing me to capture the full spectrum of mental health treatment facilities available in each county.

Timing and Lags. The CBP collects data during the payroll period including March 12th of each year. Following IRS processing and Census Bureau compilation, the data typically become available 18-24 months after collection. Consistent with previous studies (Swensen, 2015; Bondurant et al., 2018), I lag facility counts by one year in my regression analyses. This lag accounts for: (1) the time required for facility establishment and staffing, (2) patient awareness and access to new facilities, (3) the lag between IRS reporting and CBP publication, and (4) the time needed for potential improvements in mental health outcomes to materialize.

²The CBP provides comprehensive coverage of business establishments with paid employees, excluding most government entities, self-employed individuals, and agricultural production workers. Coverage includes approximately 98 percent of private non-farm employment.

Sample Coverage. The full panel includes 57,440 county-year observations spanning 1999 to 2016, covering 3,199 counties across all 50 states and the District of Columbia. The main regression samples are slightly smaller, 53,498 county-year observations in the preferred dose-response specification, because the one-year lag of facilities and the requirement of non-missing outcome and control data restrict the estimation sample; Table A1 tracks the samples used across exhibits. This extensive temporal and geographic coverage enables me to examine how changes in local access to mental health treatment facilities affect health outcomes while controlling for time-invariant county characteristics and common temporal shocks.

B.2 Mental Health Employment Data

To establish that facility openings correspond to actual expansion in healthcare provision, I utilize employment data from the County Business Patterns following the imputation methodology developed by Eckert et al. (2020). The CBP employment data present unique challenges: the majority of county-industry cells have suppressed values to protect establishment confidentiality, particularly in small counties or industries with few establishments.

Imputation Methodology. Eckert et al. (2020) address these limitations through a linear programming method that exploits the hierarchical adding-up constraints implicit in the CBP data structure.³ Their approach ensures that imputed values satisfy all disclosed totals and subtotals while minimizing deviations from observed patterns in the data.

Variable Construction. For my analysis, I extract employment data for mental health establishments using NAICS codes 621420 and 623220, consistent with my facility count measures. The employment figures capture the lower bound of full-time and part-time employees working in mental health treatment facilities during the payroll period that includes March 12th of each year. I construct two key measures:

1. *Mental health employment per 100,000 population:*

$$\text{MH Emp}_{ct} = \frac{\text{Total MH Employment}_{ct}}{\text{County Population}_{ct}} \times 100,000 \quad (1)$$

2. *Log mental health employment (plus one):* $\ln(\text{MH Emp}_{ct} + 1)$ to account for counties with zero mental health employment

³Eckert et al. (2020) develop a comprehensive imputation procedure that leverages the nested structure of industry classifications (2-digit through 6-digit NAICS codes) and geographic aggregations (national, state, county levels) in the CBP data. Their method ensures that imputed values satisfy all disclosed totals and subtotals while minimizing deviations from observed patterns. The authors make their imputed employment data publicly available at <https://www.fpeckert.me/cbp/>.

These employment measures serve as first-stage outcomes, verifying that facility openings translate into increased mental healthcare capacity rather than merely reflecting administrative changes or establishment reorganizations. For counties with suppressed employment values in the original CBP that could not be imputed, I assign zero employment, which represents a lower bound on actual mental health sector employment.

B.3 Vital Statistics Mortality Data

My primary outcome measures are mortality rates constructed from the Multiple Cause of Death (MCOd) data files from the National Center for Health Statistics (NCHS) for the period 1999 to 2016. These restricted access data capture all recorded deaths in the United States and provide detailed geographic identifiers at the county level, allowing me to track mortality patterns across space and time.

Data Structure and Coverage. The MCOd files contain one record for each death occurring in the United States, including information on the underlying cause of death (the disease or injury that initiated the chain of events leading directly to death) and up to 20 contributing causes. Each death record includes county of residence, allowing aggregation to the county level for analysis. The data cover all 50 states plus the District of Columbia and represent the complete census of deaths rather than a sample.

Mortality Classification System. Deaths are classified using the International Classification of Diseases, Tenth Revision (ICD-10), implemented in the United States in 1999 ([World Health Organization, 2019](#)). The ICD-10 system provides a hierarchical structure for categorizing diseases and causes of death, with alphanumeric codes ranging from broad categories (Chapter level) to specific diagnoses (four character subcategories). I use ICD-10 codes to identify deaths attributable to mental and behavioral disorders, as well as other specific causes of interest.

Mental and Behavioral Disorder Mortality. My primary outcome of interest is mortality from mental and behavioral disorders (MBD), defined from ICD-10 Chapter V (Mental and Behavioral Disorders, codes F00-F99). Table [A25](#) provides the complete classification structure. I examine deaths attributed to eleven distinct diagnostic categories:

- F00-F09: Organic mental disorders, including dementia in Alzheimer disease (F00) and vascular dementia (F01)
- F10-F19: Mental and behavioral disorders due to psychoactive substance use
- F20-F29: Schizophrenia, schizotypal and delusional disorders
- F30-F39: Mood (affective) disorders, including depressive episodes and bipolar disorder

- F40-F48: Neurotic, stress related and somatoform disorders
- F50-F59: Behavioral syndromes associated with physiological disturbances
- F60-F69: Disorders of adult personality and behavior
- F70-F79: Mental retardation (intellectual disabilities)
- F80-F89: Disorders of psychological development
- F90-F98: Behavioral and emotional disorders with onset in childhood
- F99: Unspecified mental disorder

I construct the aggregate measure of total MBD mortality by summing deaths with an underlying cause in codes F00 through F79; the remaining Chapter V blocks (F80-F99) are excluded because they are rarely recorded as the underlying cause of death. This provides the overall effect on mental health related mortality while the diagnostic subcategories allow examination of heterogeneous effects across condition types.

Overall Mortality. To capture broader health impacts beyond MBD specific causes, I also examine all cause mortality. This outcome includes deaths from all ICD-10 chapters and external causes, providing a comprehensive measure of population health. The all cause mortality measure captures both direct effects of mental health treatment (preventing MBD deaths) and indirect effects through improved management of comorbid physical health conditions, reduced substance abuse, and other pathways.

Component Causes of Mortality. To distinguish mental health related mortality effects from placebo outcomes and to understand mechanisms, I examine several specific causes of death:

Mental health related causes:

- Intentional self harm (suicide): ICD-10 codes X60 through X84
- Mental and behavioral disorders due to substance use: ICD-10 codes F10 through F19

External causes potentially related to mental health:

- Assault: ICD-10 codes X85 through Y09
- Transport accidents: ICD-10 codes V01 through V99
- Other external causes of accidental injury: ICD-10 codes W00 through X59

Placebo outcomes (not expected to respond to mental health treatment):

- Deaths from medical and surgical care complications: ICD-10 codes Y40 through Y84
- Other causes: Residual category including deaths from infectious disease, neoplasms, circulatory disease, and other natural causes not directly linked to mental health

This categorization allows me to test whether effects are specific to mental health related causes or reflect broader changes in mortality reporting or healthcare quality.

Variable Construction. For each cause of death category, I construct annual mortality rates at the county level. I first sum death counts across all relevant ICD-10 codes within each category for each county and year combination. I then normalize these counts by county population (expressed per 100,000 residents) using annual population estimates from the U.S. Census Bureau. This normalization ensures comparability across counties of different sizes and over time.

For county year cells with zero deaths in a particular category, I assign a mortality rate of zero. This approach is standard in mortality research and appropriate given that I examine aggregate county level outcomes rather than individual death records. For very small counties where mortality rates may be volatile due to small denominators, my estimates are less precise but remain unbiased under the parallel trends assumption.

Data Restrictions and Suppression. The NCHS suppresses county level data when death counts are small (typically fewer than 10 deaths) to protect privacy. Suppression is more common for rare causes of death and in small counties. For suppressed values, I cannot distinguish between zero deaths and small positive counts. Following standard practice, I treat suppressed values as missing rather than imputing values. This approach provides conservative estimates of mortality effects, as counties with suppressed data are excluded from the analysis for that particular cause and year.

For my main outcome (overall MBD mortality), data suppression is minimal due to the aggregation across the F00 to F79 codes. For specific diagnostic subcategories and component causes, suppression rates are higher, particularly in rural counties. The sample size varies across outcomes accordingly, as documented in the notes to each regression table.

Consistency Across Sample Period. The ICD-10 classification system remained stable throughout my sample period (1999 to 2016), ensuring consistent death classification over time. This stability is important for my identification strategy, as changes in coding practices could confound temporal variation in mortality rates with changes in facility availability. The NCHS made minor updates to coding rules during this period, but these changes were incremental and did not affect the major cause of death categories I examine.

B.4 Social Security Disability Data

To examine whether mental health facility access affects disability insurance enrollment, I obtain county-level data on Supplemental Security Income (SSI) recipients from the Social Security Administration's SSI State and County Data publication. These data provide annual counts of SSI recipients by county and age group, published in December of each year.

Data Structure and Processing. The SSA publishes disability data in Excel format with non-standard formatting that requires careful processing. The raw files contain multiple header rows and use special characters to indicate suppressed values. For each year from 2009 to 2016, I implement the following processing steps:

1. Import data starting from row 4 to skip header information
2. Extract three key columns: State name (Column A), County name (Column B), ANSI/FIPS Code (Column C), and Recipients aged 18-64 (Column H)
3. Clean FIPS codes by removing commas, spaces, and non-numeric characters
4. Clean recipient counts by removing commas and special characters (#, b, -)
5. Recode suppressed values (indicated by special characters or missing data) to zero, representing a lower bound on actual disability enrollment
6. Convert string variables to numeric format
7. Split 5-digit FIPS codes into 2-digit state codes and 3-digit county codes

Sample Restrictions. The SSA disability data are available from 2009 onwards, restricting this portion of my analysis to the period 2009-2016. I focus on recipients aged 18-64 as this working-age population is most directly affected by labor market conditions and mental health treatment availability. The final disability panel contains 25,592 county-year observations across 3,199 counties.

Suppressed Values. The SSA suppresses county-level data when recipient counts are small (typically fewer than 10 recipients) to protect individual privacy. Suppression is more common in rural counties with small populations. By coding suppressed values as zero, I obtain a lower bound on disability insurance enrollment. This conservative approach biases results toward zero, making it more difficult to detect treatment effects.

Outcome Measures. I construct two measures of disability insurance enrollment:

1. *SSI recipients per 100,000 population*: Raw count normalized by county population

2. *Log SSI recipients (plus one)*: $\ln(\text{SSI}_{ct} + 1)$ to handle counties with zero reported recipients

B.5 Medicare Part D Data

To investigate whether mental health facility access affects medication treatment for mental health conditions, I obtain prescription drug data from the Centers for Medicare and Medicaid Services (CMS) Medicare Part D Prescribers by Geography and Drug dataset. These data provide aggregate information on prescription drug claims filled by Medicare Part D beneficiaries, broken down by prescriber geography and generic drug name.

Data Availability and Geographic Level. The Medicare Part D geographic data are publicly available beginning in 2013, limiting this analysis to 2013-2016. CMS provides these data at multiple geographic levels: national, state, and metropolitan statistical area. For consistency with my facility-level analysis, I aggregate prescriptions to the state level. State-level aggregation provides sufficient statistical power while avoiding the extensive data suppression that occurs at finer geographic levels.

Drug Selection. I focus on antidepressant medications, which represent the most commonly prescribed class of psychotropic medications and serve as a key treatment modality for depression and anxiety disorders. Following clinical practice guidelines and the comparative-efficacy evidence in [Cipriani et al. \(2018\)](#), I identify nine commonly prescribed antidepressants:

- Selective Serotonin Reuptake Inhibitors (SSRIs): Sertraline, Escitalopram, Fluoxetine, Citalopram
- Serotonin-Norepinephrine Reuptake Inhibitors (SNRIs): Duloxetine, Venlafaxine
- Atypical Antidepressants: Bupropion, Trazodone
- Tricyclic Antidepressants: Amitriptyline

These medications account for the vast majority of antidepressant prescriptions in the United States during the study period.

Variable Construction. For each state-year, I sum total prescription claims across all nine antidepressant medications. The key outcome variable is:

$$\text{Antidepressant Claims}_{st} = \sum_{d \in \text{Antidepressants}} \text{Total Claims}_{dst} \quad (2)$$

where s indexes states, t indexes years, and d indexes the nine antidepressant drugs. I also construct a per-capita measure by dividing total claims by state population to account for differences in state size.

Data Processing. The CMS data require several processing steps:

1. Import annual CSV files for years 2013-2016
2. Filter to state-level observations (excluding national and MSA aggregates)
3. Select observations where the generic drug name matches one of the nine antidepressants using case-insensitive string matching
4. Convert state FIPS codes from string to numeric format
5. Collapse to state-year level by summing prescription claims across all antidepressant types

Sample Coverage. The final Medicare Part D panel contains 204 state-year observations (51 states/DC $\times$ 4 years) covering 2013-2016. This state-level analysis complements the county-level facility analysis and provides evidence on treatment intensity margins.

Limitations. Several limitations should be noted. First, Medicare Part D covers only beneficiaries aged 65 and older plus some younger disabled individuals, not the full population. Second, the data capture only filled prescriptions, not prescriptions written but not filled. Third, state-level aggregation may mask heterogeneity in treatment patterns across counties within states. Despite these limitations, the Medicare Part D data provide valuable evidence on whether facility expansion affects pharmaceutical treatment of mental health conditions.

B.6 Additional Data Sources

Demographic and Economic Controls. I incorporate time-varying county-level controls from several sources to account for differences between counties that experience changes in mental health treatment facility access and those that do not. From the National Institute for Health Surveillance (SEER) ([Surveillance, Epidemiology, and End Results](#), SEER), I obtain population counts by age group (under 18, 18-64, 65 and over), which are crucial for controlling for demographic factors that affect both mental health treatment needs and mortality patterns. From the Regional Economic Information System (REIS) ([Bureau of Economic Analysis](#), BEA), I obtain measures of economic conditions including per capita net earnings, welfare receipts, and employment-to-population ratios.

Urban-Rural Classification. I use the 1993 rural-urban continuum codes from the U.S. Department of Agriculture to classify counties along the urban-rural spectrum.⁴ This classification is time-invariant,

⁴The rural-urban continuum codes classify counties into nine categories based on population size and adjacency to metropolitan areas. I use the 1993 vintage to avoid endogenous reclassification during the sample period.

reflecting longstanding geographic characteristics that may influence both mental health facility location decisions and health outcomes. Access to mental health facilities may vary systematically with urbanization levels due to differences in population density, provider supply, and treatment-seeking behavior.

Health Insurance Coverage. To examine heterogeneous treatment effects by baseline insurance coverage, I incorporate county-level health insurance data from the Small Area Health Insurance Estimates (SAHIE) program administered by the U.S. Census Bureau (U.S. Census Bureau, 2021). The SAHIE provides model-based estimates of health insurance coverage for all counties, utilizing data from the American Community Survey, administrative records, and demographic information to produce reliable estimates even for areas with small populations. I use 2013 baseline uninsurance rates to capture pre-Affordable Care Act variation in insurance coverage. This baseline measure reflects longstanding differences in healthcare access that predate both Medicaid expansion and the individual mandate.

Medicaid Expansion. To explore policy interactions, I incorporate information on state Medicaid expansion decisions under the Affordable Care Act. Following Carey et al. (2020), I classify states based on their expansion status and timing. As of 2016, 31 states plus the District of Columbia had expanded Medicaid eligibility to 138 percent of the federal poverty level, while 19 states had not expanded.⁵ This policy variation allows me to explore whether mental health facility effects differ based on the insurance coverage environment, as Medicaid expansion substantially increased coverage for mental health and substance abuse treatment services (Maclean and Saloner, 2019).

The combination of baseline uninsurance rates and Medicaid expansion timing creates rich variation that helps identify how insurance access modulates the relationship between treatment facility availability and health outcomes. I do not separately document these data sources in detail here, as their construction and use are standard in the health economics literature studying ACA impacts.

⁵See Table A26 for detailed information on expansion timing by state.

C Additional Details on Empirical Strategy

This section provides additional details on the identification strategy, estimation methods, and robustness checks employed in the main analysis.

C.1 Identification Strategy

The identification strategy exploits within-county variation in mental health treatment facility presence to estimate their effect on mortality outcomes. This approach builds on established research examining the health effects of treatment access (Swensen, 2015; Bondurant et al., 2018; Messel et al., 2023). The empirical strategy leverages temporal within-county variation from facility openings and closings while accounting for time-invariant county characteristics and aggregate shocks.

Baseline TWFE Specification. The baseline specification estimates the relationship between mental health treatment facilities and mortality using a two-way fixed effects (TWFE) model:

$$Y_{ct} = \alpha_c + \eta_t + \beta \text{Facilities}_{c,t-1} + \mathbf{X}_{ct}' \theta + \varepsilon_{ct} \quad (3)$$

where Y_{ct} represents the mortality rate (per 100,000 residents) in county c in year t . The main explanatory variable, $\text{Facilities}_{c,t-1}$, measures the number of mental health treatment facilities in county c in year $t - 1$, allowing for delayed mortality responses to changes in treatment access.⁶ The model includes county fixed effects (α_c) to absorb time-invariant county characteristics that might correlate with both facility locations and mortality outcomes, and year fixed effects (η_t) to control for common temporal shocks. The vector $\mathbf{X}_{ct}$ includes demographic and economic controls. Standard errors are clustered at the county level to account for serial correlation in outcomes within counties over time.

Identification Assumptions. The TWFE estimator identifies the causal effect of mental health facilities under the parallel trends assumption: in the absence of facility changes, mortality trends would have evolved similarly across counties with different facility trajectories. This assumption is stronger than required for simple cross-sectional comparisons but weaker than assuming selection on observables, as county fixed effects absorb all time-invariant confounders. The inclusion of state-by-year fixed effects further relaxes this assumption by allowing for state-specific temporal shocks to mortality and mental health policy.

Threats to Identification. The primary threat to identification is that facility location decisions might re-

⁶The one-year lag structure follows Swensen (2015) and helps mitigate potential simultaneity bias between facility locations and current-period health outcomes. This specification also accounts for the lag between IRS reporting and CBP publication, as discussed in Appendix Section B.1.

spond to anticipated changes in local mortality patterns. Several features of my empirical design address this concern. First, the one-year lag between facility counts and outcomes reduces simultaneity bias. Second, mental health treatment facilities typically require substantial lead time for planning, construction, and staffing, making it unlikely that facility openings respond quickly to local mortality shocks. Third, I show in event study analyses (see below and Appendix Section C.2) that pre-treatment mortality trends are parallel between counties that do and do not experience facility changes, providing support for the parallel trends assumption.

C.2 Event Study Specifications

To examine the dynamic effects of mental health facility changes and test the parallel trends assumption, I estimate event study specifications that trace out mortality responses in the years before and after facility openings.

TWFE Event Study. The TWFE event study specification is:

$$Y_{ct} = \alpha_c + \eta_t + \gamma_{st} + \sum_{\substack{k=-5 \\ k \neq -1}}^5 \beta_k \mathbf{1}(\text{RelTime}_{ct} = k) + \mathbf{X}'_{ct} \boldsymbol{\theta} + \varepsilon_{ct} \quad (4)$$

where $\text{RelTime}_{ct} = t - T_c^*$ measures years relative to the first treatment year T_c^* for county c . The coefficients β_k trace out the temporal path of mortality responses relative to the year before facility opening ($k = -1$), which serves as the baseline period. I include leads up to five years before treatment ($k = -5, \dots, -2$) to test for pre-treatment differences in trends, and lags up to five years after treatment ($k = 0, \dots, 5$) to examine treatment effect dynamics.

Sample Restrictions. Following Messel et al. (2023), I restrict the event study sample to ensure clean identification. Treatment counties experience exactly one increase in mental health establishments during the sample period, with no decreases before or after this increase. Control counties (never-treated) maintain constant establishment counts throughout. This restriction eliminates counties with multiple treatment episodes or reversals, which could confound the interpretation of dynamic effects.⁷

Contemporaneous Event Study. I also estimate a contemporaneous event study specification to examine anticipatory effects:

⁷Counties with multiple increases or any pre-treatment changes are excluded from the event study sample but retained in the main TWFE analysis, which uses all available variation. Approximately 85 percent of county-year observations are included in the event study sample.

$$Y_{ct} = \alpha_c + \eta_t + \gamma_{st} + \sum_{\substack{k=-5 \\ k \neq -1}}^5 \beta_k \mathbf{1}(\text{RelTime}_{ct} = -k) + \mathbf{X}_{ct}' \boldsymbol{\theta} + \varepsilon_{ct} \quad (5)$$

The key difference is that the event time is reversed: negative values of k now correspond to post-treatment periods, while positive values correspond to pre-treatment periods. This specification tests for anticipatory responses to facility changes. Following [Malani and Reif \(2015\)](#) and [Angrist and Pischke \(2009\)](#), examining both lagged and contemporaneous effects provides a more complete picture of treatment dynamics.

C.3 Inverse Probability Weighting

To address potential selection bias from non-random facility location decisions, I implement inverse probability weighting (IPW) following [Fischer et al. \(2024\)](#). The IPW approach reweights observations to create balance between treated and control counties on observable characteristics, reducing bias from selection on observables.

Propensity Score Estimation. I estimate the propensity score, the probability of receiving treatment conditional on observables, using a probit model:

$$\Pr(\text{Treated}_c = 1 | \mathbf{Z}_c) = \Phi(\mathbf{Z}_c' \boldsymbol{\delta}) \quad (6)$$

where $\Phi(\cdot)$ is the standard normal cumulative distribution function, and $\mathbf{Z}_c$ includes pre-treatment county characteristics that might influence facility location decisions: baseline population, age distribution, income levels, employment rates, urbanicity, and existing healthcare infrastructure. The propensity score $\hat{p}_c = \hat{\Phi}(\mathbf{Z}_c' \hat{\boldsymbol{\delta}})$ captures the predicted probability that county c experiences a facility opening based on these observables.

IPW Estimation. I construct inverse probability weights as:

$$w_c = \frac{\text{Treated}_c}{\hat{p}_c} + \frac{1 - \text{Treated}_c}{1 - \hat{p}_c} \quad (7)$$

These weights upweight observations that are underrepresented in their treatment group given their characteristics. Counties that receive treatment despite having characteristics typical of control counties receive higher weight, and vice versa. The IPW-weighted regression is:

$$Y_{ct} = \alpha_c + \eta_t + \gamma_{st} + \beta \text{Facilities}_{c,t-1} + \mathbf{X}'_{ct} \theta + \varepsilon_{ct} \quad (8)$$

weighted by w_c . Under the assumption of selection on observables, the IPW estimator recovers the average treatment effect on the treated (ATT).

Balance Diagnostics. I assess covariate balance by comparing standardized differences in pre-treatment characteristics between treated and control counties before and after weighting. The IPW procedure substantially improves balance: all standardized differences fall below 0.1 after weighting, compared to several exceeding 0.25 before weighting. This improved balance increases confidence that the IPW estimates reflect causal effects rather than selection bias.

C.4 Sun and Abraham (2021) Estimator

Recent econometric research has shown that two-way fixed effects estimators can produce biased estimates of treatment effects when treatment timing varies across units and treatment effects are heterogeneous (Goodman-Bacon, 2021; Sun and Abraham, 2021). The bias arises because TWFE uses already-treated units as controls for newly treated units, which can lead to negative weights on some treatment effects.

Sun-Abraham Methodology. I address this concern by implementing the interaction-weighted estimator proposed by Sun and Abraham (2021). This estimator separates treatment effects by cohort (defined by initial treatment timing) and estimates cohort-specific effects using only never-treated and not-yet-treated units as controls. The Sun-Abraham specification is:

$$Y_{ct} = \alpha_c + \eta_t + \sum_{g \in G} \sum_{\substack{k=-5 \\ k \neq -1}}^5 \beta_{g,k} \mathbf{1}(G_c = g) \times \mathbf{1}(\text{RelTime}_{ct} = k) + \mathbf{X}'_{ct} \theta + \varepsilon_{ct} \quad (9)$$

where G is the set of treatment cohorts, G_c denotes county c 's cohort, and $\beta_{g,k}$ captures the effect for cohort g at relative time k . The aggregate treatment effect at relative time k is:

$$\beta_k = \sum_{g \in G} \omega_{g,k} \beta_{g,k} \quad (10)$$

where $\omega_{g,k}$ are weights proportional to cohort size. This approach ensures that estimates are not contaminated by comparisons between already-treated and newly-treated units.

Implementation Details. I implement the Sun-Abraham estimator using the `eventstudyinteract` command in Stata (Sun and Abraham, 2021). Treatment cohorts are defined by the first year a county experiences a facility increase. Never-treated counties (those with constant facility counts throughout the sample period) serve as the primary control group. I estimate cohort-specific effects for relative times $k \in \{-5, \dots, 5\}$, excluding $k = -1$ as the baseline period. Standard errors are clustered at the county level to account for serial correlation.

Comparison to TWFE. In practice, I find that Sun-Abraham estimates are very similar to TWFE estimates, suggesting limited heterogeneity in treatment effects across cohorts or that the timing of treatment is sufficiently dispersed that already-treated units do not heavily influence TWFE estimates. This similarity provides reassurance that the main TWFE results are not driven by the econometric issues identified by Goodman-Bacon (2021) and Sun and Abraham (2021).

C.5 Robustness Checks

I conduct several analyses to validate the causal interpretation of my main findings and assess the sensitivity of results to alternative specifications.

Progressive Specification Checks. As shown in Table A3, the estimated effects are robust to increasingly demanding specifications. The impact remains stable and statistically significant when progressively adding state-by-year fixed effects and socioeconomic controls. Column 1 includes only county and year fixed effects, yielding a coefficient of -1.740 deaths per 100,000 (SE = 0.220). Column 2 adds state-by-year fixed effects to account for state-specific policy changes and economic trends, with the coefficient changing to -1.373 (SE = 0.224). Columns 3 and 4 sequentially add demographic controls (age distribution) and economic controls (employment rates, per capita earnings, transfer payments), with the full specification (Column 4) indicating a coefficient of -1.559 (SE = 0.255). The stability of estimates across these specifications suggests that observed time-varying characteristics do not heavily confound the facility-mortality relationship.

Event Studies with Additional Controls. The event study analysis in Figure A16 provides crucial robustness checks by testing for pre-existing trends. Panel A augments the baseline specification with lagged mortality rates to account for pre-existing trends in health outcomes. Panel B includes lagged controls for mental health establishments to address potential dynamic selection in facility placement. Both specifications maintain the core controls from my primary event study analysis: county and year fixed effects, state-by-year fixed effects, and demographic and economic characteristics. Across both specifications, I observe no significant pre-trends in the years preceding facility changes (all pre-treatment coefficients are small and statistically insignificant), while the treatment effects emerge sharply after facility establishment and persist over time. The parallel patterns across both specifications with lagged

outcome controls (Panel A) and lagged establishment controls (Panel B) provide strong support for my identification strategy and confirm that results are not driven by anticipatory effects or pre-existing divergent trends.

Controlling for General Healthcare Infrastructure. A key concern is that mental health facility locations might correlate with broader healthcare infrastructure expansion. To address this, Table A18 presents estimates that explicitly control for hospital establishments. The results remain stable: an additional mental health facility reduces MBD mortality by 0.079 percent (Column 1), nearly identical to my baseline estimates. This robustness extends to diagnostic subcategories, with particularly strong effects for organic mental disorders (F00-F09) and mood disorders (F30-F39). Importantly, controlling for general healthcare access does not meaningfully alter my conclusions, suggesting that mental health facilities have distinct effects beyond overall healthcare infrastructure and are not merely proxying for broader improvements in medical access.

Placebo Outcomes: Unrelated Mortality Causes. To rule out that my findings reflect broader local health trends or changes in mortality reporting practices, I examine causes of death that should be unrelated to mental health treatment access. Table A19 investigates three placebo outcomes: non-mental-health-related deaths, transport accidents, and deaths from medical/surgical care complications. The estimates are small in magnitude and statistically insignificant across all outcomes. For instance, the effect on non-MBD mortality is -0.029 (SE = 0.074, $p = 0.69$), and the effect on transport accidents is 0.004 (SE = 0.014, $p = 0.77$). These null effects on unrelated mortality measures provide strong evidence against the hypothesis that results are driven by spurious correlation with broader mortality trends, changes in death certificate coding, or general improvements in healthcare quality. The specificity of effects to mental health-related mortality, combined with the hospital control analysis and event studies, strongly supports a causal interpretation of the main results.

D Additional Details on Welfare Analysis

This section provides additional technical details on the welfare analysis presented in Section 9, including the VSL methodology, cost data sources, alternative VSLY calculations, and sensitivity to discount rate assumptions.

D.1 Value of Statistical Life (VSL) Approach

The VSL represents the rate at which individuals are willing to trade wealth for reductions in mortality risk. It is derived from observed behavior in markets with mortality risk, such as labor markets where workers accept wage premiums for dangerous jobs, or consumer markets where individuals pay more for safety features. The VSL is not the value of a specific identified life, but rather the aggregated willingness to pay for small reductions in mortality risk across a population.

EPA Baseline Estimate. I use the Environmental Protection Agency’s 2014 VSL estimate of \$10 million as my baseline (Agency, 2010). This estimate is derived from a meta analysis of labor market studies examining compensating wage differentials for occupational mortality risk. The EPA regularly updates its VSL estimate for regulatory impact analyses, and the \$10 million value (in 2014 dollars) represents the agency’s central tendency estimate during my sample period.

Sensitivity Range. To account for uncertainty in VSL estimates, I consider a range from \$7 million to \$13 million. This range reflects the distribution of VSL estimates in the academic literature (Viscusi, 2018; Kniesner et al., 2012). The lower bound (\$7 million) represents more conservative estimates from stated preference studies and some labor market analyses, while the upper bound (\$13 million) captures higher estimates from recent hedonic wage studies and benefit transfer adjustments for income growth over time.

Application to Mortality Reductions. The welfare benefit of preventing θ deaths per facility per year is calculated as:

$$B = \theta \times \text{VSL} \quad (11)$$

From Table 3, Column 4, each mental health facility reduces deaths by 1.559 per 100,000 residents. Given the average county population of 97,051, this translates to:

$$\theta = \left| \hat{\beta} \right| \times \frac{\bar{P}}{100,000} = 1.559 \times \frac{97,051}{100,000} = 1.51 \text{ lives per facility per year} \quad (12)$$

At the baseline VSL of \$10 million, annual benefits per facility are:

$$B_{\text{baseline}} = 1.51 \times \$10,000,000 = \$15,100,000 \quad (13)$$

Limitations and Conservative Nature. Several factors suggest that VSL-based estimates may understate true welfare gains. First, VSL estimates are typically derived from working-age populations, while mental health-related mortality affects a broader age distribution. Second, VSL does not capture the value of reduced morbidity, improved quality of life, or positive externalities to family members and communities. Third, the VSL approach assumes that all prevented deaths have equal value, while deaths at younger ages (with more remaining life-years) may warrant higher valuations. Despite these limitations, the VSL approach provides a transparent, widely accepted framework for monetizing mortality reductions that facilitates comparison with other health interventions and policy analyses.

D.2 Facility Cost Data

Facility cost estimates are drawn from the Montana Legislative Children, Families, Health, and Human Services Interim Committee’s “HJR 16 Study: State-Operated Institutions Building and Operating a 16-Bed Inpatient Facility” (May 2014).⁸ This study provides detailed cost projections for constructing and operating a 16-bed mental health treatment facility, based on proposals from multiple architectural firms and operational data from existing facilities.

Capital Costs. The study reports construction cost estimates ranging from \$4.5 million to \$7.6 million for a 16-bed secure residential treatment facility. These costs include:

- Building construction and site work
- Specialized infrastructure (security systems, safety features)
- Furniture, fixtures, and equipment
- Architectural and engineering fees
- Contingency reserves (typically 10-15% of construction costs)

The range reflects variation in site characteristics, construction materials, and regulatory requirements. I use the midpoint estimate of \$6.05 million for baseline calculations and report results across the full range for robustness.

⁸Available at: <https://archive.legmt.gov/content/Committees/Interim/2013-2014/Children-Family/Committee-Topics/HJR16/hjr16-building-operating-16-bed-facilities-may2014.pdf>

Operating Costs. Annual operating costs are estimated at \$3.0 million per 16-bed facility, based on staffing models and operational data from comparable facilities. Major cost components include:

- Personnel (psychiatrists, nurses, counselors, administrative staff): \$2.1 million (70%)
- Pharmaceuticals and medical supplies: \$450,000 (15%)
- Facility maintenance and utilities: \$300,000 (10%)
- Other operating expenses (food, transportation, etc.): \$150,000 (5%)

These estimates are conservative in that they reflect secure residential treatment settings with higher staffing ratios and operational requirements than typical outpatient mental health facilities. Many facilities in my sample provide outpatient services with lower per-facility operating costs.

Annualization of Capital Costs. To compare one-time capital costs with annual operating costs and benefits, I annualize capital expenditures over the facility's expected lifetime. Following standard practice in infrastructure cost-benefit analysis, I assume a 20-year facility lifetime and a 3 percent real discount rate. The annuity factor is:

$$a_{T,r} = \frac{1 - (1 + r)^{-T}}{r} = \frac{1 - (1.03)^{-20}}{0.03} = 14.88 \quad (14)$$

The annualized capital cost is then:

$$C_{\text{capital}}^{\text{annual}} = \frac{C_{\text{capital}}^{\text{total}}}{a_{T,r}} \quad (15)$$

For the midpoint capital cost of \$6.05 million, this yields an annual equivalent of \$407,000. Combined with the \$3.0 million annual operating cost, total annual costs are \$3.407 million per facility.

Applicability to Sample Facilities. The Montana cost data provide a reasonable benchmark for mental health treatment facilities in my sample. While facility sizes vary (the Montana study focuses on 16-bed facilities), cost estimates scale approximately linearly with capacity for facilities of similar type. Moreover, using these standardized cost estimates avoids endogeneity concerns that might arise from using actual facility-specific costs, which could correlate with local economic conditions or treatment effectiveness.

D.3 Value of Statistical Life-Year (VSLY) Alternative

As an alternative to the VSL approach, I calculate benefits using the Value of Statistical Life-Year (VSLY), which may better capture the age distribution of prevented deaths. The VSLY represents will-

ingness to pay for an additional year of life expectancy and is particularly appropriate when comparing interventions with different age profiles of mortality reduction.

Methodology. The total value of preventing a death is:

$$V = \text{VSLY} \times \sum_{j=1}^T \left(\frac{1}{1+r} \right)^{j-1} \quad (16)$$

where T is remaining life expectancy at the average age of death, r is the discount rate, and the sum represents the present value of future life-years. For a remaining life expectancy of T years and discount rate r , this simplifies to:

$$V = \text{VSLY} \times \frac{1 - (1+r)^{-T}}{r} \quad (17)$$

Parameter Assumptions. Mental health-related mortality in my sample affects individuals across a broad age range, with substantial mortality in middle age (45-65 years). For individuals dying at age 55, remaining life expectancy is approximately 25 years (based on actuarial tables). I use this as my baseline assumption, with sensitivity analysis considering 20 to 30 years.

For the VSLY, I consider three estimates:

- Conservative: \$100,000 per life-year
- Baseline: \$250,000 per life-year
- Upper bound: \$400,000 per life-year

These values are consistent with VSL estimates of \$2-10 million divided by remaining life expectancy of 25-40 years, as recommended by [Murphy and Topel \(2006\)](#) and applied in recent health economics research ([Cutler and McClellan, 2001](#)).

Results. At the baseline VSLY of \$250,000 and 25 years remaining life expectancy, the present value (discounted at 3% annually) of preventing one death is:

$$V = \$250,000 \times \frac{1 - (1.03)^{-25}}{0.03} = \$250,000 \times 17.41 = \$4,353,000 \quad (18)$$

For 1.51 lives saved per facility per year, total annual benefits are:

$$B_{\text{VSLY}} = 1.51 \times \$4,353,000 = \$6,573,000 \quad (19)$$

Against midpoint annual costs of \$3.407 million, this yields net benefits of \$3.166 million per facility per year. While lower than VSL-based estimates (reflecting the discounting of future life-years), the VSLY approach still indicates substantial positive net benefits.

Sensitivity to Life Expectancy. The VSLY-based benefits are sensitive to assumptions about remaining life expectancy:

- 20 years: Net benefits = \$1.79 million per facility
- 25 years (baseline): Net benefits = \$3.17 million per facility
- 30 years: Net benefits = \$4.35 million per facility

All scenarios yield positive net benefits, confirming that results are robust to alternative assumptions about the age distribution of prevented deaths.

D.4 Sensitivity to Discount Rates

The choice of discount rate affects both the annualization of capital costs and the present value of future life-years in VSLY calculations. I examine sensitivity to discount rates ranging from 2 percent to 7 percent, spanning the range commonly used in regulatory cost benefit analyses.

Impact on Capital Costs. The discount rate directly affects the annuity factor used to convert one time capital expenditures into annual equivalents. At lower discount rates, the annuity factor is larger, reducing the annual equivalent cost of capital investments. At higher rates, annualized capital costs increase. Using the midpoint capital cost of \$6.05 million from the Montana study:

Discount Rate	Annuity Factor	Annual Capital	Total Annual Cost
2%	16.35	\$370,000	\$3,370,000
3% (baseline)	14.88	\$407,000	\$3,407,000
5%	12.46	\$486,000	\$3,486,000
7%	10.59	\$571,000	\$3,571,000

where total annual cost includes \$3.0 million in operating expenses plus the annualized capital cost. The variation in total annual costs is modest (6 percent range from lowest to highest), suggesting that discount rate uncertainty has limited impact on cost estimates. The baseline 3 percent discount rate used in Table 9 yields an annualized capital cost of \$407,000, producing total annual costs of \$3.407 million as reported in the main welfare analysis.

Impact on VSLY Calculations. The discount rate more substantially affects VSLY based benefit estimates, as higher discount rates reduce the present value of future life years. For baseline VSLY of \$250,000 and 25 year life expectancy, the present value of preventing one death varies significantly with the discount rate:

Discount Rate	PV Factor	Value per Life	Net Benefit per Facility
2%	19.52	\$4,880,000	\$4,000,000
3% (baseline)	17.41	\$4,353,000	\$3,166,000
5%	14.09	\$3,523,000	\$1,911,000
7%	11.65	\$2,913,000	\$831,000

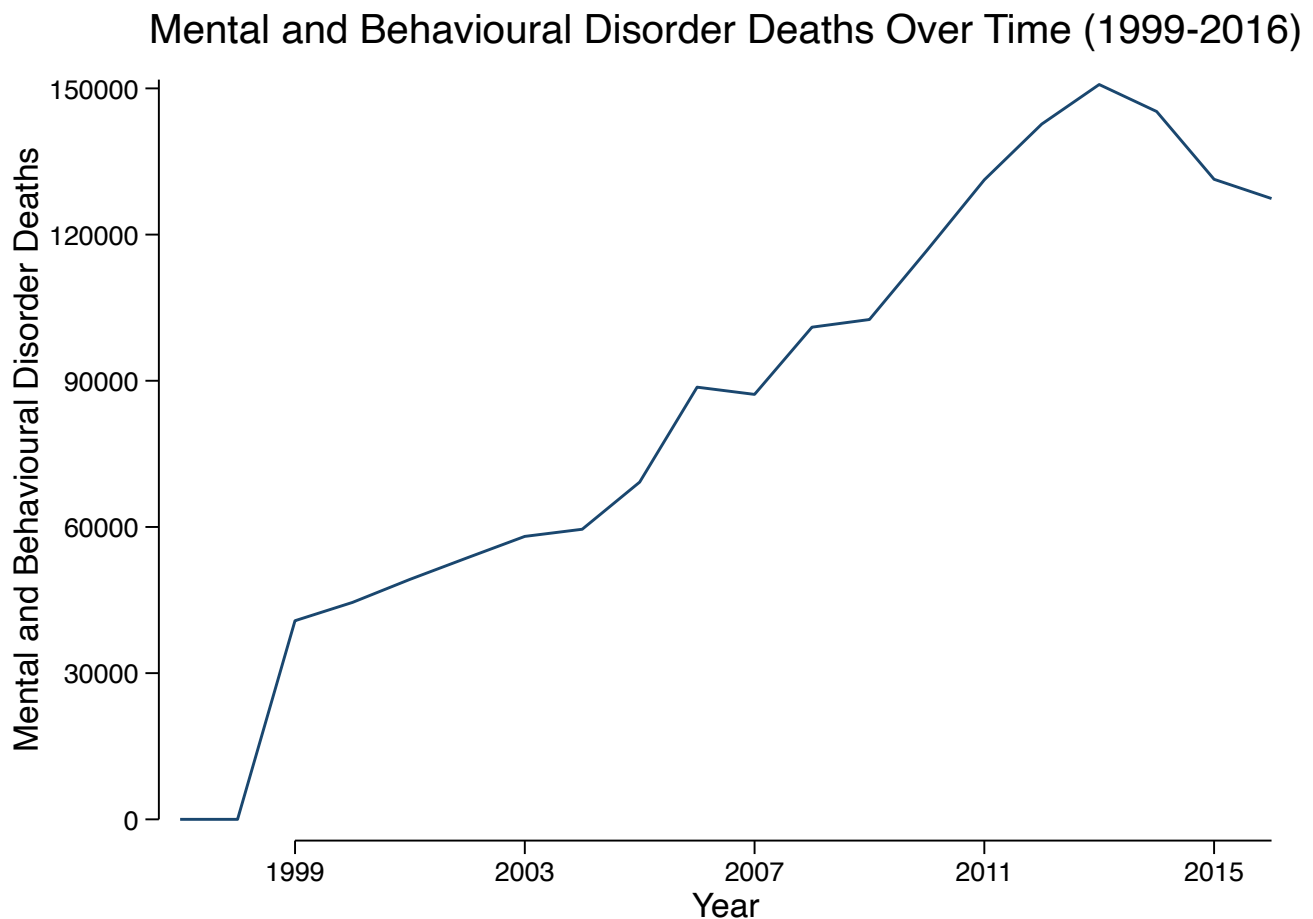
where the present value factor equals $(1 - (1 + r)^{-T})/r$ for discount rate r and life expectancy $T = 25$ years. Net benefit per facility equals the value per life times 1.51 lives saved per facility (from Table 9) minus total annual costs of \$3.407 million. While net benefits decline with higher discount rates (reflecting lower present value of future life years), all scenarios remain positive. Even at the conservative 7 percent discount rate, which is higher than typical rates used for long term health investments, mental health facilities generate net benefits exceeding \$800,000 per facility per year.

Comparison to VSL Approach. Importantly, the VSL approach used in the main analysis (Table 9) does not require explicit discounting of future mortality reductions, as VSL estimates already embed individuals' time preferences and mortality risk profiles. The VSLY approach requires explicit discounting and is therefore more sensitive to discount rate assumptions. The robustness of conclusions across both approaches and across different discount rates strengthens confidence in the finding of positive net social benefits from mental health facility access.

E Appendix Figures

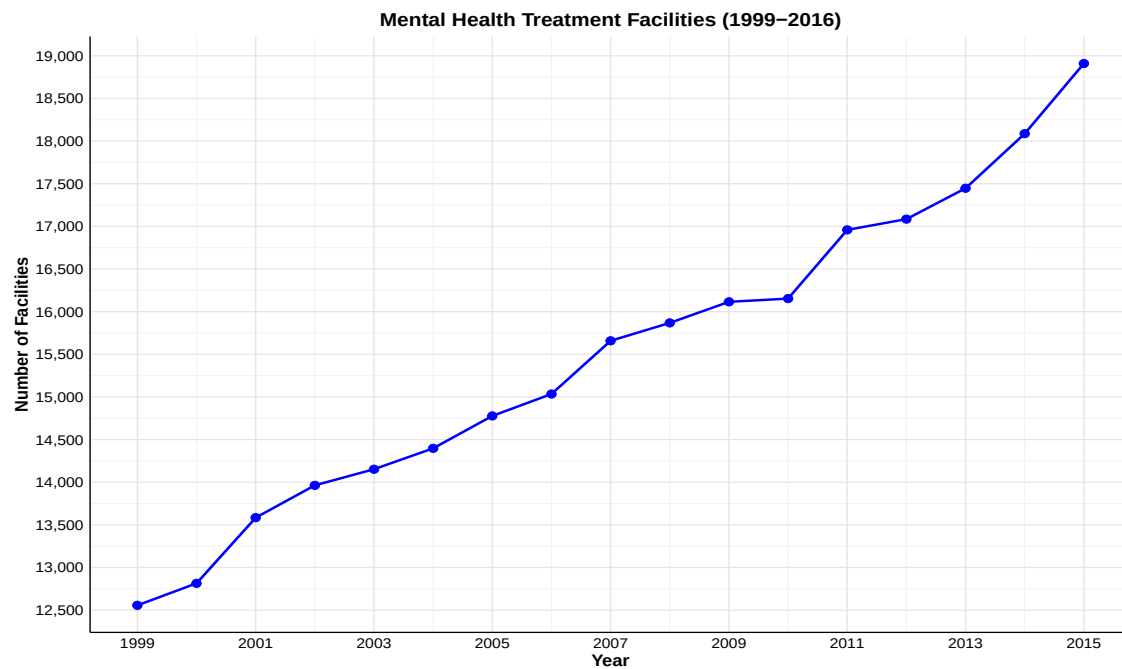
A note on units in the event-study exhibits. The Sun-Abraham event studies in this appendix that predate the revised draft report effects on death *counts*; where an axis label in those exhibits reads “rate,” the underlying outcome is the count of deaths. The revised headline exhibits in the main text report effects both on counts and on population-weighted deaths per 100,000, and Section 5 of the main text explains the roles of the two sets of units. Percent changes, which are invariant to this distinction up to the population response documented in Table A7, are unaffected.

Figure A1: Trend of Mental and Behavioral Disorder Deaths, 1999-2016



Notes: This figure plots the aggregate annual number of deaths from Mental and Behavioral Disorders (MBDs) in the United States from 1999 to 2016. Data are from the restricted-access Multiple Cause of Death (MCOD) files from the National Center for Health Statistics (NCHS). The MBD mortality measure is an aggregate of deaths with an underlying cause falling under the International Classification of Diseases, 10th Revision (ICD-10) codes F00-F79. The y-axis represents the total count of deaths, not a rate.

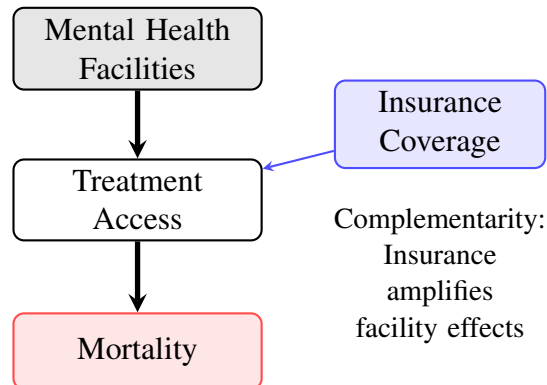
Figure A2: Temporal Variation of Mental Health Treatment Facilities



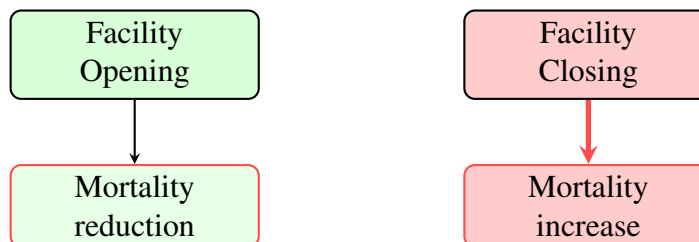
Note: Data on the county's mental health facility treatment comes from the County Business Patterns. A county is designated as having access to mental health services if it has any establishment that provides these services at a given year. The sample is restricted to 1999-2016. The figure presents mean establishment trend over time. A county is designated as having "net access" if it goes from having no mental health treatment facilities to having some for the rest of the sample period. Section 4.1 presents more information on these data.

Figure A3: Conceptual Framework. Panel A illustrates how mental health facilities affect mortality through treatment access, with insurance coverage potentially acting as a complementary input. Panel B depicts potential asymmetry in facility changes, where closures may generate larger mortality effects than openings due to disruption costs and adjustment frictions. Panel C shows three hypothesized mechanisms: facilities as gateways to disability programs, enablers of prescription access, and sources of mental health employment.

Panel A: How Mental Health Facilities Affect Mortality



Panel B: Asymmetric Effects



Panel C: Mechanisms

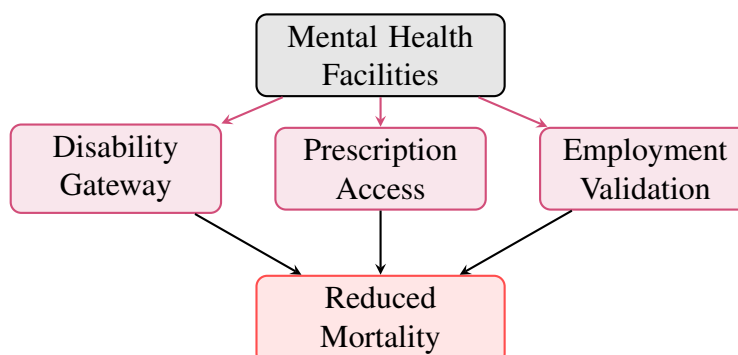
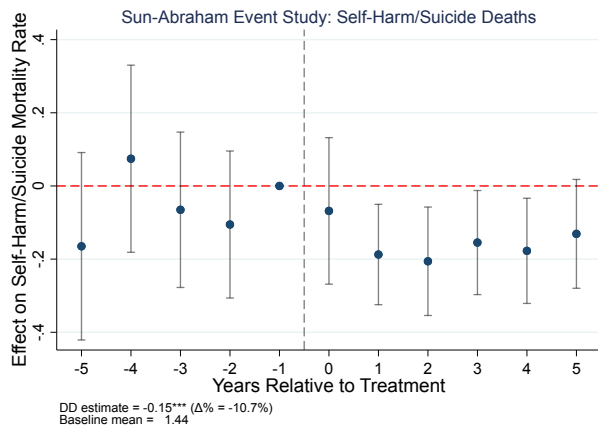
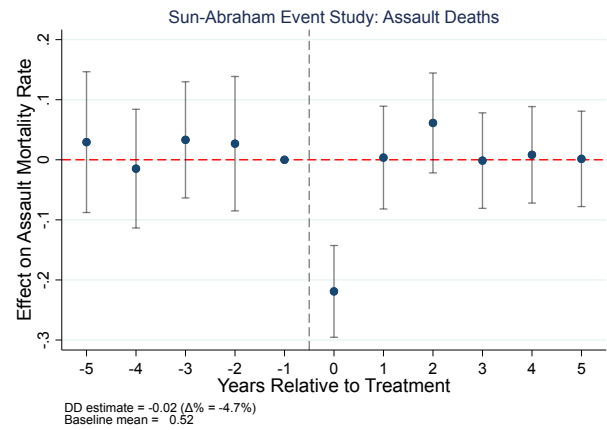


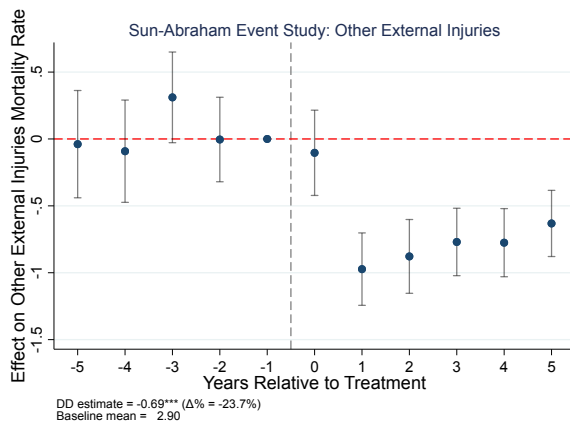
Figure A4: Sun-Abraham Event Study Estimates for Component Causes of Mortality



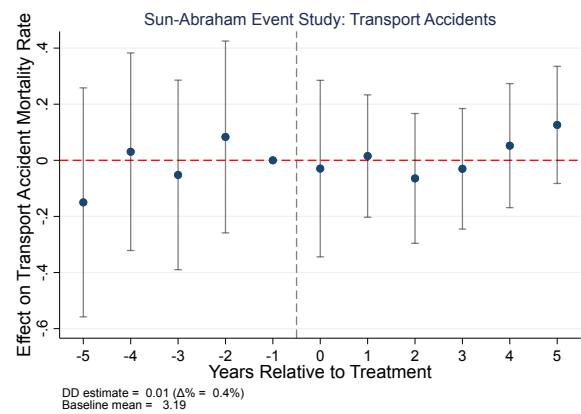
(a) Self-Harm/Suicide Deaths



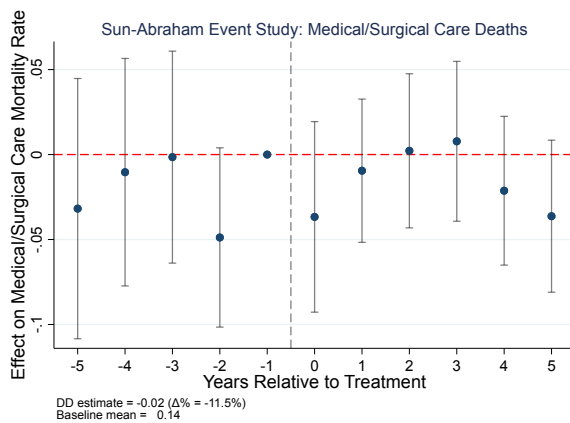
(b) Assault Deaths



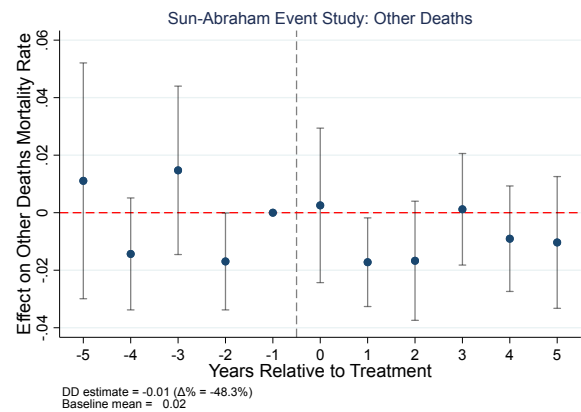
(c) Other External Injuries



(d) Transport Accidents



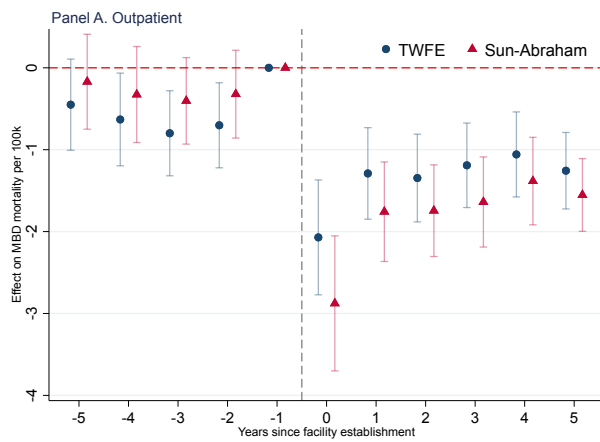
(e) Medical/Surgical Care Deaths



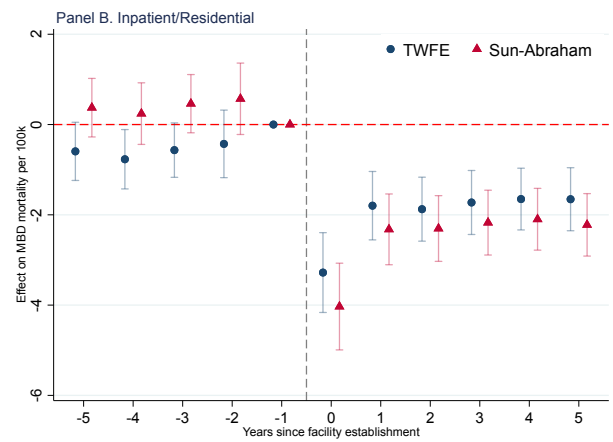
(f) Other Deaths

Notes: This figure presents event study estimates using the [Sun and Abraham \(2021\)](#) interaction-weighted estimator for component causes of mortality. Each panel shows point estimates and 95% confidence intervals for the effect of mental health treatment facility openings on cause-specific mortality rates. The horizontal axis measures years relative to the first facility opening in a county, with $t = -1$ (one year before treatment) as the omitted reference period indicated by the vertical dashed line. The outcome variables are deaths per 100,000 county residents from each specific cause. The sample is restricted to counties with a single facility opening and no pre-treatment changes in facility counts, plus never-treated counties as the control group. All specifications include county fixed effects, year fixed effects, state-by-year fixed effects, demographic and economic controls. Standard errors are clustered at the county level. The average treatment effect shown in each panel note is calculated as the mean of coefficients from $t = 0$ to $t = 5$. Panel (c) is Other external causes of accidental injury, and panel (f) is Other (skin, blood, residual codes). See Table A20, and A19 for coefficient estimates and baseline means.

Figure A5: Effect of Outpatient versus Inpatient Facility Establishment on Mortality



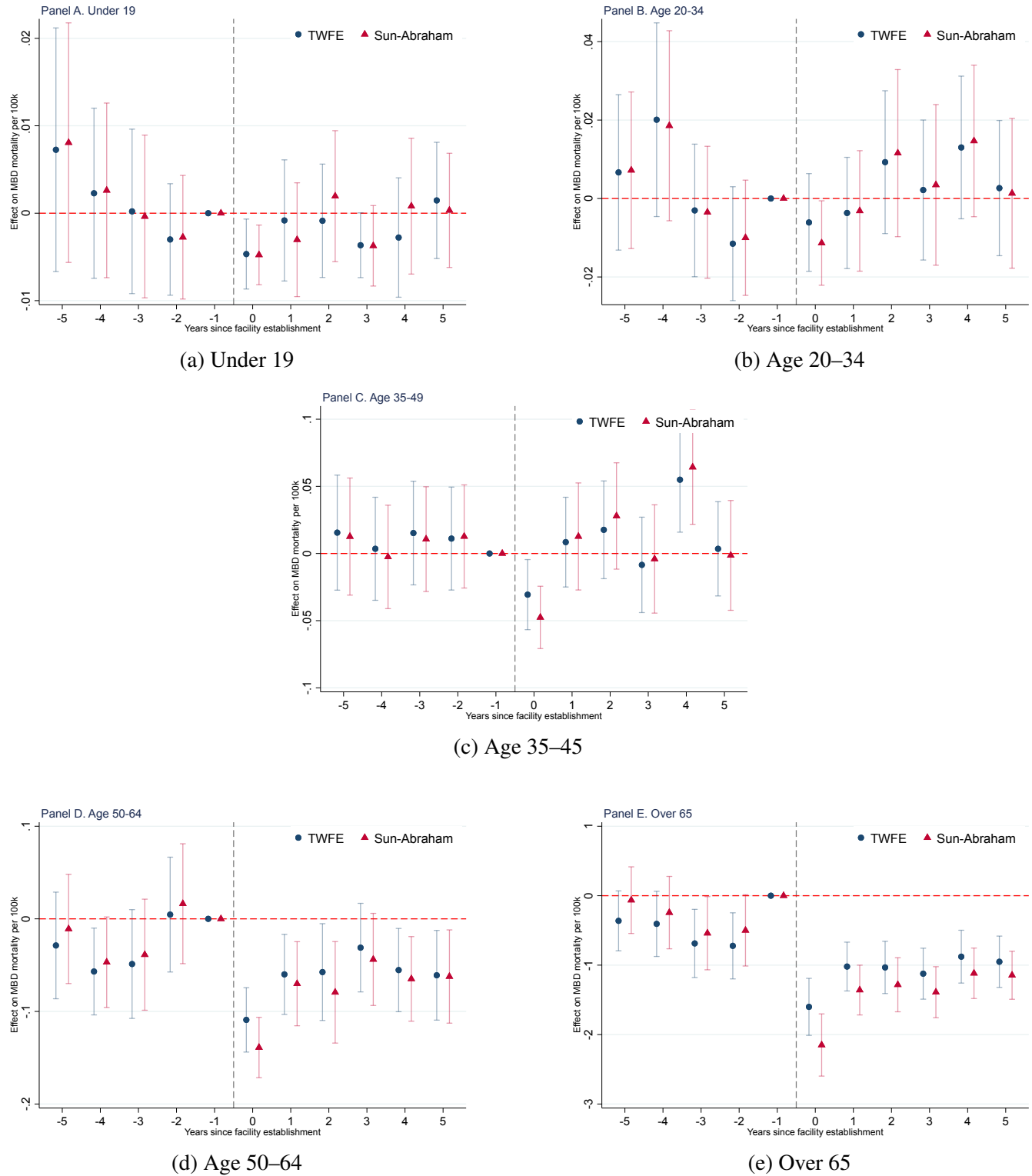
(a) Outpatient Facilities (NAICS 621420)



(b) Inpatient/Residential Facilities (NAICS 623220)

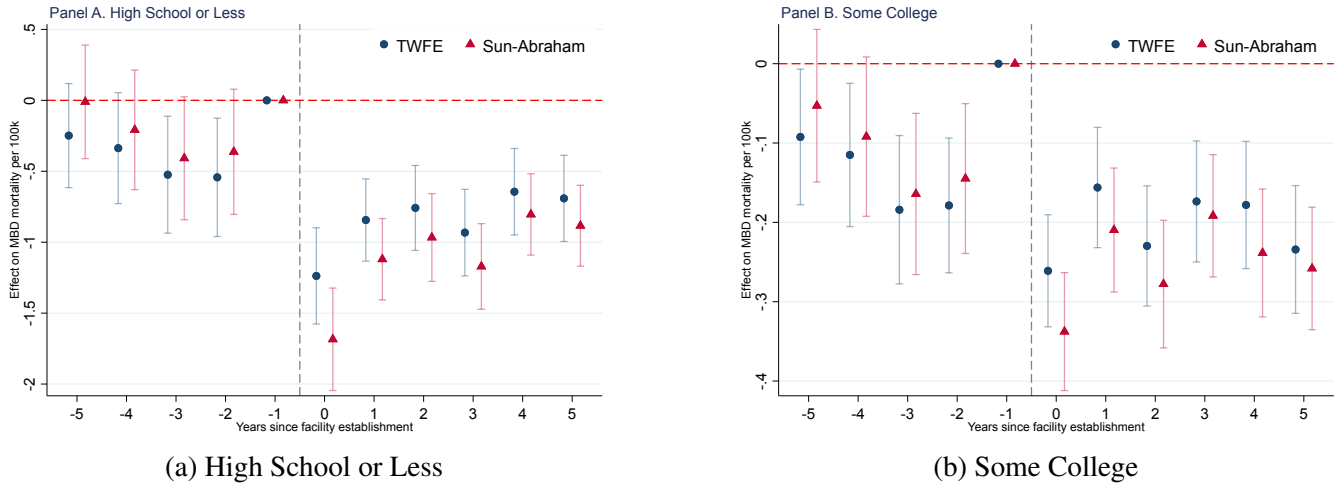
Notes: These figures plot event study coefficients from equation (3) for the effect of different facility types on MBD mortality rates. Circles represent two-way fixed effects (TWFE) estimates; triangles represent Sun and Abraham (2020) estimates. The vertical line at $t = -1$ denotes the year before facility establishment (normalized to zero). Bars represent 95 percent confidence intervals. The TWFE average represents the mean effect over periods 0–5 relative to period -1. The Sun-Abraham average represents the weighted average of cohort-specific treatment effects. Standard errors clustered at the county level in parentheses. See Table A4 for full regression results.

Figure A6: Effect of Mental Health Facility Establishment on Age-Specific Mortality



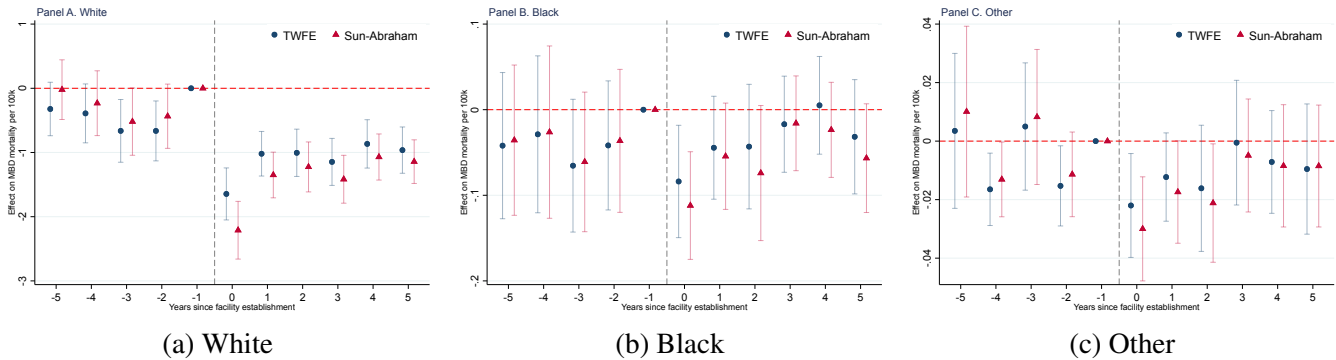
Notes: These figures plot event study coefficients from equation (3) for age-specific MBD mortality rates. Circles represent two-way fixed effects (TWFE) estimates; triangles represent Sun and Abraham (2020) estimates. The vertical line at $t = -1$ denotes the year before facility establishment (normalized to zero). Bars represent 95 percent confidence intervals. The TWFE average represents the mean effect over periods 0–5 relative to period -1. The Sun-Abraham average represents the weighted average of cohort-specific treatment effects. Standard errors clustered at the county level in parentheses. See Table A5 for full regression results.

Figure A7: Effect of Mental Health Facility Establishment on Mortality by Education Level



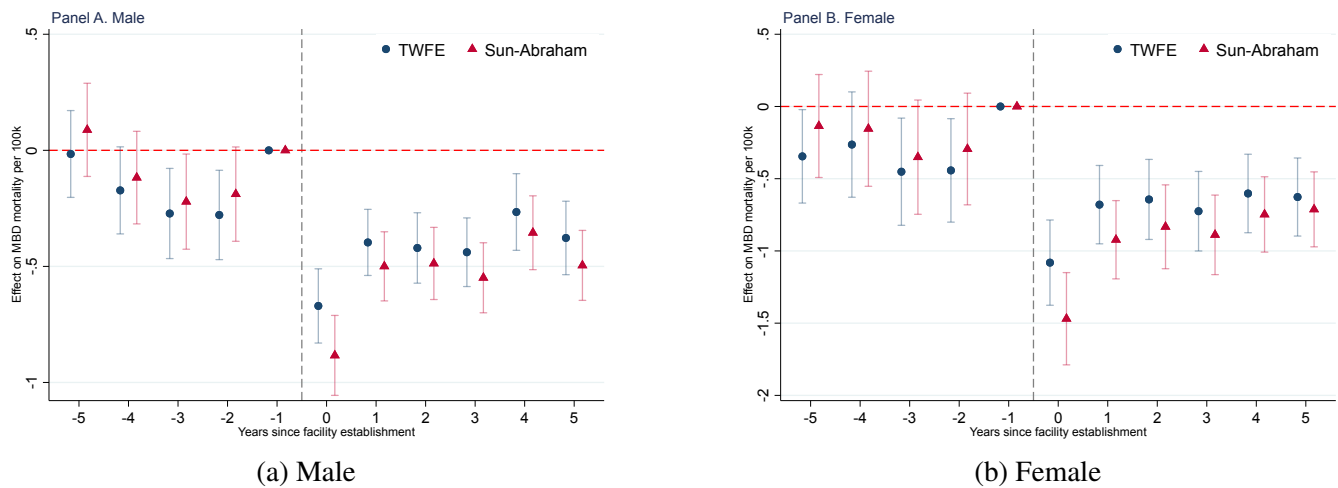
Notes: These figures plot event study coefficients from equation (3) for education-specific MBD mortality rates. Circles represent two-way fixed effects (TWFE) estimates; triangles represent Sun and Abraham (2020) estimates. The vertical line at $t = -1$ denotes the year before facility establishment (normalized to zero). Bars represent 95 percent confidence intervals. The TWFE average represents the mean effect over periods 0–5 relative to period -1. The Sun-Abraham average represents the weighted average of cohort-specific treatment effects. Standard errors clustered at the county level in parentheses. See Table A8 for full regression results.

Figure A8: Effect of Mental Health Facility Establishment on Mortality by Race



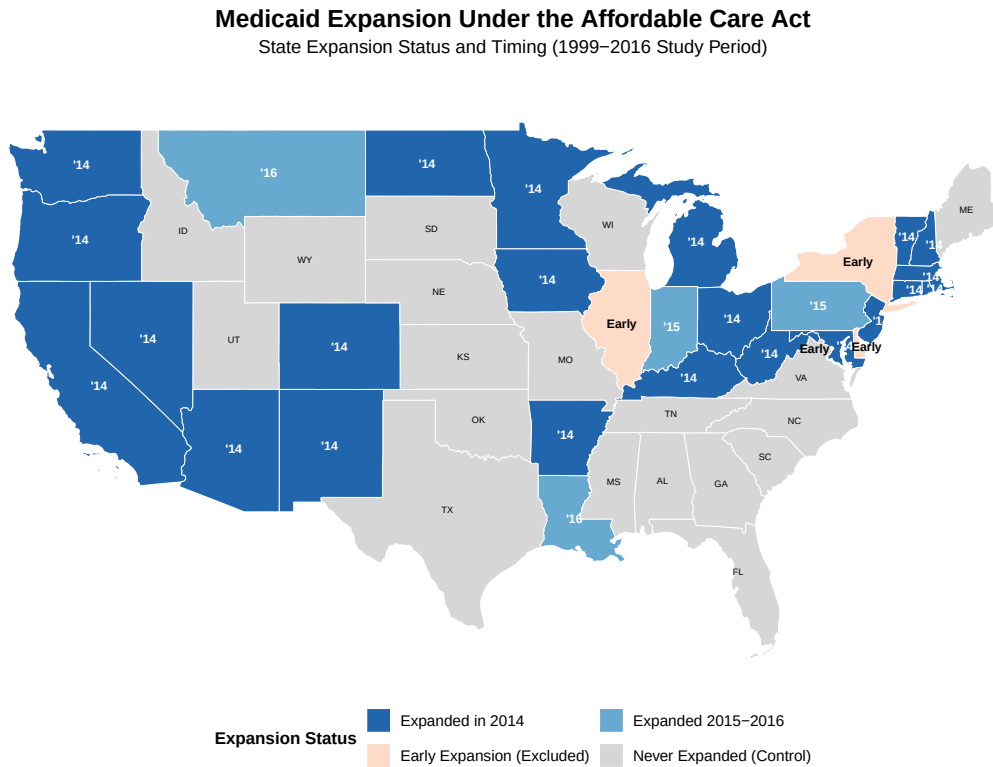
Notes: These figures plot event study coefficients from equation (3) for race-specific MBD mortality rates. Circles represent two-way fixed effects (TWFE) estimates; triangles represent Sun and Abraham (2020) estimates. The vertical line at $t = -1$ denotes the year before facility establishment (normalized to zero). Bars represent 95 percent confidence intervals. The TWFE average represents the mean effect over periods 0–5 relative to period -1. The Sun-Abraham average represents the weighted average of cohort-specific treatment effects. Standard errors clustered at the county level in parentheses. See Table A9 for full regression results.

Figure A9: Effect of Mental Health Facility Establishment on Mortality by Gender



Notes: These figures plot event study coefficients from equation (3) for gender-specific MBD mortality rates. Circles represent two-way fixed effects (TWFE) estimates; triangles represent Sun and Abraham (2020) estimates. The vertical line at $t = -1$ denotes the year before facility establishment (normalized to zero). Bars represent 95 percent confidence intervals. The TWFE average represents the mean effect over periods 0–5 relative to period -1. The Sun-Abraham average represents the weighted average of cohort-specific treatment effects. Standard errors clustered at the county level in parentheses. See Table A10 for full regression results.

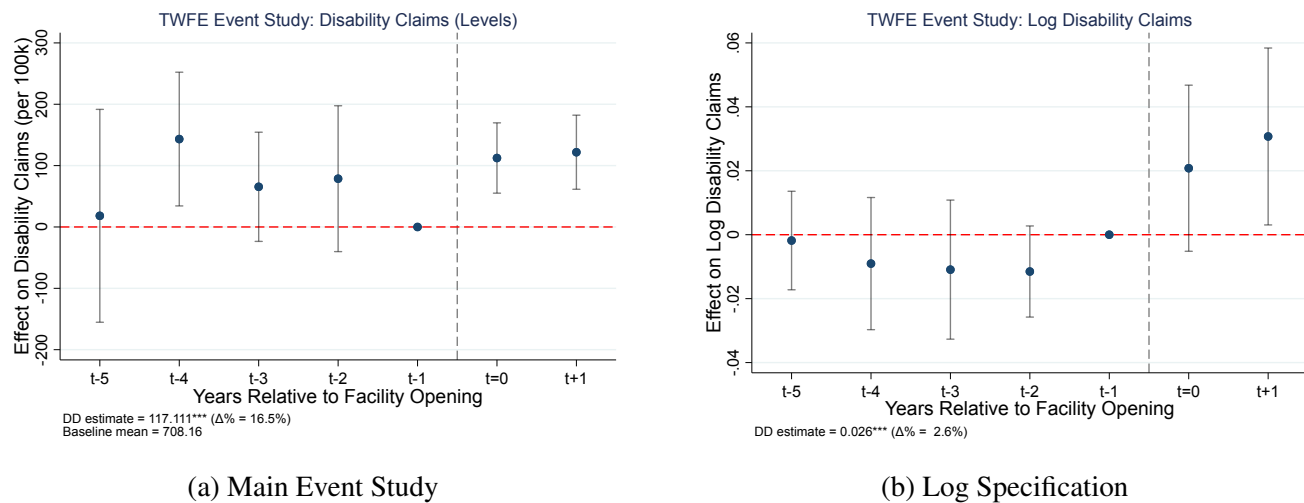
Figure A10: Medicaid Expansion Status and Timing Under the ACA



Notes: Main treatment states expanded Medicaid in January 2014. Early expansion states (CA, CT, DC, MA, MN, NJ, VT, WA) expanded before 2014 and are excluded from main analysis. Later expansion states expanded in 2015–2016. Control states never expanded during the study period. Alaska excluded from main analysis (expanded September 2015). Source: Kaiser Family Foundation, NBER, and author's analysis.

Notes: This figure illustrates the state-level timing of Medicaid expansion under the Affordable Care Act (ACA). States are categorized based on their expansion status and timing. The main treatment group consists of states that expanded in 2014. "Early Expansion" states adopted Medicaid expansion prior to 2014 under Section 1115 waivers and are excluded from the main analysis. "Control" states did not expand Medicaid during the study period. Year labels on the map indicate the year of expansion. Data on expansion timing are compiled from the Kaiser Family Foundation and the National Bureau of Economic Research (NBER).

Figure A11: Mechanism: Dynamic Effects of Mental Health Treatment Facilities on Disability Claims



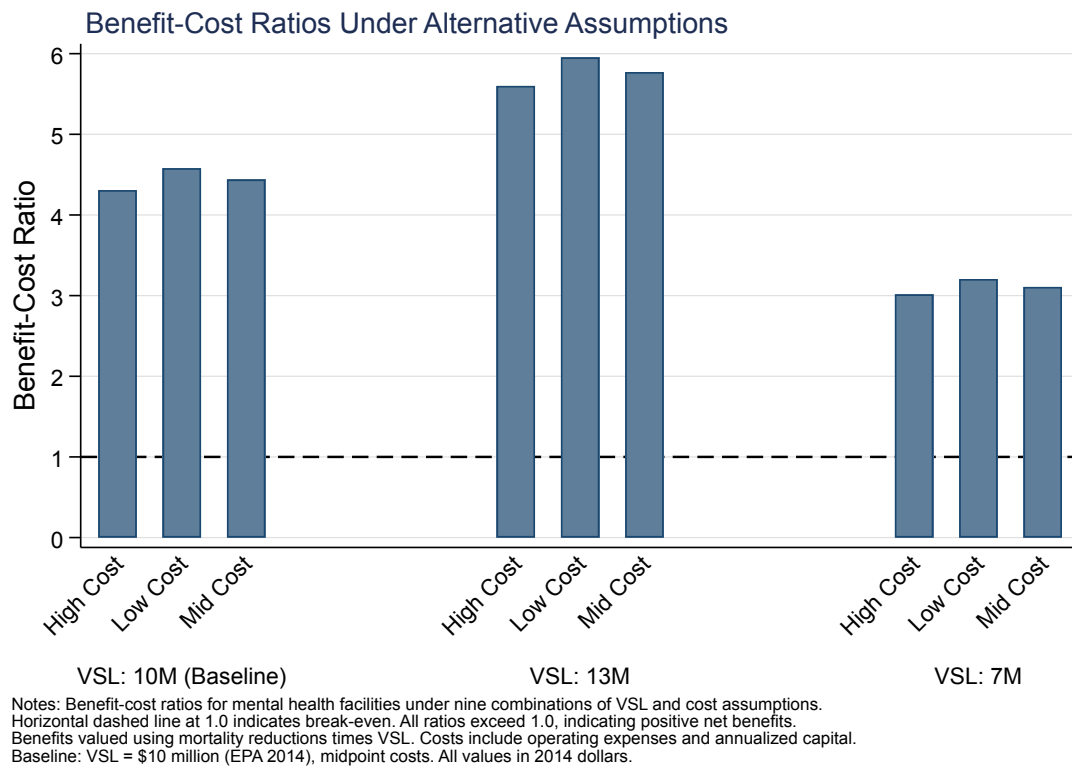
Notes: The estimates implement a two-way fixed effects (TWFE) event study estimator where the outcome variable is SSI disability recipients aged 18–64 per 100,000 county residents obtained from Social Security Administration county-level data. Panel (a) examines the effects in levels. Panel (b) displays the log specification. Both specifications include county fixed effects, year fixed effects, and state-by-year fixed effects. The observations are weighted by inverse propensity score weights based on county-level demographic characteristics. Additional controls include demographic composition (share under 19) and labor market conditions (employment-to-population ratio). The standard errors are clustered at the county level. The data combines County Business Patterns (CBP) for mental health facility openings with SSA disability data for the period 2009–2016. The figures display coefficient estimates and 95% confidence intervals, with $t = -1$ as the omitted reference period. The vertical dashed line indicates the treatment period ($t = 0$). The sample is restricted to counties with a single facility opening and no pre-treatment changes in facility counts, plus never-treated counties as the control group. Negative coefficients indicate that mental health facilities reduce disability claims. The time window spans five pre-treatment periods ($t = -5$ to $t = -1$) and two post-treatment periods ($t = 0$ to $t = 1$), constrained by the availability of SSA disability data through 2016.

Figure A12: Mechanism: Dynamic Effects of Mental Health Facilities and Medicaid Expansion on Antidepressant Prescriptions



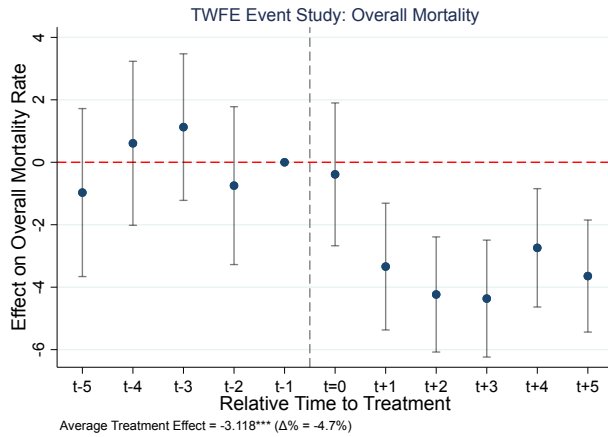
Notes: The estimates implement a two-way fixed effects (TWFE) interaction event study estimator where the outcome variable is antidepressant claims per 100,000 population obtained from Medicare Part D prescriber data. The interaction specification compares the effect of mental health facilities in Medicaid expansion states versus non-expansion states, before and after expansion. Panel (a) examines the effects in levels. Panel (b) displays the log specification. Both specifications include state fixed effects and year fixed effects. Additional controls include demographic composition (share under 19) and labor market conditions (employment-to-population ratio). The standard errors are clustered at the state level. The data combines state-level aggregates of County Business Patterns (CBP) for mental health facilities with Medicare Part D geographic prescriber data for the period 2013–2016. The figures display interaction coefficient estimates and 95% confidence intervals, with $t = -1$ (2013) as the omitted reference period. The vertical dashed line indicates $t = 0$ (2014), when Medicaid expansion took effect in participating states. Medicaid expansion states are those that adopted expansion between 2014 and 2016. The antidepressant category includes commonly prescribed SSRIs, SNRIs, and tricyclic antidepressants (sertraline, escitalopram, trazodone, bupropion, fluoxetine, citalopram, duloxetine, venlafaxine, amitriptyline). The limited pre-period reflects data availability constraints; 2013 is the first year of Medicare Part D geographic prescriber data. The event study captures the dynamic complementarity between mental health facility access and insurance coverage expansion. Positive coefficients indicate that facility effects on prescription utilization are larger in expansion states post-expansion, consistent with increased demand for mental health services when insurance barriers are reduced.

Figure A13: Benefit-Cost Ratios Under Alternative Assumptions

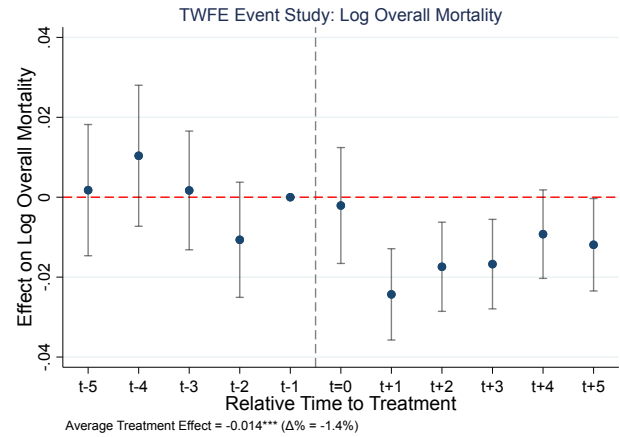


Notes: This figure displays benefit-cost ratios for mental health facilities under nine combinations of Value of Statistical Life (VSL) and cost assumptions. The horizontal dashed line at 1.0 indicates break-even. All ratios exceed 1.0, indicating positive net benefits across all scenarios. Benefits are valued using mortality reductions times VSL. Costs include operating expenses and annualized capital expenditures. The baseline scenario uses VSL = \$10 million (EPA 2014 estimate) and midpoint facility costs. All monetary values are in 2014 dollars.

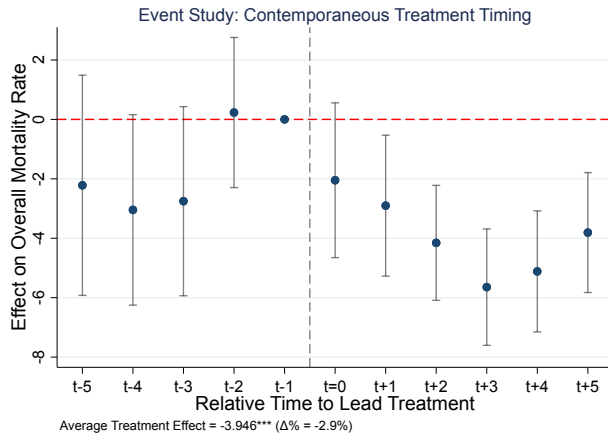
Figure A14: Robustness: Two-Way Fixed Effects Event Study Estimates for Overall Mortality



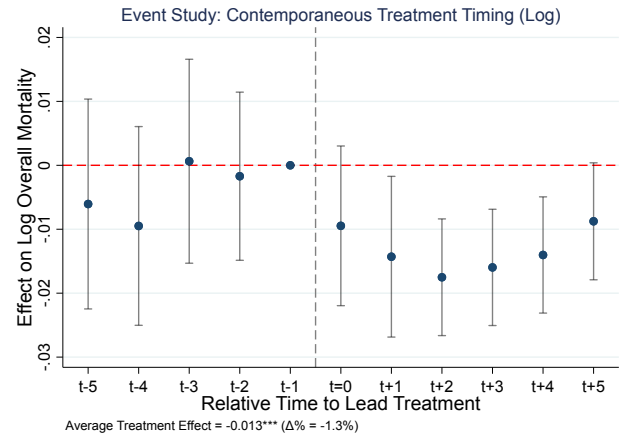
(a) Main Specification



(b) Log Specification



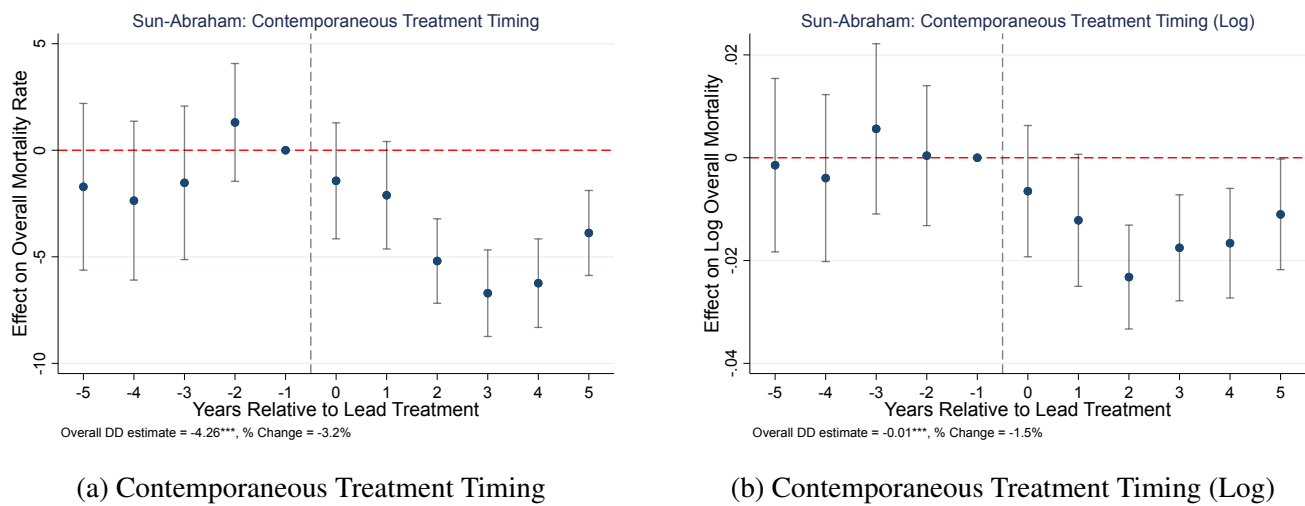
(c) Contemporaneous Treatment Timing



(d) Contemporaneous Treatment Timing (Log)

Notes: This figure presents two-way fixed effects (TWFE) event study estimates as a robustness check for the Sun-Abraham specifications shown in Figure 4. Panel (a) shows the main specification using the mortality rate from all causes per 100,000 county residents. Panel (b) presents the log specification. Panels (c) and (d) use contemporaneous treatment timing (lead specification) to test for anticipatory effects, where relative time is defined using the lead of facility openings. Points represent coefficient estimates for each period relative to $t = -1$ (the omitted reference category). All specifications include county fixed effects, year fixed effects, state-by-year fixed effects, demographic controls (share of population under 19), labor market controls (employment-to-population ratio), and additional county-level covariates including earnings per capita, transfer payments per capita, hospital availability, population density, rural status indicators, and age distribution. The sample is restricted to counties with a single facility opening and no pre-treatment changes in facility counts, plus never-treated counties as controls. Inverse probability weights are applied based on 1999 demographic characteristics. Standard errors are clustered at the county level. Data combine County Business Patterns and Vital Statistics mortality data for 1999–2016. The vertical dashed line indicates the treatment period.

Figure A15: Sun-Abraham Event Study: Contemporaneous Treatment Timing for Overall Mortality

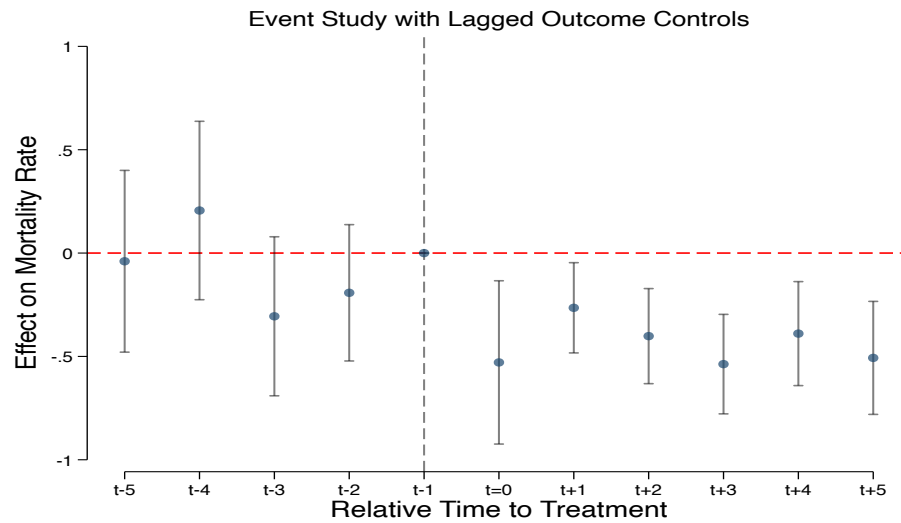


(a) Contemporaneous Treatment Timing

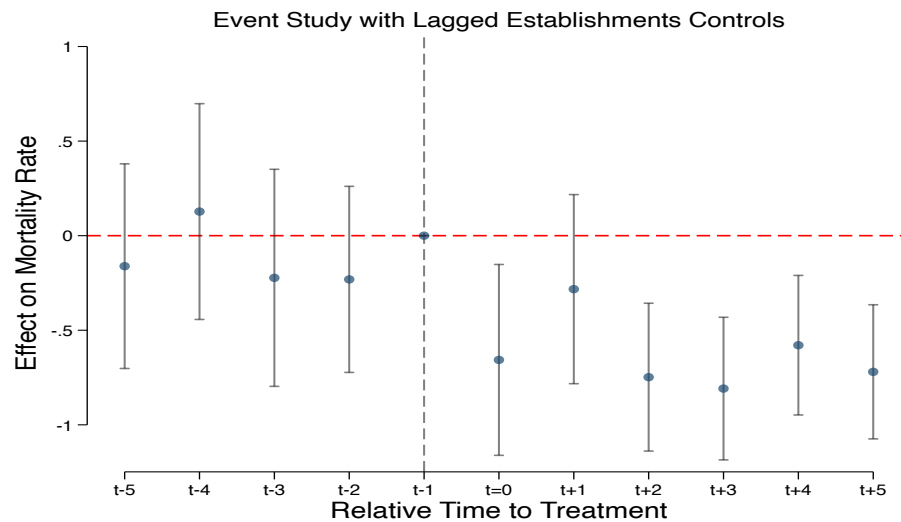
(b) Contemporaneous Treatment Timing (Log)

Notes: This figure presents [Sun and Abraham \(2021\)](#) event study estimates using contemporaneous treatment timing as an alternative specification to address potential anticipatory effects. The relative time is defined using the lead of facility openings, testing whether effects occur at the time of treatment rather than in anticipation. Panel (a) uses the mortality rate from all causes per 100,000 county residents. Panel (b) presents the log specification. The sample, controls, and estimation approach are identical to Figure 4, but the timing is shifted forward by one period. This specification helps distinguish between effects that occur contemporaneously with facility openings versus those that may reflect anticipatory behavioral changes. Points represent interaction-weighted estimates and 95% confidence intervals for each period relative to $t = -1$, with the vertical dashed line indicating the treatment period. Standard errors are clustered at the county level. Data: 1999–2016.

Figure A16: Robustness Checks: Event Studies with Alternative Controls



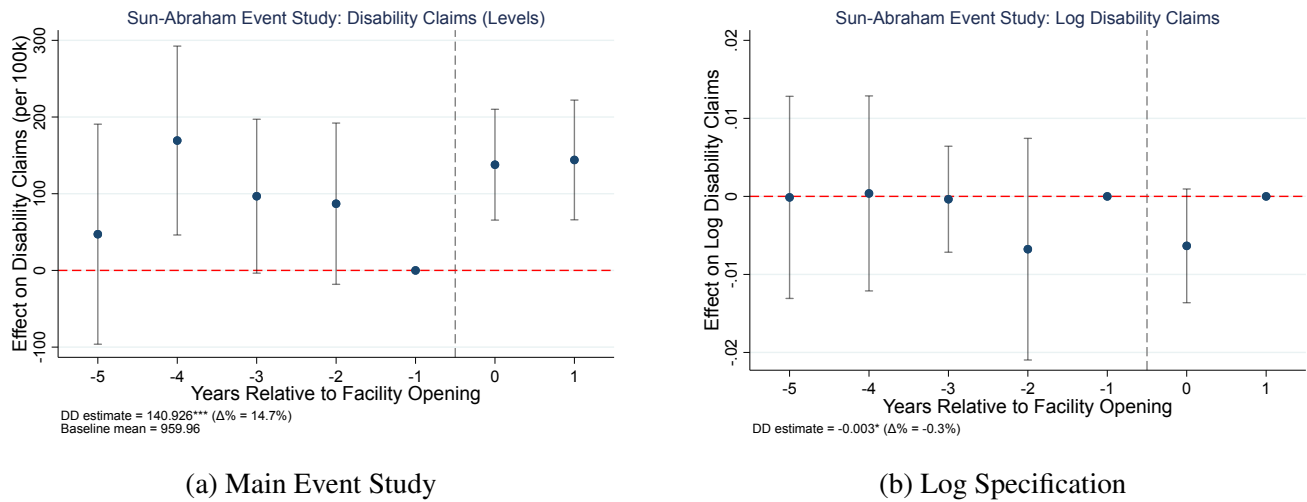
(a) Event Study with Lagged Outcome Controls



(b) Event Study with Lagged Establishments Controls

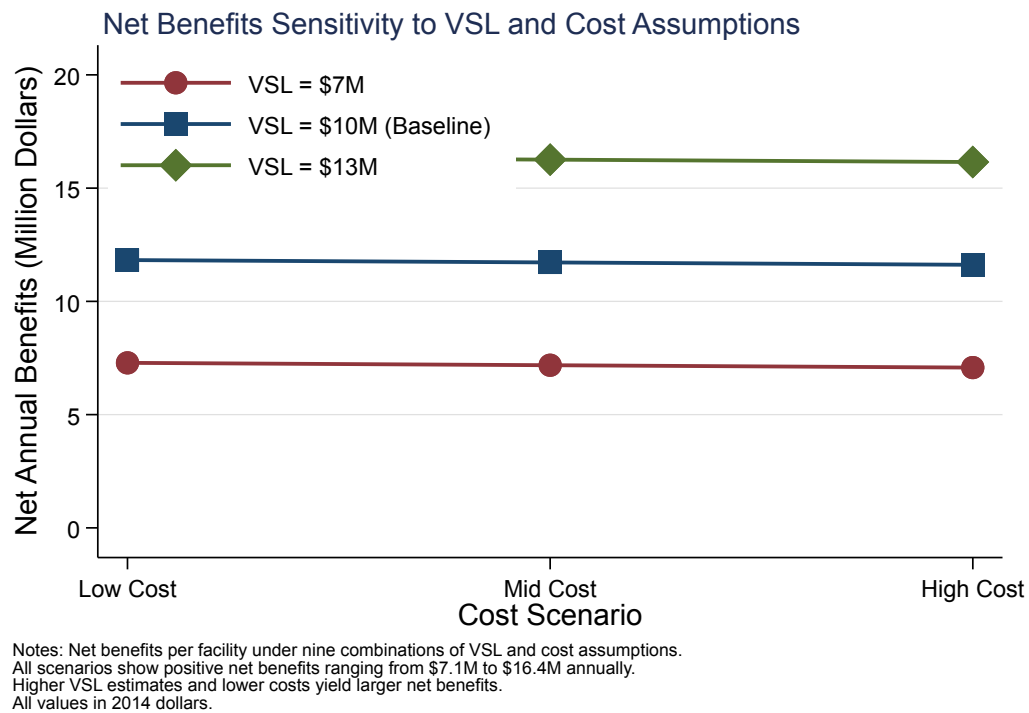
Notes: These figures present robustness checks for our main event study analysis. Panel (a) augments our baseline specification by including two lags of the outcome variable (mortality rate) as additional controls. Panel (b) includes two lags of mental health establishments as controls. Both specifications maintain the core controls from our main analysis: county and year fixed-effects (Panel a) or county, year, and state-by-year fixed effects (Panel b), demographic characteristics, and economic conditions. All observations are weighted by 1999 county-level demographic characteristics, with standard errors clustered at the county level. The data combines County Business Patterns (CBP) and Vital Statistics mortality data for the period 1999-2016. The vertical dashed lines indicate the timing of treatment, while horizontal lines at zero provide reference points. Point estimates and 95% confidence intervals are displayed. These robustness checks help address potential concerns about pre-existing trends in either mortality rates or mental health establishment counts affecting our main results.

Figure A17: Dynamic Effects of Mental Health Treatment Facilities on Disability Claims: Sun-Abraham Estimates



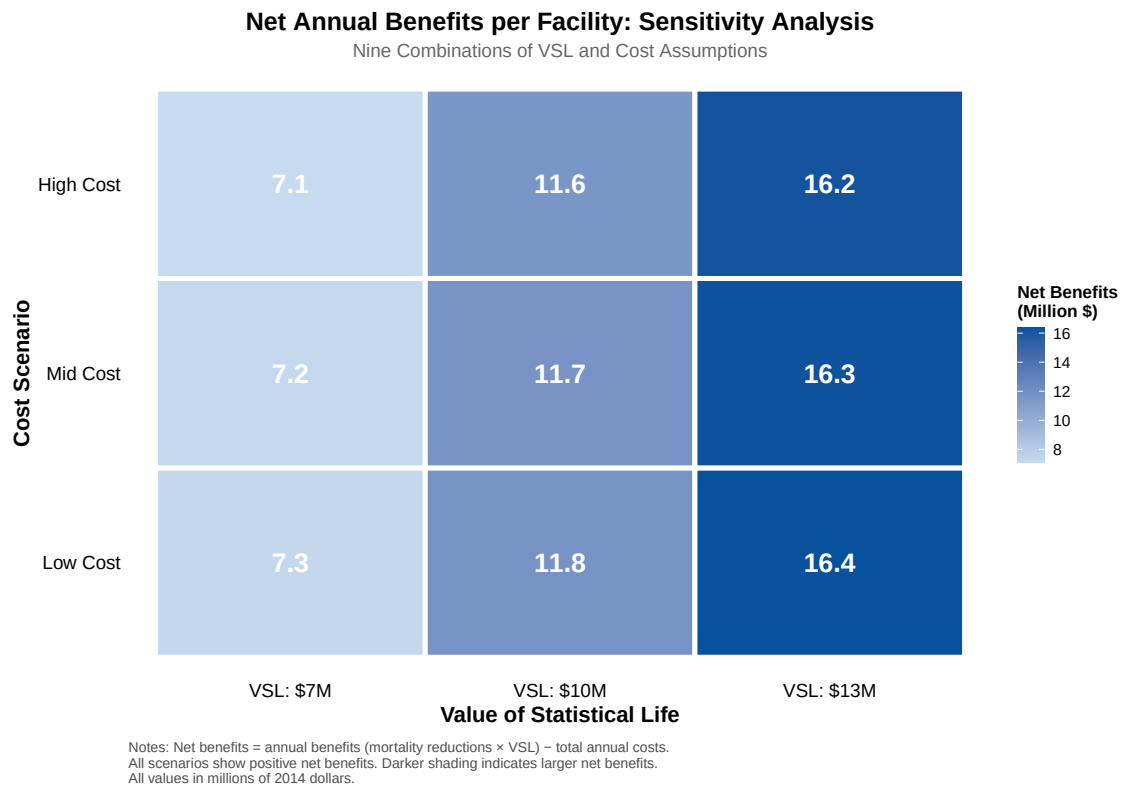
Notes: The estimates implement the [Sun and Abraham \(2021\)](#) heterogeneity-robust event study estimator where the outcome variable is SSI disability recipients aged 18–64 per 100,000 county residents obtained from Social Security Administration county-level data. Panel (a) examines the effects in levels. Panel (b) displays the log specification. Both specifications include county fixed effects, year fixed effects, and state-by-year fixed effects. The observations are weighted by inverse propensity score weights based on county-level demographic characteristics. Additional controls include demographic composition (share under 19) and labor market conditions (employment-to-population ratio). The standard errors are clustered at the county level. The data combines County Business Patterns (CBP) for mental health facility openings with SSA disability data for the period 2009–2016. The figures display interaction-weighted estimates and 95% confidence intervals, with $t = -1$ as the omitted reference period. The vertical dashed line indicates the treatment period ($t = 0$). The sample is restricted to counties with a single facility opening and no pre-treatment changes in facility counts, plus never-treated counties as the control group. These estimates account for potential heterogeneity in treatment effects across cohorts, addressing limitations of traditional two-way fixed effects models. The time window spans five pre-treatment periods ($t = -5$ to $t = -1$) and two post-treatment periods ($t = 0$ to $t = 1$), constrained by the availability of SSA disability data through 2016. In Panel (b), the coefficient for $t = 1$ is not estimated due to insufficient observations with positive disability rates after log transformation, reflecting the sparse nature of disability claims data at the county-year level.

Figure A18: Sensitivity of Net Benefits to VSL and Cost Assumptions



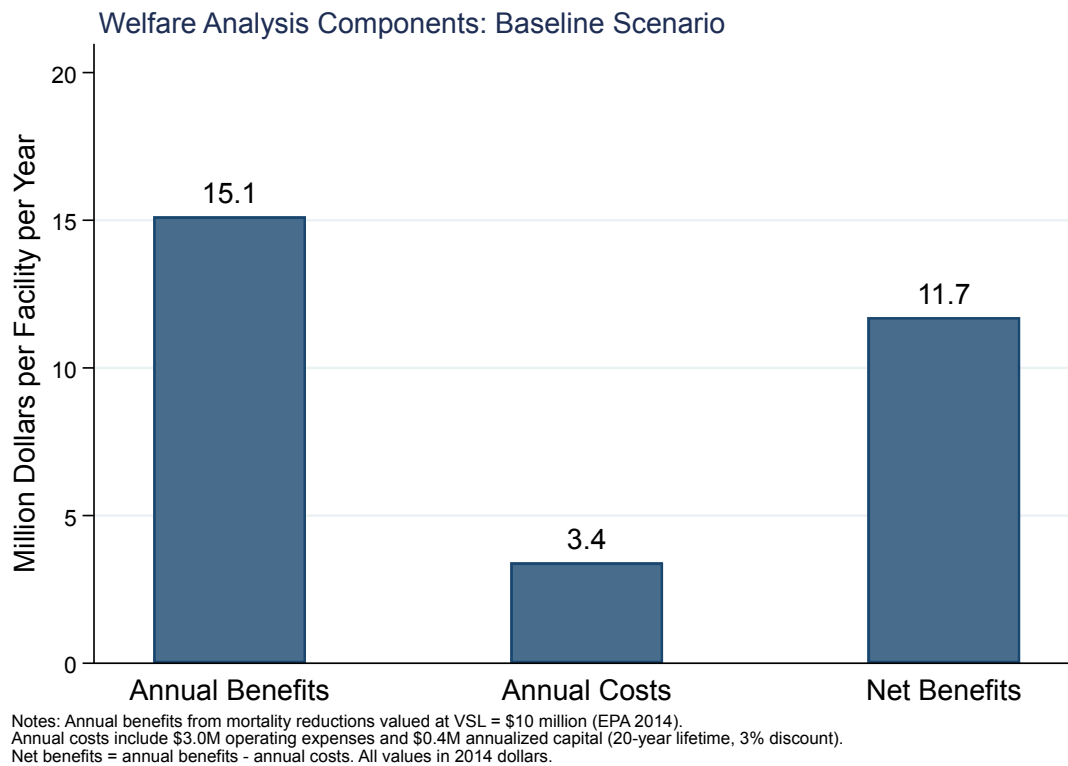
Notes: This figure displays net annual benefits per facility across nine combinations of Value of Statistical Life (VSL) and cost assumptions. Each line represents a different VSL level: \$7 million (conservative), \$10 million (baseline, EPA 2014), and \$13 million (upper bound). The x-axis shows three cost scenarios: low (\$3.30M per year), mid (\$3.41M per year), and high (\$3.51M per year), reflecting operating costs plus annualized capital expenditures. All nine scenarios yield positive net benefits ranging from \$7.1 million to \$16.4 million annually, demonstrating robustness to alternative assumptions. Results are more sensitive to VSL variation than to cost variation. All monetary values are in 2014 dollars.

Figure A19: Net Benefits Heatmap: VSL and Cost Combinations



Notes: This heatmap visualizes net annual benefits per facility under nine combinations of VSL and cost assumptions. Darker shading indicates larger net benefits. All scenarios show positive net benefits, demonstrating robustness to alternative assumptions. Values range from \$7.1 million (VSL = \$7M, high cost) to \$16.4 million (VSL = \$13M, low cost) annually. All monetary values are in 2014 dollars.

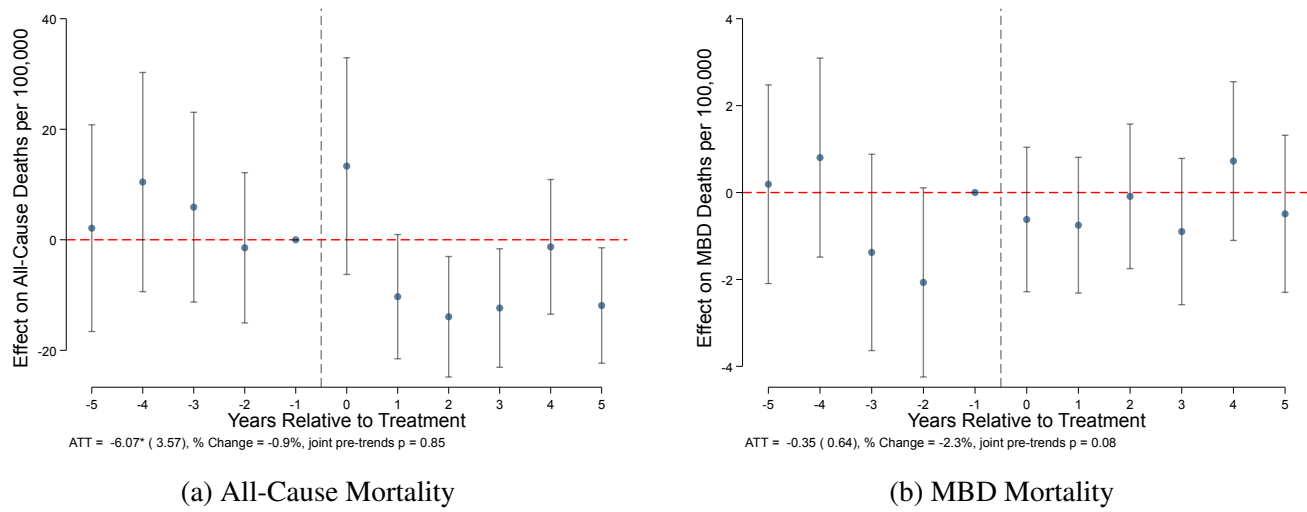
Figure A20: Welfare Analysis Components: Baseline Scenario



Notes: This figure decomposes the welfare analysis into its constituent components under baseline assumptions. Annual benefits (\$15.1 million) represent mortality reductions valued at VSL = \$10 million (EPA 2014 estimate). Annual costs (\$3.4 million) include \$3.0 million in operating expenses and \$0.4 million in annualized capital costs, based on construction costs of \$4.5-7.6 million amortized over a 20-year facility lifetime at a 3 percent discount rate (Montana Legislative HJR 16 Study, May 2014). Net benefits equal annual benefits minus annual costs, yielding \$11.7 million per facility per year. The figure illustrates that mortality-related benefits alone substantially exceed facility costs, with a benefit-cost ratio of 4.44. All monetary values are in 2014 dollars.

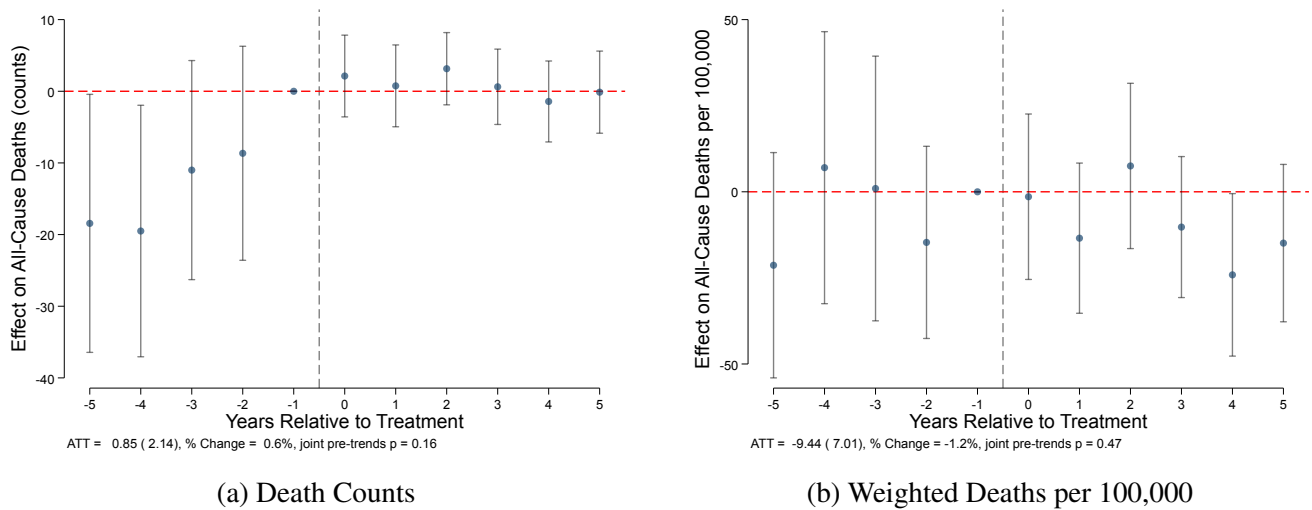
Additional Event-Study Evidence for the Revised Analysis

Figure A21: Headline Event Studies Without Controls



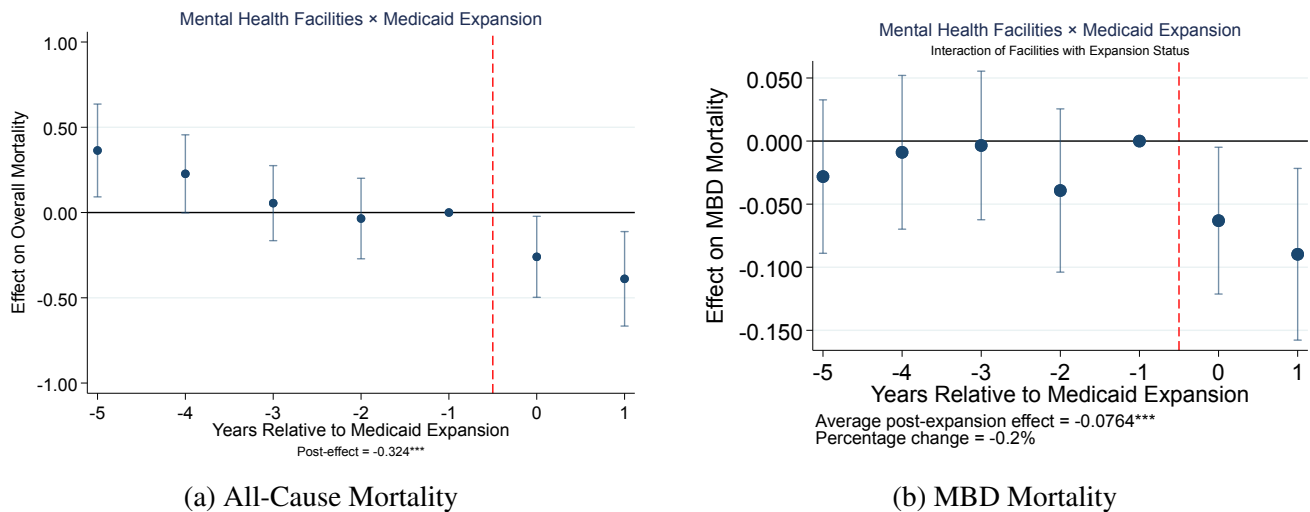
Notes: Sun and Abraham (2021) event studies of a first facility opening on population-weighted deaths per 100,000 with no covariates, county and state-by-year fixed effects, county-clustered standard errors, and 95% confidence intervals. These specifications complement the parsimonious-control versions in the main text; the all-cause estimate is similar in magnitude with the cleanest pre-period balance of any specification (joint pre-trends $p = 0.85$). Each panel's note reports the post-period ATT, percent change, and joint pre-trends p -value.

Figure A22: Closings Event Studies: All-Cause Mortality



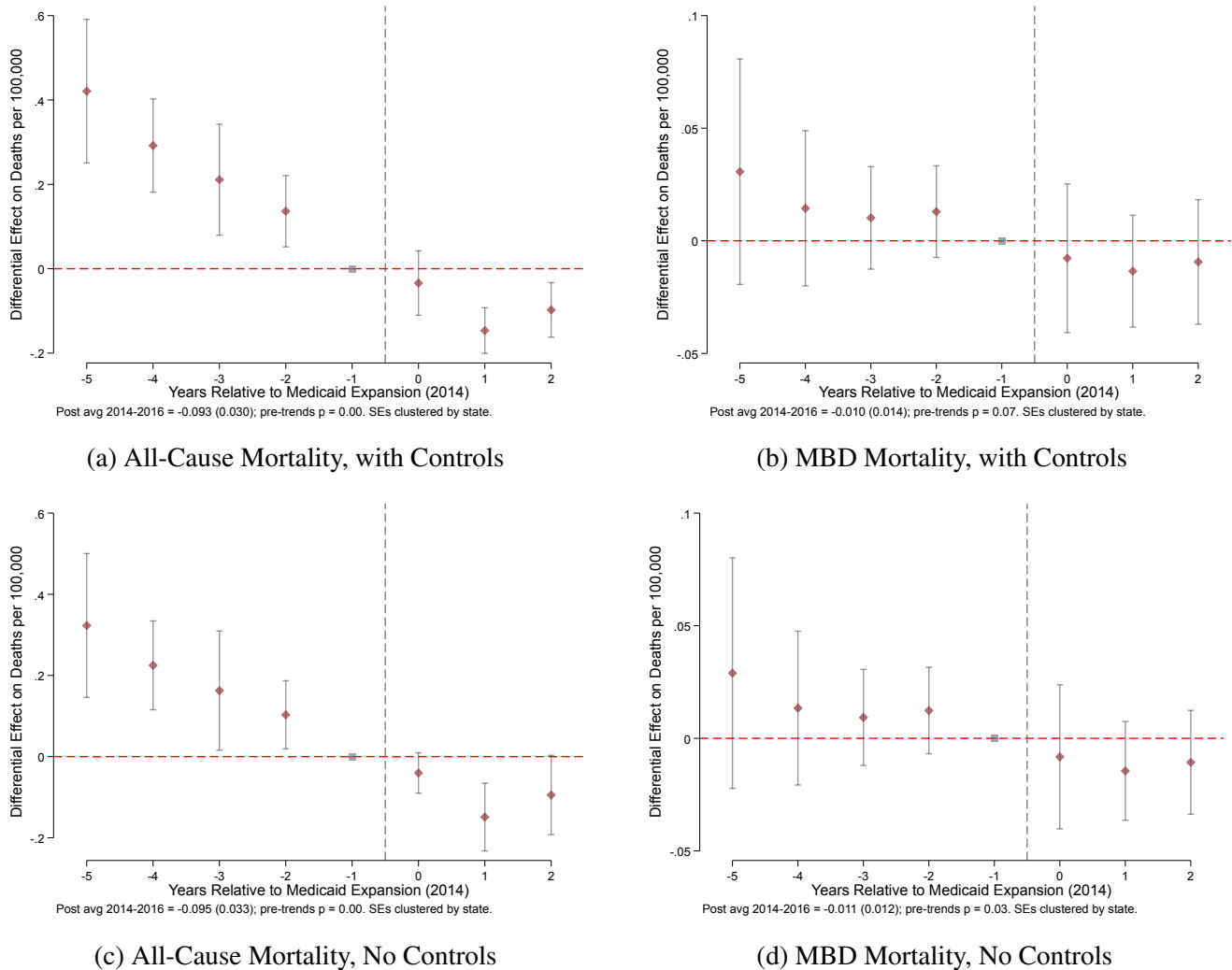
Notes: Mirror of the openings event-study design for facility closings: the sample includes counties experiencing exactly one facility decrease and no increases, plus never-changing controls, with the closing year as the event, county and state-by-year fixed effects, parsimonious controls, and county-clustered standard errors. The estimates are uninformative in both units (counts: ATT 0.85, $p = 0.69$; weighted per 100,000: ATT -9.44, $p = 0.18$), reflecting the scarcity of clean single-closure counties; the confidence intervals admit both substantial harms from closings and zero. These figures document the state of the evidence transparently rather than supporting a causal claim about closures.

Figure A23: Dynamic Facility-Expansion Interactions: Original Design



Notes: This figure reproduces the dynamic exhibit of earlier drafts of this paper exactly. Each panel plots coefficients on the interaction of facility counts with expansion-state event-time dummies, where event time is defined relative to each state's own Medicaid expansion year and $t = -1$ is the omitted reference period; the estimation window is 2009–2015 (–5 to +1) and the regressions include the baseline facility effect. Outcomes are annual death counts. Panel (a) includes county, year, and state-by-year fixed effects with demographic and economic controls and displays 95% confidence intervals; Panel (b) includes county, year, and state-by-year fixed effects without additional controls and displays 90% confidence intervals, as in the original exhibit. Standard errors are clustered by county. As reported in the panel notes, the post-expansion differential averages –0.324 all-cause and –0.076 MBD deaths per facility over the expansion year and the year after. The population-weighted per-100,000 variant with state-clustered standard errors is reported in Figure A24. Data: County Business Patterns and Vital Statistics, 2009–2015.

Figure A24: Dynamic Facility-Expansion Interactions Around Medicaid Expansion



Notes: Event studies of the differential effect of lagged facility counts in Medicaid expansion versus non-expansion states by year relative to 2014, on population-weighted deaths per 100,000 over 2009–2016, with county and state-by-year fixed effects and standard errors clustered by state; 2013 is the omitted reference year (gray marker). Each panel’s note reports the 2014–2016 post-expansion average and the joint pre-trends p-value. For all-cause mortality the post-expansion differential is negative and significant (-0.093 , SE 0.030 with controls) but the joint test rejects parallel pre-trends ($p < 0.01$), driven by secular drift in the facility-mortality gradient; for MBD mortality pre-trends are marginal ($p = 0.07$) and the post differential is insignificant. As discussed in the main text, these dynamic estimates are treated as suggestive, and the complementarity conclusion rests on the levels specifications.

F Appendix Tables

Table A1: Analysis Samples Across Exhibits

Analysis (exhibit)	Counties	County-years	Sample definition
Summary statistics (Table 1)	3,199	57,440	Full panel: counties with CBP and Vital Statistics data, 1999–2016
Dose-response panel (Table 3)	2,972	53,498 ^a	Lagged facilities and non-missing outcome and controls
Openings and closings (Table 6)	2,974	53,517 ^b	Year-over-year facility changes defined
Facility-type panel (Table A4)	2,973	53,502 ^c	Lagged facility-type counts non-missing
Event studies, weighted rates (Figures 4b and 5b)	–	26,395	Single facility increase, no prior changes, plus never-treated controls
Event studies, counts (Figure 4a; Table 5)	1,319	23,742	Event sample with non-suppressed cause-specific death counts
ACA interactions (Tables 7 and 8)	2,439	43,886	Excludes early-expansion states; SAHIE uninsurance data required
Disability panel (online appendix)	3,199	25,592	SSA data availability, 2009–2016

Notes: This table records the estimation sample used by each set of exhibits and the restriction that generates it; all samples cover 1999–2016. ^a 53,499 in the columns with demographic and economic controls. ^b 53,499 in the columns with demographic and economic controls. ^c 53,484 in the column with economic controls.

Table A2: Baseline Model - Main Effects of Access to Mental Health Treatment Facilities on Mortality Related to Mental Health and Behavioral Disorders

	All MBD Deaths (1)	F00-F09 (2)	F10-F19 (3)	F20-F29 (4)	F30-F39 (5)	F40-F48 (6)	F50-F59 (7)	F60-F69 (8)	F70-F79 (9)
Number of Facilities	-0.06647** (0.02928) [0.21]	-0.06146** (0.02843) [0.22]	-0.00254 (0.00368) [0.09]	-0.00059 (0.00059) [0.26]	-0.00025 (0.00079) [0.07]	-0.00052** (0.00026) [0.76]	-0.00059 (0.00043) [0.46]	0.00001 (0.00003) [0.48]	-0.00052 (0.00056) [0.26]
County FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.509	0.523	0.100	0.028	0.028	0.013	0.004	0.003	0.088
N	53,498	53,498	53,498	53,498	53,498	53,498	53,498	53,498	53,498

Notes: This table reports the effects of mental health treatment facilities on mortality rates from mental and behavioral disorders (MBD). Each coefficient represents a separate regression following Equation 1. The dependent variable is deaths per 100,000 county residents. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* p<.10 ** p<.05 *** p<.01). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. All specifications include county and year fixed effects to control for time-invariant county characteristics and common temporal shocks. The sample covers 1998-2016 with 53,498 county-year observations. Column headers denote the dependent variable. Dependent variables are coded using ICD-10 categories: F00-F09 (organic mental disorders), F10-F19 (substance use disorders), F20-F29 (schizophrenia and psychotic disorders), F30-F39 (mood disorders), F40-F48 (anxiety disorders), F50-F59 (behavioral syndromes), F60-F69 (personality disorders), and F70-F79 (intellectual disabilities). Column 1 reports effects on total MBD mortality, while columns 2-9 show effects for specific diagnostic categories. Mortality rates and facility counts are measured at the county-year level. See Section 4 for more details on the construction of the variables and data sources. The sample is restricted to 1999-2016.

Table A3: Main Effects - Full Specification

	(1)	(2)	(3)	(4)
Number of Facilities	-0.06647** (0.02928) [-0.21]	-0.06484*** (0.02353) [-0.21]	-0.07897*** (0.02969) [-0.25]	-0.07898*** (0.02974) [-0.25]
Fixed Effects				
County	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
State-by-Year	No	Yes	Yes	Yes
Controls				
Demographic	No	No	Yes	Yes
Economic	No	No	No	Yes
Adj. R ²	0.509	0.529	0.529	0.529
N	53,498	53,498	53,498	53,480

Notes: This table reports the effects of mental health treatment facilities on mortality rates from mental and behavioral disorders (MBD). Each column represents a separate estimation of Equation 2, where the dependent variable is deaths per 100,000 county residents. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* p<.10 ** p<.05 *** p<.01). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. All specifications include county and year fixed effects. Column (1) presents the baseline specification. Column (2) adds state-by-year fixed effects to account for state-specific temporal trends. Column (3) introduces demographic controls. Column (4) adds economic controls. See Section 4 for more details on the construction of the variables and data sources. The sample is restricted to 1999-2016.

Table A4: Heterogeneity by Facility Type: Effects on Mental Health Outcomes

	(1)	(2)	(3)	(4)
<i>Panel A: Outpatient mental health facilities</i>				
Number of facilities	-0.081*	-0.108***	-0.126***	-0.126***
	(0.048)	(0.036)	(0.043)	(0.043)
	[-0.3%]	[-0.3%]	[-0.4%]	[-0.4%]
<i>Panel B: Residential mental health facilities</i>				
Number of facilities	-0.070	-0.060	-0.090*	-0.090*
	(0.049)	(0.037)	(0.051)	(0.051)
	[-0.2%]	[-0.2%]	[-0.3%]	[-0.3%]
Fixed Effects				
County	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
State-by-Year	No	Yes	Yes	Yes
Controls				
Demographic	No	No	Yes	Yes
Economic	No	No	No	Yes
Counties	2,973	2,973	2,973	2,972
Observations	56,471	56,471	56,471	56,452
Mean of dep. var.	31.561	31.561	31.561	31.561

Notes: This table reports the differential effects of outpatient and residential mental health treatment facilities on mortality rates from mental and behavioral disorders (MBD). Panel A presents estimates for outpatient mental health and substance abuse centers (NAICS 621420), while Panel B presents estimates for residential mental health and substance abuse facilities (NAICS 623220). Each column represents a separate estimation, where the dependent variable is deaths per 100,000 county residents. The main explanatory variables are the lagged number of each facility type in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable (calculated as coefficient/pre-treatment mean $\times$ 100). All specifications include county and year fixed effects. Column (1) presents the baseline specification. Column (2) adds state-by-year fixed effects to account for state-specific temporal trends. Column (3) introduces demographic controls. Column (4) adds economic controls. The sample is restricted to 1999-2016.

Table A5: Effects of Treatment Facilities by Age Groups

	All Ages (1)	Under 19 (2)	20-34 (3)	35-49 (4)	50-64 (5)	Over 65 (6)
Number of Facilities	-0.07898*** (0.02974) [-0.26]	0.00001 (0.00019) [0.02]	-0.00154** (0.00069) [-0.74]	-0.00101 (0.00174) [-0.12]	-0.00112 (0.00264) [-0.07]	-0.07524*** (0.02837) [-0.27]
Fixed Effects						
County	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes
State-by-Year	Yes	Yes	Yes	Yes	Yes	Yes
Controls						
Demographic	Yes	Yes	Yes	Yes	Yes	Yes
Economic	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R ²	0.536	0.536	0.536	0.536	0.536	0.536
N	53,480	53,480	53,480	53,480	53,480	53,480

Notes: This table reports the effects of mental health treatment facilities on mortality rates from mental and behavioral disorders (MBD) by age groups. Each coefficient represents a separate estimation of Equation 2, where the dependent variable is deaths per 100,000 county residents within each age group. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* p<.10 ** p<.05 *** p<.01). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. All specifications include county, year, and state-by-year fixed effects, as well as demographic and economic controls. Column (1) presents estimates for the entire population, while columns (2)-(6) show estimates for specific age groups: under 19, 20-34, 35-49, 50-64, and over 65 years old, respectively. See Section 4 for more details on the construction of the variables and data sources. The sample is restricted to 1999-2016.

Table A6: Specification Grid: Event-Study ATTs by Outcome, Units, and Fixed Effects

Outcome	Death counts		Weighted deaths per 100,000	
	Year FE	State-by-Year FE	Year FE	State-by-Year FE
All causes	−3.442*** (0.689) [0.29]	−2.979*** (0.712) [0.30]	−7.314** (3.244) [0.23]	−6.562** (3.299) [0.14]
Mental health related	−2.222*** (0.225) [0.45]	−1.777*** (0.213) [0.29]	−2.248*** (0.835) [0.51]	−0.604 (0.813) [0.11]
Placebo causes	−0.012 (0.062) [0.96]	0.013 (0.065) [0.96]	−1.148*** (0.376) [0.96]	−0.919** (0.396) [0.91]
Transport accidents	0.012 (0.060) [0.92]	0.040 (0.063) [0.93]	−0.970*** (0.362) [0.92]	−0.758** (0.382) [0.88]
Chronic and other internal	−1.208** (0.586) [0.45]	−1.215** (0.615) [0.52]	−3.917 (3.074) [0.39]	−5.039 (3.138) [0.36]

Notes: Each cell reports the [Sun and Abraham \(2021\)](#) post-period average treatment effect of a county's first facility on the row outcome, under the column's units and fixed effects, with the full control set. Standard errors clustered by county in parentheses; joint pre-trend test p-values in brackets (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). Count regressions are unweighted; per-100,000 regressions are population weighted. The all-cause effect is negative and significant in every cell. Cause-level rate contrasts partly reflect the post-opening population decline documented in [Table A7](#), which is why the cause decomposition in the main text is conducted on counts.

Table A7: Population Response to a First Facility Opening

	Population (persons)		Log population	
	Year FE	State-by-Year FE	Year FE	State-by-Year FE
SA post-period ATT	−341.6*** (95.6)	−324.5*** (117.8)	−0.0160*** (0.0031)	−0.0135*** (0.0032)
Joint pre-trends p	0.984	0.952	0.584	0.308
Pre-period mean	18,152	18,152		

Notes: Sun and Abraham (2021) post-period average treatment effects of a county's first mental health treatment facility on county population, event sample, no additional controls, standard errors clustered by county (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). Populations decline by roughly 1.4 percent after a first facility opens, with no differential pre-trends. Because death counts decline proportionally more (about 3 percent), per-capita mortality falls on net; the population response implies that count-based estimates are the cleaner basis for cause-level decomposition, while population-weighted per-capita estimates remain informative about effects per person.

Table A8: Effects of Treatment Facilities by Education Level

	All Levels (1)	High School or Less (2)	Some College (3)
Number of Facilities	-0.07898*** (0.02974) [-0.26]	-0.09526*** (0.02152) [-0.43]	0.00315 (0.00563) [0.08]
Fixed Effects			
County	Yes	Yes	Yes
Year	Yes	Yes	Yes
State-by-Year	Yes	Yes	Yes
Controls			
Demographic	Yes	Yes	Yes
Economic	Yes	Yes	Yes
Adj. R ²	0.213	0.213	0.213
N	50,512	50,512	50,512

Notes: This table reports the effects of mental health treatment facilities on mortality rates from mental and behavioral disorders (MBD) by education level. Each coefficient represents a separate estimation of Equation 2, where the dependent variable is deaths per 100,000 county residents within each education group. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* $p < .10$ ** $p < .05$ *** $p < .01$). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. All specifications include county, year, and state-by-year fixed effects, as well as demographic and economic controls. Column (1) presents estimates for all education levels combined, while columns (2)-(3) show estimates by education level: high school or less and some college, respectively. See Section 4 for more details on the construction of the variables and data sources. The sample is restricted to 1999-2016.

Table A9: Effects of Treatment Facilities by Race

	All Races (1)	White (2)	Black (3)	Other Races (4)
Number of Facilities	-0.07898*** (0.02974) [-0.26]	-0.09896*** (0.02737) [-0.35]	0.01708 (0.01048) [0.98]	0.00290 (0.00310) [0.84]
Fixed Effects				
County	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
State-by-Year	Yes	Yes	Yes	Yes
Controls				
Demographic	Yes	Yes	Yes	Yes
Economic	Yes	Yes	Yes	Yes
Adj. R ²	0.290	0.290	0.290	0.290
N	53,480	53,480	53,480	53,480

Notes: This table reports the effects of mental health treatment facilities on mortality rates from mental and behavioral disorders (MBD) by race. Each coefficient represents a separate estimation of Equation 2, where the dependent variable is deaths per 100,000 county residents within each racial group. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* p<.10 ** p<.05 *** p<.01). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. All specifications include county, year, and state-by-year fixed effects, as well as demographic and economic controls. Column (1) presents estimates for all races combined, while columns (2)-(4) show estimates by racial group: White, Black, and other races, respectively. See Section 4 for more details on the construction of the variables and data sources. The sample is restricted to 1999-2016.

Table A10: Effects of Treatment Facilities by Gender

	All (1)	Male (2)	Female (3)
Number of Facilities	-0.07898*** (0.02974) [-0.26]	-0.02919*** (0.01090) [-0.26]	-0.04978** (0.01993) [-0.25]
Fixed Effects			
County	Yes	Yes	Yes
Year	Yes	Yes	Yes
State-by-Year	Yes	Yes	Yes
Controls			
Demographic	Yes	Yes	Yes
Economic	Yes	Yes	Yes
Adj. R ²	0.467	0.467	0.467
N	53,480	53,480	53,480

Notes: This table reports the effects of mental health treatment facilities on mortality rates from mental and behavioral disorders (MBD) by gender. Each coefficient represents a separate estimation of Equation 2, where the dependent variable is deaths per 100,000 county residents within each gender group. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* p<.10 ** p<.05 *** p<.01). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. All specifications include county, year, and state-by-year fixed effects, as well as demographic and economic controls. Column (1) presents estimates for all groups combined, while columns (2)-(3) show estimates by gender: male and female, respectively. See Section 4 for more details on the construction of the variables and data sources. The sample is restricted to 1999-2016.

Table A11: Mechanism: Effect of Mental Health Facilities on Disability Claims: TWFE Estimates

	(1)	(2)	(3)	(4)
Mental Health Facilities	8.89042 (2.41119)*** [1.38%]	6.62725 (1.99641)*** [1.03%]	8.98873 (2.41433)*** [1.39%]	8.87719 (2.43903)*** [1.37%]
<i>Fixed Effects:</i>				
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State × Year FE	No	Yes	Yes	Yes
<i>Controls:</i>				
Demographics	No	No	Yes	Yes
Economic	No	No	No	Yes
Observations	21,140	21,140	20,811	20,804
Adjusted R ²	0.630	0.774	0.785	0.787

Notes: This table examines disability claims as a mechanism through which mental health facilities affect population wellbeing. Data from Social Security Administration county-level SSI recipient counts (2009-2016). The outcome is SSI disability recipients aged 18-64 per 100,000 population. Negative coefficients indicate that mental health facilities reduce disability claims, consistent with improved mental health treatment preventing disability. Standard errors clustered at county level in parentheses. Percentage changes relative to pre-2014 mean shown in brackets. *** p<0.01, ** p<0.05, * p<0.10.

Table A12: Mechanism: Mental Health Facilities and Medicaid Expansion on Antidepressant Prescriptions: TWFE Estimates

	(1) Basic	(2) Year FE	(3) Demographics	(4) Full
MH Facilities $\times$ Post $\times$ Expansion	279.586 (54.667)***	279.586 (54.667)***	276.995 (49.234)***	217.894 (64.759)***
<i>Fixed Effects:</i>				
State FE	Yes	Yes	Yes	Yes
Year FE	No	Yes	Yes	Yes
<i>Controls:</i>				
Demographics	No	No	Yes	Yes
Economic	No	No	No	Yes
Observations	144	144	144	144
Adjusted R ²	0.967	0.967	0.968	0.971
Number of States	36	36	36	36

Notes: This table presents state-level evidence on the prescription drug mechanism. The outcome is antidepressant claims per 100,000 population using Medicare Part D data (2013-2016). The triple interaction coefficient captures the differential effect of mental health facilities in Medicaid expansion states after expansion, relative to non-expansion states. Medicaid expansion states are those that expanded between 2014-2016. Demographics include share under 19. Economic controls include employment-population ratio. Standard errors clustered at the state level in parentheses. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A13: Summary Statistics: Treatment and Control Counties

	Full Sample		Control		Treatment		p-value
	Mean	(SD)	Mean	(SD)	Mean	(SD)	
<i>Panel A: Mental Health Treatment Facilities</i>							
Total facilities	0.69	(0.95)	0.00	(0.00)	0.92	(1.00)	0.000***
Facilities per 100,000	3.03	(4.84)	0.00	(0.00)	4.04	(5.21)	0.000***
Outpatient facilities	0.52	(0.78)	0.00	(0.03)	0.70	(0.83)	0.000***
Outpatient per 100,000	2.34	(4.14)	0.00	(0.18)	3.11	(4.52)	0.000***
Inpatient facilities	0.17	(0.50)	0.00	(0.03)	0.23	(0.56)	0.000***
Inpatient per 100,000	0.72	(2.46)	0.00	(0.15)	0.95	(2.79)	0.000***
Annual openings (%)	0.03	(0.17)	0.00	(0.00)	0.04	(0.19)	0.000***
Annual closings (%)	0.04	(0.20)	0.00	(0.00)	0.06	(0.23)	0.000***
<i>Panel B: Mental and Behavioral Disorder Deaths (per 100,000)</i>							
All ages	30.82	(27.52)	30.40	(31.51)	30.96	(26.06)	0.554
Age < 19	0.03	(0.39)	0.02	(0.47)	0.03	(0.36)	0.017**
Ages 20–34	0.20	(1.15)	0.20	(1.45)	0.20	(1.02)	0.955
Ages 35–49	0.79	(2.42)	0.83	(3.15)	0.78	(2.13)	0.331
Ages 50–64	1.56	(3.46)	1.57	(4.40)	1.56	(3.08)	0.794
Ages 65+	28.23	(26.55)	27.77	(30.42)	28.39	(25.13)	0.512
<i>Panel C: MBD Deaths by Race (per 100,000)</i>							
White	28.43	(26.61)	27.27	(30.26)	28.82	(25.26)	0.106
Black	2.15	(5.51)	2.82	(6.85)	1.92	(4.97)	0.002***
Other races	0.24	(1.96)	0.30	(2.75)	0.22	(1.62)	0.207
<i>Panel D: MBD Deaths by Gender (per 100,000)</i>							
Male	11.10	(11.94)	11.00	(14.05)	11.13	(11.14)	0.701
Female	19.72	(19.56)	19.40	(22.93)	19.83	(18.29)	0.522
<i>Panel E: County Characteristics</i>							
Population (100,000s)	0.28	(0.20)	0.17	(0.13)	0.31	(0.20)	0.000***
Employment-population ratio	0.47	(0.13)	0.46	(0.15)	0.47	(0.12)	0.772
Earnings per capita	17.54	(5.98)	17.19	(7.04)	17.66	(5.58)	0.203
Transfers per capita	6.48	(2.33)	6.53	(2.34)	6.46	(2.33)	0.395
Population growth (%)	0.32	(1.40)	0.19	(1.66)	0.36	(1.30)	0.015**
Population density	50.62	(49.53)	31.32	(28.52)	57.01	(53.21)	0.000***
Urban county	0.19	(0.39)	0.15	(0.35)	0.21	(0.41)	0.051*
Population share under 19 (%)	27.13	(3.43)	27.24	(3.92)	27.09	(3.26)	0.473
Population share 20–34 (%)	17.57	(3.15)	17.18	(3.55)	17.69	(2.99)	0.021**
Population share 35–49 (%)	20.50	(2.46)	20.25	(2.53)	20.58	(2.42)	0.008***
Population share 50–64 (%)	18.98	(3.08)	19.01	(3.30)	18.97	(3.00)	0.761
Population share 65+ (%)	15.83	(3.72)	16.31	(4.29)	15.67	(3.50)	0.011**
Counties	1,323		565		758		
County-year observations	26,455		11,295		15,160		

Notes: This table presents summary statistics for counties included in the event study analysis. Treatment counties are those that experience a single mental health facility opening during the sample period with no pre-treatment changes. Control counties never receive a mental health facility. All statistics are weighted by county population. Mental and Behavioral Disorder (MBD) deaths correspond to ICD-10 codes F10–F99. The p-value column reports the significance of the difference in means between treatment and control groups, calculated from regressions with standard errors clustered at the county level. Urban counties are those with Rural-Urban Continuum Codes 1–3. *** p<0.01, ** p<0.05, * p<0.10.

Table A14: Asymmetric Effects of Mental Health Facilities on Disability Claims

	(1)	(2)	(3)	(4)
<i>Panel A: Effect of Facility Openings</i>				
Facilities Opened	1.87502 (2.03998) [0.30%]	1.87502 (2.04198) [0.30%]	7.90303*** (1.99865) [1.25%]	8.09271*** (1.95310) [1.28%]
<i>Panel B: Effect of Facility Closings</i>				
Facilities Closed	-5.88172** (2.91666) [-0.93%]	-5.88172** (2.91952) [-0.93%]	-3.65638 (2.92752) [-0.58%]	-3.32336 (2.89561) [-0.53%]
<i>Asymmetry Test</i>				
H ₀ : Opening effect = -Closing effect	p-value = 0.182			
<i>Fixed Effects:</i>				
County FE	Yes	Yes	Yes	Yes
Year FE	Yes	Yes	Yes	Yes
State × Year FE	No	Yes	Yes	Yes
<i>Controls:</i>				
Demographics	No	No	Yes	Yes
Economic	No	No	No	Yes
Pre-2014 Mean			633.0	
Observations	25,535	25,535	23,792	23,784
Counties			2,973	
Adjusted R ²			0.642	

Notes: This table presents asymmetric effects of mental health facility openings versus closings on disability claims using SSA county-level data (2009-2016). The dependent variable is SSI disability recipients aged 18-64 per 100,000 population. Panel A shows the effect of facility openings (increases in the number of facilities), while Panel B shows the effect of facility closings (decreases in the number of facilities). The asymmetry test examines whether the magnitude of opening effects equals the negative of closing effects. A significant p-value indicates asymmetric effects, suggesting that preventing closures may be more important than facilitating openings for disability prevention. Demographic controls include share under 19. Economic controls include employment-population ratio. Standard errors clustered at the county level in parentheses. Percentage changes relative to pre-2014 mean in square brackets. * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.

Table A15: Effects of Treatment Facilities on Different MBD Outcome and Sub-category: Full Specification

	All MBD-Deaths	F00-F09	F10-F19	F20-F29	F30-F39	F40-F48	F50-F59	F60-F69	F70-F79
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Number of Facilities	-0.07898*** (0.02974)	-0.07747*** (0.02841)	-0.00047 (0.00385)	0.00022 (0.00068)	-0.00028 (0.00087)	-0.00038 (0.00028)	-0.00057 (0.00048)	0.00002 (0.00003)	-0.00027 (0.00057)
Ratio									
Adj. R ²	0.529	0.545	0.106	0.024	0.025	0.013	0.000	-0.003	0.085
N	53,480	53,480	53,480	53,480	53,480	53,480	53,480	53,480	53,480
Fixed Effects									
County	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-by-Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls									
Demographic	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Economic	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table replicates Table A2 using our preferred specification that includes all fixed effects and controls. Heteroskedasticity robust standard errors clustered by county in parentheses (* $p < .10$ ** $p < .05$ *** $p < .01$). The ratio of point estimate and mean is in square brackets. Each column represents a separate regression. All specifications include county and year fixed effects, state-by-year fixed effects, and demographic and economic controls.

Table A16: Sun and Abraham (2021) Estimates of Mental Health Treatment Facility Effects

	(1)	(2)	(3)	(4)
Number of Facilities	-1.800*** (0.170) [-70.6%]	-1.474*** (0.169) [-57.8%]	-1.308*** (0.164) [-51.3%]	-1.287*** (0.163) [-50.5%]
Fixed Effects				
County	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
State-by-Year	No	Yes	Yes	Yes
Controls				
Demographic	No	No	Yes	Yes
Economic	No	No	No	Yes
Adj. R ²	0.760	0.792	0.735	0.735
Observations	27,338	27,238	26,395	26,395
Number of Counties	1,368	1,363	1,320	1,320
Number of Treated Counties	870	868	840	840
Number of Control Counties	649	645	626	626

Notes: This table reports Sun and Abraham (2021) interaction-weighted estimates of the effect of mental health treatment facilities on mortality rates from mental and behavioral disorders (MBD). Each column represents a separate model specification. The dependent variable is deaths per 100,000 county residents. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. Model (1) includes only county and year fixed effects. Model (2) adds state-by-year fixed effects. Model (3) introduces demographic controls. Model (4) adds economic controls. See Section 4 for more details on the construction of the variables and data sources. The sample period is 1999-2016.

Table A17: Dynamic Effects of Mental Health Treatment Facilities on External Cause Mortality: [Sun and Abraham \(2021\)](#) Estimates

Panel A: Dynamic Treatment Effects	
t = -5	0.151 (0.285)
t = -4	0.435 (0.301)
t = -3	-0.029 (0.282)
t = -2	-0.233 (0.260)
t = 0	-0.409 (0.274)
t = 1	-1.114*** (0.189)
t = 2	-1.049*** (0.193)
t = 3	-1.110*** (0.190)
t = 4	-0.796*** (0.193)
t = 5	-0.999*** (0.190)
Panel B: Sample Information	
Number of Treated Counties	783
Number of Control Counties	585
Total Counties	1,319
Observations	23,742
Pre-treatment Mean	2.55
Panel C: Specification	
County FE	Yes
Year FE	Yes
State $\times$ Year FE	Yes
Demographic Controls	Yes
Economic Controls	Yes

Notes: This table presents dynamic treatment effect estimates of mental health treatment facilities on external cause mortality rates using the methodology developed by [Sun and Abraham \(2021\)](#). The estimates account for treatment effect heterogeneity in the presence of staggered treatment timing and multiple treatment cohorts. Panel A reports event-study coefficients for periods $t = -5$ to $t = 5$ relative to treatment timing, with $t = -1$ as the omitted reference period. Estimates are interaction-weighted averages across all treatment cohorts. Standard errors (in parentheses) are clustered at the county level. Treatment counties are defined as those that experience exactly one increase in mental health facilities during the sample period, with no pre-treatment changes. Control counties never experience changes in facility numbers. The sample spans 1999-2016 and includes demographic and economic controls of age, employment-to-population ratio, earnings per capita, transfer payments per capita, hospital presence, population density, rural-urban classification. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table A18: Effects of Treatment Facilities on Different MBD Outcome and Sub-category: Robustness Check with Hospital Controls

	All MBD-Deaths	F00-F09	F10-F19	F20-F29	F30-F39	F40-F48	F50-F59	F60-F69	F70-F79
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
Number of Facilities	-0.07928*** (0.02760)	-0.07979*** (0.02666)	0.00025 (0.00422)	0.00053 (0.00075)	0.00033 (0.00096)	-0.00029 (0.00031)	-0.00052 (0.00053)	0.00003 (0.00004)	-0.00009 (0.00063)
Ratio									
Adj. R ²	0.503	0.526	0.097	0.025	0.024	0.013	-0.001	-0.002	0.089
N	50,512	50,512	50,512	50,512	50,512	50,512	50,512	50,512	50,512
Fixed Effects									
County	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
State-by-Year	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls									
Hospital Establishments	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Demographic	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Economic	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes

Notes: This table replicates Table A2 with additional controls for hospital establishments. Heteroskedasticity robust standard errors clustered by county in parentheses (* p<.10 ** p<.05 *** p<.01). The ratio of point estimate and mean is in square brackets. Each column represents a separate regression. All specifications include county and year fixed effects, state-by-year fixed effects, demographic and economic controls, and controls for the number of hospital establishments in each county.

Table A19: Effects of Treatment Facilities on Other Mortality Outcomes

	Other Deaths (1)	Transport Accidents (2)	Medical/Surgical Care Deaths (3)
Number of Facilities	0.00290 (0.00310) [0.82]	-0.00112 (0.00943) [-0.01]	-0.00091 (0.00125) [-0.08]
Fixed Effects			
County	Yes	Yes	Yes
Year	Yes	Yes	Yes
State-by-Year	Yes	Yes	Yes
Controls			
Demographic	Yes	Yes	Yes
Economic	Yes	Yes	Yes
Adj. R ²	0.290	0.275	0.022
N	53,480	53,480	53,480

Notes: This table reports the effects of mental health treatment facilities on other mortality rates unrelated to mental and behavioral disorders (MBD). Each coefficient represents a separate estimation of Equation 2, where the dependent variable is deaths per 100,000 county residents. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* p<.10 ** p<.05 *** p<.01). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. All specifications include county, year, and state-by-year fixed effects, as well as demographic and economic controls. Column (1) presents estimates for other unclassified deaths, column (2) for deaths from transport accidents, and column (3) for deaths related to medical/surgical care. See Section 4 for more details on the construction of the variables and data sources. The sample is restricted to 1998-2016.

Table A20: Effects of Treatment Facilities on Similar Deaths to MBD

	Assault Deaths (1)	Self-Harm Deaths (2)	Other External Injuries (3)
Number of Facilities	-0.01075** (0.00502) [-0.28]	-0.03083*** (0.00732) [-0.23]	-0.07288*** (0.01668) [-0.26]
Fixed Effects			
County	Yes	Yes	Yes
Year	Yes	Yes	Yes
State-by-Year	Yes	Yes	Yes
Controls			
Demographic	Yes	Yes	Yes
Economic	Yes	Yes	Yes
Adj. R ²	0.270	0.174	0.280
N	53,480	53,480	53,480

Notes: This table reports the effects of mental health treatment facilities on mortality rates from causes similar to mental and behavioral disorders (MBD). Each coefficient represents a separate estimation of Equation 2, where the dependent variable is deaths per 100,000 county residents. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* p<.10 ** p<.05 *** p<.01). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. All specifications include county, year, and state-by-year fixed effects, as well as demographic and economic controls. Column (1) presents estimates for deaths due to assault, column (2) for intentional self-harm deaths, and column (3) for other external injury deaths. See Section 4 for more details on the construction of the variables and data sources. The sample is restricted to 1998-2016.

Table A21: Robustness Check: Effects of Mental Health Treatment Facilities on MBD Mortality Pre-ACA (1999-2013)

	(1)	(2)	(3)	(4)
Dependent Variable: MBD Mortality Rate (per 100,000)				
Number of Facilities	-0.0301 (0.0328) [-0.11%]	-0.0409 (0.0255) [-0.15%]	-0.0535* (0.0307) [-0.19%]	-0.0535* (0.0308) [-0.19%]
Fixed Effects				
County	Yes	Yes	Yes	Yes
Year	Yes	Yes	Yes	Yes
State-by-Year	No	Yes	Yes	Yes
Controls				
Demographic	No	No	Yes	Yes
Economic	No	No	No	Yes
Adj. R ²	0.505	0.523	0.524	0.524
Observations	47,552	47,552	47,552	47,536

Notes: This table reports the robustness check of mental health treatment facilities' effects on mortality rates from mental and behavioral disorders (MBD), restricting the sample to pre-Affordable Care Act years (1999-2013). Each column represents a separate estimation of Equation 2, where the dependent variable is deaths per 100,000 county residents. The main explanatory variable is the number of treatment facilities in a county. Heteroskedasticity-robust standard errors clustered at the county level are reported in parentheses (* p<.10 ** p<.05 *** p<.01). Numbers in square brackets represent the percentage change relative to the pre-treatment mean of the dependent variable. Column (1) presents the baseline specification with county and year fixed effects. Column (2) adds state-by-year fixed effects. Column (3) introduces demographic controls. Column (4) adds economic controls. This robustness check confirms that the main results are not driven by the post-ACA period implementation.

Table A22: Decomposition of Overall Mortality Effects by Cause of Death

Cause of Death	Coefficient	Std. Error	Baseline Mean (per 100k)	Share of Baseline (%)	Contribution to Reduction (%)
Overall Mortality	-1.5586***	(0.2551)	927.41	100.00	–
MBD Deaths	-0.0790***	(0.0297)	31.56	3.40	5.07
Self-Harm/Suicide Deaths	-0.0308***	(0.0073)	13.23	1.43	1.98
Assault Deaths	-0.0107**	(0.0050)	3.80	0.41	0.69
Other External Injuries	-0.0729***	(0.0167)	27.52	2.97	4.68
Transport Accidents	-0.0011	(0.0094)	21.86	2.36	0.07
Medical/Surgical Care Deaths	-0.0009	(0.0012)	1.10	0.12	0.06
Other Deaths	0.0029	(0.0031)	0.36	0.04	-0.19

Notes: This table presents the decomposition of treatment effects on overall mortality by underlying cause of death. Column (1) shows the outcome variable. Column (2) reports point estimates from separate regressions where the dependent variable is deaths per 100,000 county residents from each cause. Column (3) reports heteroskedasticity-robust standard errors clustered at the county level. Column (4) shows the pre-treatment mean mortality rate for each cause. Column (5) reports each cause's share of total baseline mortality, calculated as the ratio of the cause-specific baseline mean to the overall baseline mortality rate. Column (6) shows each component cause's contribution to the overall mortality reduction, calculated as the ratio of the cause-specific coefficient to the overall mortality coefficient. All regressions include county fixed effects, year fixed effects, state-by-year fixed effects, and controls for economic and demographics.

(* p<.10 ** p<.05 *** p<.01)

Table A23: Sensitivity Analysis: Net Benefits Under Alternative Assumptions

VSL Assumption	Cost Scenario		
	Low Cost	Mid Cost	High Cost
\$7 million	\$ 7,285,695	\$ 7,181,511	\$ 7,077,326
\$10 million (baseline)	\$11,823,480	\$11,719,296	\$11,615,111
\$13 million	\$16,361,265	\$16,257,081	\$16,152,897

Notes: This table reports net annual benefits per facility under nine combinations of VSL and cost assumptions. Each cell shows annual benefit (lives saved times VSL) minus total annual cost (operating + annualized capital). Low cost: 3.30M per year (operating + annualized capital at lower bound). Mid cost: 3.41M per year (operating + annualized capital at midpoint). High cost: 3.51M per year (operating + annualized capital at upper bound). Lives saved per facility per year: 1.51 (point estimate from Table 3, Column 4). All nine scenarios show positive net benefits, indicating that mental health facilities generate welfare gains under conservative, baseline, and optimistic assumptions. All values in 2014 dollars.

Table A24: Aggregate National Welfare Effects of Mental Health Facilities

Effect	Annual Estimate
Average annual facility stock	14,928
Lives saved per year	22,541
Total annual benefits (VSL = \$10M)	\$225.4 billion
Total annual costs	\$50.9 billion
Net annual benefits	\$174.6 billion

Notes: This table aggregates the facility-level welfare effects to the national level. The sample contains 268,699 facility-year observations over the eighteen-year panel (1999-2016), corresponding to an average stock of approximately 14,928 facilities operating in a given year across the sample counties. Lives saved per year equal the average annual facility stock times lives saved per facility per year (1.51). Total annual benefits value these lives at the baseline VSL of \$10 million. Total annual costs apply the midpoint total annual cost of \$3.41 million per facility to the average annual stock. Net annual benefits equal total benefits minus total costs, equivalent to roughly \$550 per capita. As a check on magnitudes, 22,541 averted deaths represent just under one percent of annual deaths in the sample, consistent with the reduced-form estimates aggregated over the observed facility stock. These estimates represent the aggregate welfare effects across the sample of counties and provide a lower bound for national effects, as the sample does not include all U.S. counties and excludes morbidity reductions, quality-of-life improvements, and other positive externalities of mental health treatment. All values in 2014 dollars.

Table A25: Classification of Mental and Behavioral Disorders Related Mortality

Category Code	Description	Classification Group
F00-F09	Organic Mental Disorders ^a	Primary Disorders
F10-F19	Mental and Behavioral Disorders due to Psychoactive Substance Use	Substance-Related
F20-F29	Schizophrenia, Schizotypal and Delusional Disorders	Psychotic Disorders
F30-F39	Mood [Affective] Disorders	Affective Disorders
F40-F48	Neurotic, Stress-related and Somatoform Disorders	Anxiety-Related
F50-F59	Behavioral Syndromes Associated with Physiological Disturbances	Physiological
F60-F69	Disorders of Adult Personality and Behavior	Personality
F70-F79	Mental Retardation	Developmental
F80-F89	Disorders of Psychological Development	Developmental
F90-F98	Behavioral and Emotional Disorders with Childhood Onset	Early-Onset
F99	Unspecified Mental Disorder	Other

^a Including dementia in Alzheimer's disease (F00*) and other diseases classified elsewhere (F02*)

Note: Classification based on ICD-10 Chapter V: Mental and Behavioral Disorders (World Health Organization, 2019).

Asterisk (*) categories indicate special coding for certain conditions.

Mental and Behavioural Disorders (ICD-10 Section V)

Table A26: State Medicaid Expansion Status and Timing Under the ACA

State	Expansion Status	Implementation Date
Arizona	Expansion	January 2014
Arkansas	Expansion	January 2014
California	Expansion	January 2014
Colorado	Expansion	January 2014
Connecticut	Early Expansion	January 2014 ^a
Delaware	Early Expansion	January 2014 ^a
Hawaii	Expansion	January 2014
Illinois	Expansion	January 2014
Iowa	Expansion	January 2014
Kentucky	Expansion	January 2014
Maryland	Expansion	January 2014
Massachusetts	Early Expansion	January 2014 ^a
Michigan	Expansion	January 2014
Minnesota	Expansion	January 2014
Nevada	Expansion	January 2014
New Hampshire	Expansion	January 2014
New Jersey	Expansion	January 2014
New Mexico	Expansion	January 2014
New York	Early Expansion	January 2014 ^a
North Dakota	Expansion	January 2014
Ohio	Expansion	January 2014
Oregon	Expansion	January 2014
Rhode Island	Expansion	January 2014
Vermont	Early Expansion	January 2014 ^a
Washington	Expansion	January 2014
West Virginia	Expansion	January 2014
Indiana	Expansion	February 2015
New Hampshire	Expansion	August 2015
Pennsylvania	Expansion	January 2015
Louisiana	Expansion	July 2016
Montana	Expansion	January 2016

^a These states implemented early expansion under Section 1115 waivers prior to 2014.

Notes: This table shows the timing of Medicaid expansion adoption under the Affordable Care Act. Early expansion states implemented expansion through Section 1115 waivers prior to 2014. Large expansion states (California, Iowa, Minnesota, Hawaii, Indiana, Maryland, Connecticut, Wisconsin) experienced particularly large increases in Medicaid enrollment following expansion. Data sources: Medicaid expansion dates from Kaiser Family Foundation and state websites. Early expansion classification follows NEJM appendix [Miller and Wherry \(2017\)](#). Large expansion states identified following [Carey et al. \(2020\)](#).

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